PROBLEMS SHEET 4, PART 2 • OFFICIAL SOLUTIONS

Department of Mathematics • UC3M Aerospace Engineering

Chapter 4 Problem Set, Part 2: Complex and Repeated Eigenvalues, Spirals and Improper Nodes

Complete, step-by-step solutions to the eight exercises of the second Chapter 4 sheet: phase portraits and general solutions from complex eigen-data, canonical forms with Q = [u | w], independence of real and imaginary parts, building matrices from eigen-data, repeated eigenvalues with the Jordan chain, the nilpotent case and the full qualitative catalogue of linear portraits.

Source Document:	sources/cuatrimestre-1/04-advanced-maths/unit-04-systems-of-odes/problemas/probls_ch4-ii_2627.pdf

EXERCISE 1

Eight Spiral and Centre Portraits from Complex Eigen-data

Original statement · probls_ch4-ii_2627.pdf

Complex Eigenvalues • Spirals and Centres • Eight Portraits

STATEMENT

Draw the phase portrait for the equation $d\mathbf{x}/dt = A\mathbf{x}$, when the eigenvalues ($\lambda_\pm$) and eigenvectors ($\boldsymbol{\eta}_\pm$) of A are as given. The sign of $\dot{x}$ is indicated when $x = 0$ and $y > 0$.

- (i) $\lambda_\pm = 1 \pm 3i$ with $\boldsymbol{\eta}_\pm = (1,\ 2 \mp i)$, $\dot{x} < 0$;
- (ii) $\lambda_\pm = \pm 3i$ with $\boldsymbol{\eta}_\pm = (1 \pm 2i,\ 1 \mp 3i)$, $\dot{x} < 0$;
- (iii) $\lambda_\pm = -2 \pm i$ with $\boldsymbol{\eta}_\pm = (1 \mp i,\ 3 \pm i)$, $\dot{x} > 0$;
- (iv) $\lambda_\pm = -1 \pm i$ with $\boldsymbol{\eta}_\pm = (1,\ \pm i)$, $\dot{x} > 0$;
- (v) $\lambda_\pm = 2 \pm 2i$ with $\boldsymbol{\eta}_\pm = (\pm 3i,\ 5 \mp 4i)$, $\dot{x} < 0$;
- (vi) $\lambda_\pm = 5 \pm 3i$ with $\boldsymbol{\eta}_\pm = (2 \pm 5i,\ \mp i)$, $\dot{x} < 0$;
- (vii) $\lambda_\pm = \pm 7i$ with $\boldsymbol{\eta}_\pm = (1 \pm i,\ -1 \pm 2i)$, $\dot{x} > 0$;
- (viii) $\lambda_\pm = -13 \pm 17i$ with $\boldsymbol{\eta}_\pm = (\pm 6i - 8,\ 4 \mp 5i)$, $\dot{x} > 0$.

PHASE 1: SETUP AND CLASSIFICATION

In each item $\lambda_{\pm} = \rho \pm i\omega$ with $\omega > 0$ and $\boldsymbol{\eta}_{+} = \mathbf{u} + i\mathbf{w}$ is the eigenvector of $\lambda_{+} = \rho + i\omega$ (the second eigenvector $\boldsymbol{\eta}_{-} = \bar{\boldsymbol{\eta}}_{+}$ carries no new information). The type is fixed by ρ: $\rho < 0$ stable spiral, $\rho > 0$ unstable spiral, $\rho = 0$ centre (Topic 4 Sec. 7.6). The sense of rotation follows from the vector field at $(0,1)^T$: $\dot{\mathbf{x}} = A(0,1)^T = (a_{12}, a_{22})^T$, and $\dot{x} = a_{12} > 0$ means clockwise, $\dot{x} < 0$ counter-clockwise (Topic 4 Sec. 7.8). The sign of $\dot{x}$ given in the statement is therefore the sense of rotation:

Item	ρ	ω	**Type**	**Given sign of $\dot{x}$ at $(0,1)$**	**Sense**
(i)	1	3	unstable spiral	< 0	counter-clockwise
(ii)	0	3	centre	< 0	counter-clockwise
(iii)	-2	1	stable spiral	> 0	clockwise
(iv)	-1	1	stable spiral	> 0	clockwise
(v)	2	2	unstable spiral	< 0	counter-clockwise
(vi)	5	3	unstable spiral	< 0	counter-clockwise
(vii)	0	7	centre	> 0	clockwise
(viii)	-13	17	stable spiral	> 0	clockwise

PHASE 2: FORMULATION

For each item the eigenvector gives $\mathbf{u} = \operatorname{Re}\boldsymbol{\eta}_{+}$ and $\mathbf{w} = \operatorname{Im}\boldsymbol{\eta}_{+}$, hence $Q = [\mathbf{u} \mid \mathbf{w}]$ and the canonical form $J = \begin{pmatrix} \rho & \omega \\ -\omega & \rho \end{pmatrix}$ (rotation clockwise in the $\mathbf{z}$-plane). The matrix $A = QJQ^{-1}$ is rebuilt, the given sign of $\dot{x}$ is checked against a_{12}, and the portrait is drawn in the $\mathbf{z}$-plane and in the (x,y)-plane ($\mathbf{x} = Q\mathbf{z}$).

METHOD OF THE CLASS NOTES [NOTES, P. 10]

Case 2 of the notes: with $\mathbf{v}_{\pm} = \mathbf{v}_1 \pm i\mathbf{v}_2$ (here $\mathbf{u} = \mathbf{v}_1, \mathbf{w} = \mathbf{v}_2$) and $Q = (\mathbf{v}_1 \mid \mathbf{v}_2)$, $\mathbf{z} = Q^{-1}\mathbf{x}$ satisfies $\dot{\mathbf{z}} = \begin{pmatrix} \rho & \omega \\ -\omega & \rho \end{pmatrix}\mathbf{z}$. In polar form $r = r(0)e^{\rho t}$ and $\theta = \theta(0) - \omega t$: clockwise in the $\mathbf{z}$-plane, a stable spiral for $\rho < 0$, an unstable spiral for $\rho > 0$ and a centre for $\rho = 0$. The sense of rotation in the (x,y)-plane is read, as in the example of the notes, from the sign of $\dot{x}$ on the positive y-axis.

PHASE 3: DERIVATION OF THE EIGHT PORTRAITS

(i) $\lambda_{\pm} = 1 \pm 3i$ with $\boldsymbol{\eta}_{\pm} = (1,\ 2 \mp i),\ \dot{x} < 0$

Split the eigenvector of $\lambda_{+} = 1 + 3i$ into real and imaginary parts, $\boldsymbol{\eta}_{+} = \begin{pmatrix} 1 \\ 2 - i \end{pmatrix} = \mathbf{u} + i\mathbf{w}$, so $\rho = 1, \omega = 3$ and

$$\mathbf{u} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \qquad \mathbf{w} = \begin{pmatrix} 0 \\ -1 \end{pmatrix}$$

The matrix is not given; it is rebuilt from $A\mathbf{u} = \rho\mathbf{u} - \omega\mathbf{w}$, $A\mathbf{w} = \omega\mathbf{u} + \rho\mathbf{w}$ (Topic 4 Eq. 7.4), which say $AQ = QJ$ with $Q = [\mathbf{u} \mid \mathbf{w}]$, so $A = QJQ^{-1}$:

$$Q = \begin{pmatrix} 1 & 0 \\ 2 & -1 \end{pmatrix}, \quad J = \begin{pmatrix} 1 & 3 \\ -3 & 1 \end{pmatrix}, \quad \det Q = -1, \quad Q^{-1} = \begin{pmatrix} 1 & 0 \\ 2 & -1 \end{pmatrix}$$

$$\begin{aligned} (QJ)_{11} &= 1 \cdot 1 + 0 \cdot (-3) = 1 \\ (QJ)_{12} &= 1 \cdot 3 + 0 \cdot 1 = 3 \\ (QJ)_{21} &= 2 \cdot 1 + (-1) \cdot (-3) = 5 \\ (QJ)_{22} &= 2 \cdot 3 + (-1) \cdot 1 = 5 \end{aligned}$$

$$QJ = \begin{pmatrix} 1 & 3 \\ 5 & 5 \end{pmatrix}$$

$$\begin{aligned} (A)_{11} &= 1 \cdot 1 + 3 \cdot 2 = 7 \\ (A)_{12} &= 1 \cdot 0 + 3 \cdot (-1) = -3 \\ (A)_{21} &= 5 \cdot 1 + 5 \cdot 2 = 15 \\ (A)_{22} &= 5 \cdot 0 + 5 \cdot (-1) = -5 \end{aligned}$$

$$A = QJQ^{-1} = \begin{pmatrix} 7 & -3 \\ 15 & -5 \end{pmatrix}$$

Check of the given sign: $\dot{x}$ at $(0, 1)$ equals $a_{12} = -3$, which is negative, in agreement with the statement ($\dot{x} < 0$).

How A acts on $\mathbf{u}$ and $\mathbf{w}$

Separating the real and imaginary parts of $A\mathbf{v} = \lambda\mathbf{v}$ gives (Topic 4, Eq. 7.4) $A\mathbf{u} = \rho\mathbf{u} - \omega\mathbf{w}$ and $A\mathbf{w} = \omega\mathbf{u} + \rho\mathbf{w}$. Verification with the numbers:

$$A\mathbf{u} = \begin{pmatrix} 1 \\ 5 \end{pmatrix}, \qquad \rho\mathbf{u} - \omega\mathbf{w} = 1\begin{pmatrix} 1 \\ 2 \end{pmatrix} - 3\begin{pmatrix} 0 \\ -1 \end{pmatrix} = \begin{pmatrix} 1 \\ 5 \end{pmatrix}$$

$$A\mathbf{w} = \begin{pmatrix} 3 \\ 5 \end{pmatrix}, \qquad \omega\mathbf{u} + \rho\mathbf{w} = 3\begin{pmatrix} 1 \\ 2 \end{pmatrix} + 1\begin{pmatrix} 0 \\ -1 \end{pmatrix} = \begin{pmatrix} 3 \\ 5 \end{pmatrix}$$

Change of coordinates

$\mathbf{u}$ and $\mathbf{w}$ are linearly independent (they are not parallel when $\omega \neq 0$; Topic 4, Lemma of Sec. 7.3). With the columns in the order $\mathbf{u}, \mathbf{w}$,

$$Q = [\mathbf{u} \mid \mathbf{w}] = \begin{pmatrix} 1 & 0 \\ 2 & -1 \end{pmatrix}$$

$\det Q = (1)(-1) - (0)(2) = -1$, which is not zero, so Q is invertible. For a 2×2 matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad-bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$ (Robinson, Appendix B):

$$Q^{-1} = \frac{1}{-1}\begin{pmatrix} -1 & 0 \\ -2 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 2 & -1 \end{pmatrix}$$

Compute $AQ = [A\mathbf{u} \mid A\mathbf{w}]$ and multiply by Q^{-1} on the left, entry by entry:

$$AQ = \begin{pmatrix} 1 & 3 \\ 5 & 5 \end{pmatrix}$$

$$\begin{aligned} (Q^{-1}AQ)_{11} &= 1 \cdot 1 + 0 \cdot 5 = 1 \\ (Q^{-1}AQ)_{12} &= 1 \cdot 3 + 0 \cdot 5 = 3 \\ (Q^{-1}AQ)_{21} &= 2 \cdot 1 + (-1) \cdot 5 = -3 \\ (Q^{-1}AQ)_{22} &= 2 \cdot 3 + (-1) \cdot 5 = 1 \end{aligned}$$

$$Q^{-1}AQ = \begin{pmatrix} 1 & 3 \\ -3 & 1 \end{pmatrix} = J = \begin{pmatrix} \rho & \omega \\ -\omega & \rho \end{pmatrix}$$

Solution in the canonical plane

$$\dot{z}_1 = z_1 + 3z_2, \qquad \dot{z}_2 = -3z_1 + z_2$$

The two equations are coupled, but they say "scale and rotate", which polar coordinates express directly: $z_1 = r\cos\theta$, $z_2 = r\sin\theta$ (Topic 4, Sec. 7.5).

The radius. From $r^2 = z_1^2 + z_2^2$ the chain rule gives $2r\dot{r} = 2z_1\dot{z}_1 + 2z_2\dot{z}_2$. Substituting $\dot{z}_1 = \rho z_1 + \omega z_2$ and $\dot{z}_2 = -\omega z_1 + \rho z_2$:

$$r\dot{r} = z_1(\rho z_1 + \omega z_2) + z_2(-\omega z_1 + \rho z_2) = \rho z_1^2 + \omega z_1 z_2 - \omega z_1 z_2 + \rho z_2^2 = \rho r^2$$

The terms in ω cancel. Dividing by $r \neq 0$ (the origin is the equilibrium): $\dot{r} = \rho r$.

The angle. Differentiate $\tan\theta = z_2/z_1$ with the chain rule on the left and the quotient rule on the right: $\sec^2\theta\,\dot{\theta} = (z_1\dot{z}_2 - z_2\dot{z}_1)/z_1^2$. Since $\sec^2\theta = r^2/z_1^2$, $\dot{\theta} = (z_1\dot{z}_2 - z_2\dot{z}_1)/r^2$, and

$$z_1\dot{z}_2 - z_2\dot{z}_1 = z_1(-\omega z_1 + \rho z_2) - z_2(\rho z_1 + \omega z_2) = -\omega z_1^2 + \rho z_1 z_2 - \rho z_1 z_2 - \omega z_2^2 = -\omega r^2 \implies \dot{\theta} = -\omega$$

Integration. $\frac{d}{dt}\left(e^{-\rho t}r\right) = e^{-\rho t}(\dot{r} - \rho r) = 0$, and Barrow's rule between 0 and t, $\left[e^{-\rho s}r(s)\right]_0^t = 0$, gives $r(t) = r_0 e^{\rho t}$. For the angle, $\theta(t) - \theta_0 = \int_0^t (-\omega)\,ds = -\omega t$. Back to Cartesian coordinates, with $c_1 = r_0\cos\theta_0 = z_1(0)$, $c_2 = r_0\sin\theta_0 = z_2(0)$ and the subtraction formulas $\cos(\theta_0 - \omega t) = \cos\theta_0\cos\omega t + \sin\theta_0\sin\omega t$, $\sin(\theta_0 - \omega t) = \sin\theta_0\cos\omega t - \cos\theta_0\sin\omega t$:

$$z_1(t) = e^{1t}\left(c_1\cos 3t + c_2\sin 3t\right), \qquad z_2(t) = e^{1t}\left(c_2\cos 3t - c_1\sin 3t\right)$$

Check by substitution (product rule and chain rule): $\dot{z}_1 = \rho z_1 + e^{\rho t}\omega\left(-c_1\sin\omega t + c_2\cos\omega t\right) = \rho z_1 + \omega z_2$ and $\dot{z}_2 = \rho z_2 + e^{\rho t}\omega\left(-c_2\sin\omega t - c_1\cos\omega t\right) = \rho z_2 - \omega z_1$, as required.

Back to the original coordinates

$\mathbf{x} = Q\mathbf{z} = z_1\mathbf{u} + z_2\mathbf{w}$, so

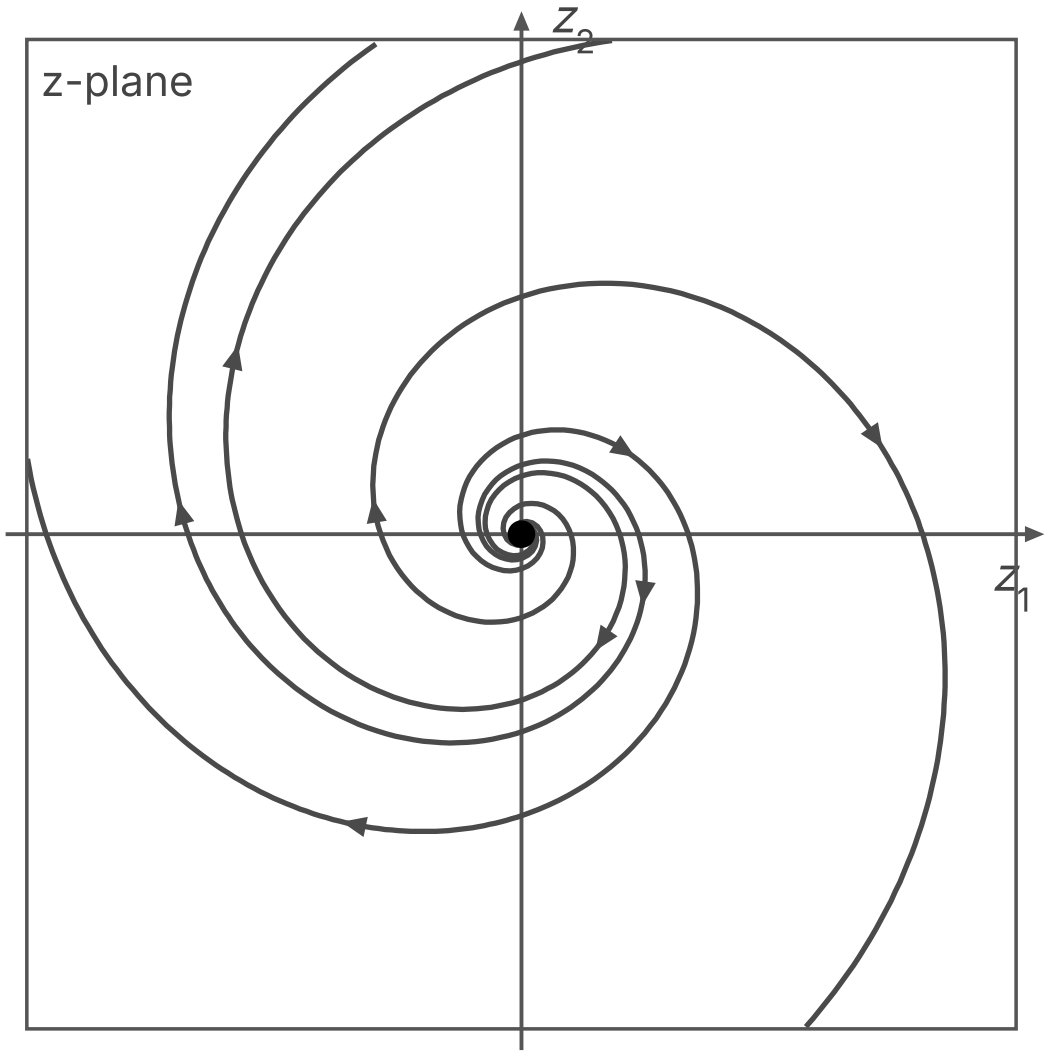
$$\mathbf{x}(t) = e^{1t}\left[(c_1\cos 3t + c_2\sin 3t)\begin{pmatrix}1\\2\end{pmatrix} + (c_2\cos 3t - c_1\sin 3t)\begin{pmatrix}0\\-1\end{pmatrix}\right]$$

$$x(t) = c_1 e^t\cos 3t + c_2 e^t\sin 3t, \qquad y(t) = c_1 e^t\sin 3t + 2c_1 e^t\cos 3t + 2c_2 e^t\sin 3t - c_2 e^t\cos 3t$$

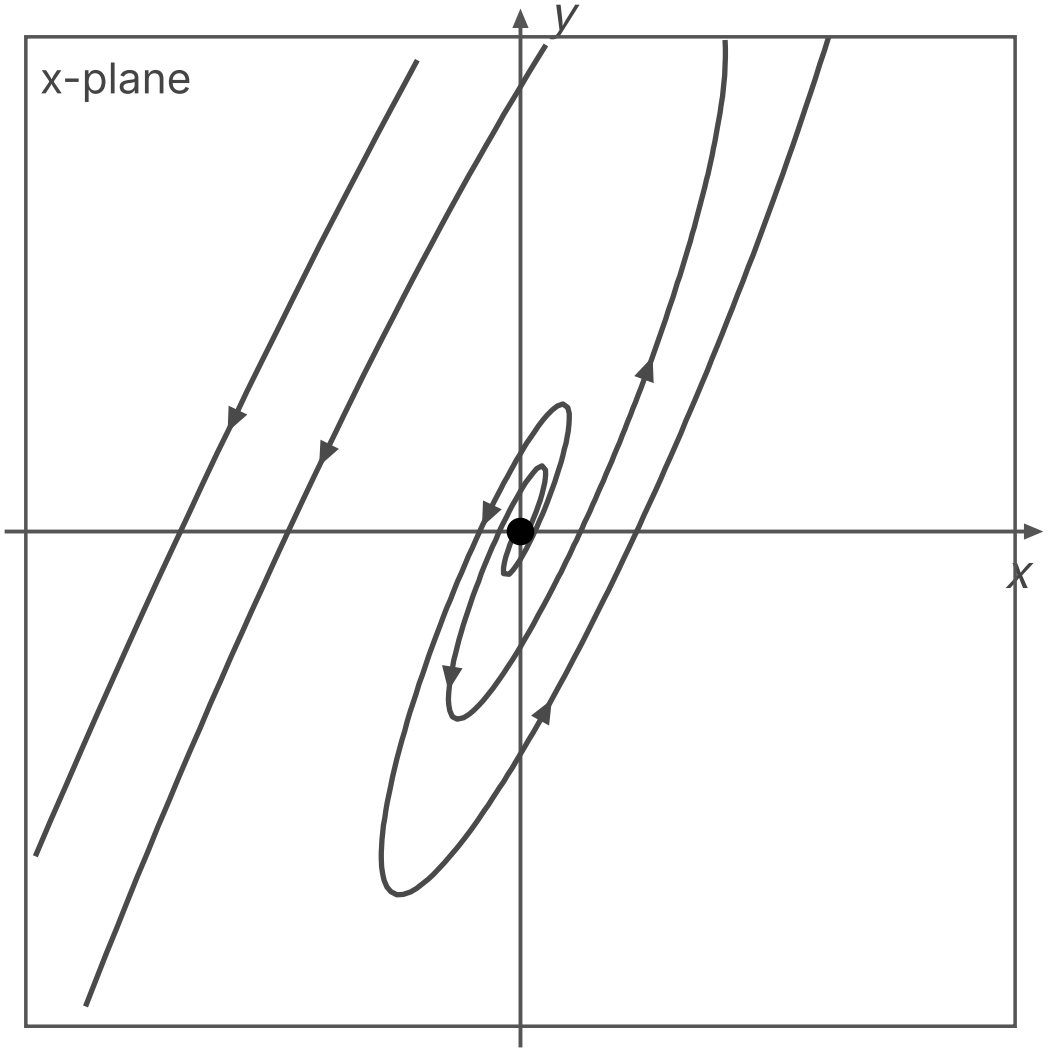
Phase portrait

$\rho = 1$ and $\omega = 3$: the origin is a **unstable spiral**. In the $\mathbf{z}$-plane $r = r_0 e^{1t}$ and $\theta = \theta_0 - 3t$: the motion is always **clockwise** ($\dot\theta = -\omega < 0$), and one full turn takes $T = 2\pi/\omega$ while the radius is multiplied by $e^{2\pi\rho/\omega} = e^{\frac{2\pi}{3}}$ (growing).

Sense of rotation in the $\mathbf{x}$-plane, tested directly on the vector field (Topic 4 Sec. 7.8): at $(0,1)^T$, $\dot{\mathbf{x}} = A(0,1)^T = \begin{pmatrix} -3 \\ -5 \end{pmatrix}$, so $\dot x = -3$ is < 0: to the left at the top of the plane: the rotation is **counter-clockwise**. Cross-check: $\det Q = -1$ is < 0, so Q reverses orientation and the clockwise motion of the $\mathbf{z}$-plane becomes counter-clockwise in the $\mathbf{x}$-plane, in agreement.



Canonical $\mathbf{z}$-plane (unstable spiral): clockwise, $\dot\theta = -\omega$.



Original (x, y)-plane (unstable spiral), counter-clockwise.

(ii) $\lambda_{\pm} = \pm 3i$ with $\boldsymbol{\eta}_{\pm} = (1 \pm 2i,\ 1 \mp 3i)$, $\dot{x} < 0$

Split the eigenvector of $\lambda_{+} = 3i$ into real and imaginary parts, $\boldsymbol{\eta}_{+} = \begin{pmatrix} 1 + 2i \\ 1 - 3i \end{pmatrix} = \mathbf{u} + i\mathbf{w}$, so $\rho = 0$, $\omega = 3$ and

$$\mathbf{u} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \qquad \mathbf{w} = \begin{pmatrix} 2 \\ -3 \end{pmatrix}$$

The matrix is not given; it is rebuilt from $A\mathbf{u} = \rho\mathbf{u} - \omega\mathbf{w}$, $A\mathbf{w} = \omega\mathbf{u} + \rho\mathbf{w}$ (Topic 4 Eq. 7.4), which say $AQ = QJ$ with $Q = [\mathbf{u} \mid \mathbf{w}]$, so $A = QJQ^{-1}$:

$$Q = \begin{pmatrix} 1 & 2 \\ 1 & -3 \end{pmatrix}, \quad J = \begin{pmatrix} 0 & 3 \\ -3 & 0 \end{pmatrix}, \quad \det Q = -5, \quad Q^{-1} = \begin{pmatrix} \frac{3}{5} & \frac{2}{5} \\ \frac{1}{5} & -\frac{1}{5} \end{pmatrix}$$

$$
\begin{aligned}
(QJ)_{11} &= 1\cdot 0 + 2\cdot(-3) = -6\\
(QJ)_{12} &= 1\cdot 3 + 2\cdot 0 = 3\\
(QJ)_{21} &= 1\cdot 0 + (-3)\cdot(-3) = 9\\
(QJ)_{22} &= 1\cdot 3 + (-3)\cdot 0 = 3
\end{aligned}
$$

$$
QJ = \begin{pmatrix} -6 & 3 \\ 9 & 3 \end{pmatrix}
$$

$$
\begin{aligned}
(A)_{11} &= (-6)\cdot\frac{3}{5} + 3\cdot\frac{1}{5} = -3\\
(A)_{12} &= (-6)\cdot\frac{2}{5} + 3\cdot(-\frac{1}{5}) = -3\\
(A)_{21} &= 9\cdot\frac{3}{5} + 3\cdot\frac{1}{5} = 6\\
(A)_{22} &= 9\cdot\frac{2}{5} + 3\cdot(-\frac{1}{5}) = 3
\end{aligned}
$$

$$
A = QJQ^{-1} = \begin{pmatrix} -3 & -3 \\ 6 & 3 \end{pmatrix}
$$

Check of the given sign: $\dot{x}$ at $(0,1)$ equals $a_{12} = -3$, which is negative, in agreement with the statement ($\dot{x} < 0$).

How A acts on $\mathbf{u}$ and $\mathbf{w}$

Separating the real and imaginary parts of $A\mathbf{v} = \lambda\mathbf{v}$ gives (Topic 4, Eq. 7.4) $A\mathbf{u} = \rho\mathbf{u} - \omega\mathbf{w}$ and $A\mathbf{w} = \omega\mathbf{u} + \rho\mathbf{w}$. Verification with the numbers:

$$
A\mathbf{u} = \begin{pmatrix} -6 \\ 9 \end{pmatrix}, \qquad \rho\mathbf{u} - \omega\mathbf{w} = 0\begin{pmatrix} 1 \\ 1 \end{pmatrix} - 3\begin{pmatrix} 2 \\ -3 \end{pmatrix} = \begin{pmatrix} -6 \\ 9 \end{pmatrix}
$$

$$
A\mathbf{w} = \begin{pmatrix} 3 \\ 3 \end{pmatrix}, \qquad \omega\mathbf{u} + \rho\mathbf{w} = 3\begin{pmatrix} 1 \\ 1 \end{pmatrix} + 0\begin{pmatrix} 2 \\ -3 \end{pmatrix} = \begin{pmatrix} 3 \\ 3 \end{pmatrix}
$$

Change of coordinates

$\mathbf{u}$ and $\mathbf{w}$ are linearly independent (they are not parallel when $\omega \neq 0$; Topic 4, Lemma of Sec. 7.3). With the columns in the order $\mathbf{u}$, $\mathbf{w}$,

$$
Q = [\mathbf{u} \mid \mathbf{w}] = \begin{pmatrix} 1 & 2 \\ 1 & -3 \end{pmatrix}
$$

$\det Q = (1)(-3) - (2)(1) = -5$, which is not zero, so Q is invertible. For a 2×2 matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad-bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$ (Robinson, Appendix B):

$$Q^{-1} = \frac{1}{-5}\begin{pmatrix} -3 & -2 \\ -1 & 1 \end{pmatrix} = \begin{pmatrix} \frac{3}{5} & \frac{2}{5} \\ \frac{1}{5} & -\frac{1}{5} \end{pmatrix}$$

Compute $AQ = [A\mathbf{u} \mid A\mathbf{w}]$ and multiply by Q^{-1} on the left, entry by entry:

$$AQ = \begin{pmatrix} -6 & 3 \\ 9 & 3 \end{pmatrix}$$

$$\begin{aligned} (Q^{-1}AQ)_{11} &= \frac{3}{5}\cdot(-6) + \frac{2}{5}\cdot 9 = 0 \\ (Q^{-1}AQ)_{12} &= \frac{3}{5}\cdot 3 + \frac{2}{5}\cdot 3 = 3 \\ (Q^{-1}AQ)_{21} &= \frac{1}{5}\cdot(-6) + (-\frac{1}{5})\cdot 9 = -3 \\ (Q^{-1}AQ)_{22} &= \frac{1}{5}\cdot 3 + (-\frac{1}{5})\cdot 3 = 0 \end{aligned}$$

$$Q^{-1}AQ = \begin{pmatrix} 0 & 3 \\ -3 & 0 \end{pmatrix} = J = \begin{pmatrix} \rho & \omega \\ -\omega & \rho \end{pmatrix}$$

Solution in the canonical plane

$$\dot{z}_1 = 3z_2, \qquad \dot{z}_2 = -3z_1$$

The two equations are coupled, but they say "scale and rotate", which polar coordinates express directly: $z_1 = r\cos\theta$, $z_2 = r\sin\theta$ (Topic 4, Sec. 7.5).

The radius. From $r^2 = z_1^2 + z_2^2$ the chain rule gives $2r\dot{r} = 2z_1\dot{z}_1 + 2z_2\dot{z}_2$. Substituting $\dot{z}_1 = \rho z_1 + \omega z_2$ and $\dot{z}_2 = -\omega z_1 + \rho z_2$:

$$r\dot{r} = z_1(\rho z_1 + \omega z_2) + z_2(-\omega z_1 + \rho z_2) = \rho z_1^2 + \omega z_1 z_2 - \omega z_1 z_2 + \rho z_2^2 = \rho r^2$$

The terms in ω cancel. Dividing by $r \neq 0$ (the origin is the equilibrium): $\dot{r} = \rho r$.

The angle. Differentiate $\tan\theta = z_2/z_1$ with the chain rule on the left and the quotient rule on the right: $\sec^2\theta\,\dot{\theta} = (z_1\dot{z}_2 - z_2\dot{z}_1)/z_1^2$. Since $\sec^2\theta = r^2/z_1^2$, $\dot{\theta} = (z_1\dot{z}_2 - z_2\dot{z}_1)/r^2$, and

$$z_1\dot{z}_2 - z_2\dot{z}_1 = z_1(-\omega z_1 + \rho z_2) - z_2(\rho z_1 + \omega z_2) = -\omega z_1^2 + \rho z_1 z_2 - \rho z_1 z_2 - \omega z_2^2 = -\omega r^2 \implies \dot{\theta} = -\omega$$

Integration. $\frac{d}{dt}\left(e^{-\rho t} r\right) = e^{-\rho t}(\dot{r} - \rho r) = 0$, and Barrow's rule between 0 and t, $\left[e^{-\rho s} r(s)\right]_0^t = 0$, gives $r(t) = r_0 e^{\rho t}$. For the angle, $\theta(t) - \theta_0 = \int_0^t (-\omega)\, ds = -\omega t$. Back to Cartesian coordinates, with $c_1 = r_0 \cos\theta_0 = z_1(0)$, $c_2 = r_0 \sin\theta_0 = z_2(0)$ and the subtraction formulas $\cos(\theta_0 - \omega t) = \cos\theta_0 \cos\omega t + \sin\theta_0 \sin\omega t$, $\sin(\theta_0 - \omega t) = \sin\theta_0 \cos\omega t - \cos\theta_0 \sin\omega t$:

$$z_1(t) = e^{0t}\left(c_1 \cos 3t + c_2 \sin 3t\right), \qquad z_2(t) = e^{0t}\left(c_2 \cos 3t - c_1 \sin 3t\right)$$

Check by substitution (product rule and chain rule): $\dot{z}_1 = \rho z_1 + e^{\rho t}\omega\left(-c_1 \sin\omega t + c_2 \cos\omega t\right) = \rho z_1 + \omega z_2$ and $\dot{z}_2 = \rho z_2 + e^{\rho t}\omega\left(-c_2 \sin\omega t - c_1 \cos\omega t\right) = \rho z_2 - \omega z_1$, as required.

Back to the original coordinates

$\mathbf{x} = Q\mathbf{z} = z_1\mathbf{u} + z_2\mathbf{w}$, so

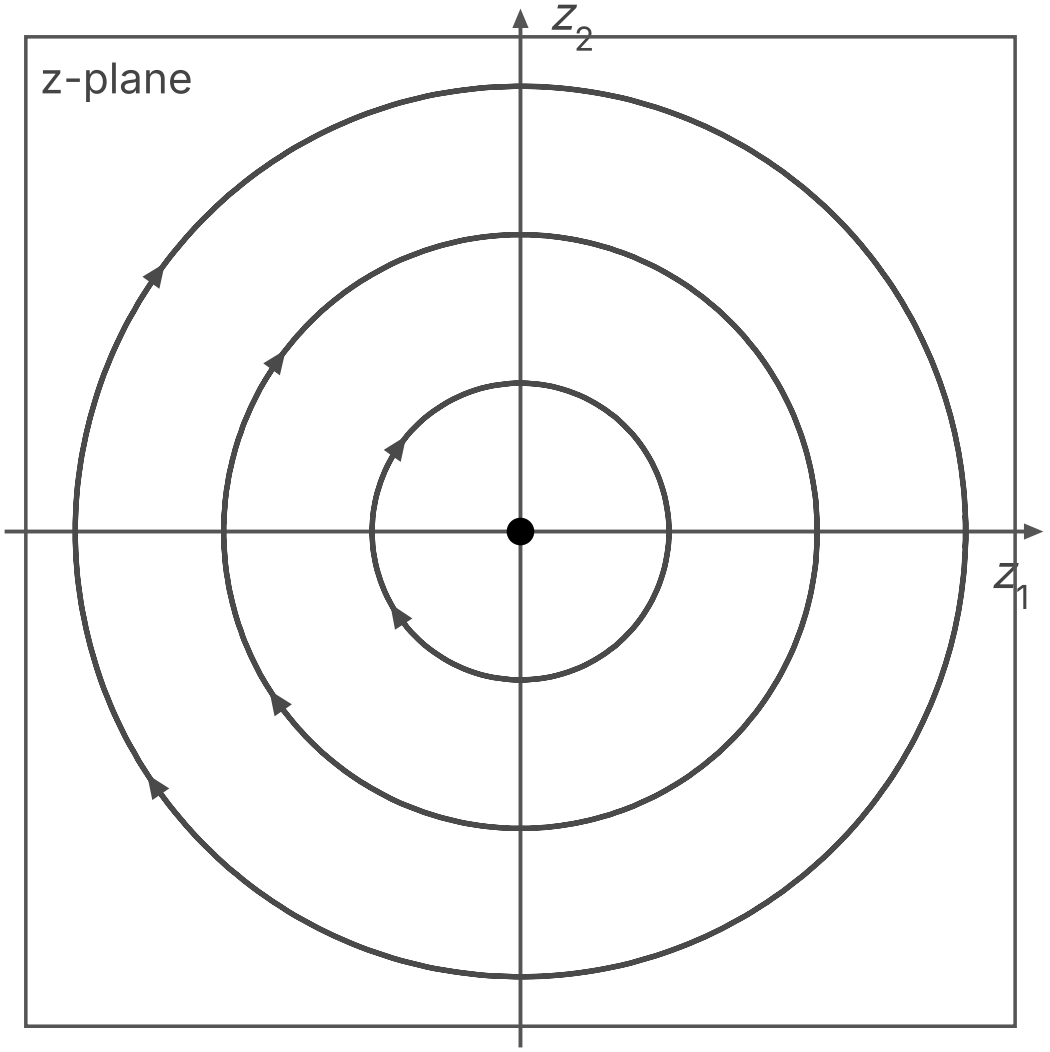
$$\mathbf{x}(t) = e^{0t}\left[\left(c_1 \cos 3t + c_2 \sin 3t\right)\begin{pmatrix}1\\1\end{pmatrix} + \left(c_2 \cos 3t - c_1 \sin 3t\right)\begin{pmatrix}2\\-3\end{pmatrix}\right]$$

$$x(t) = -2c_1 \sin 3t + c_1 \cos 3t + c_2 \sin 3t + 2c_2 \cos 3t, \qquad y(t) = 3c_1 \sin 3t + c_1 \cos 3t + c_2 \sin 3t - 3c_2 \cos 3t$$

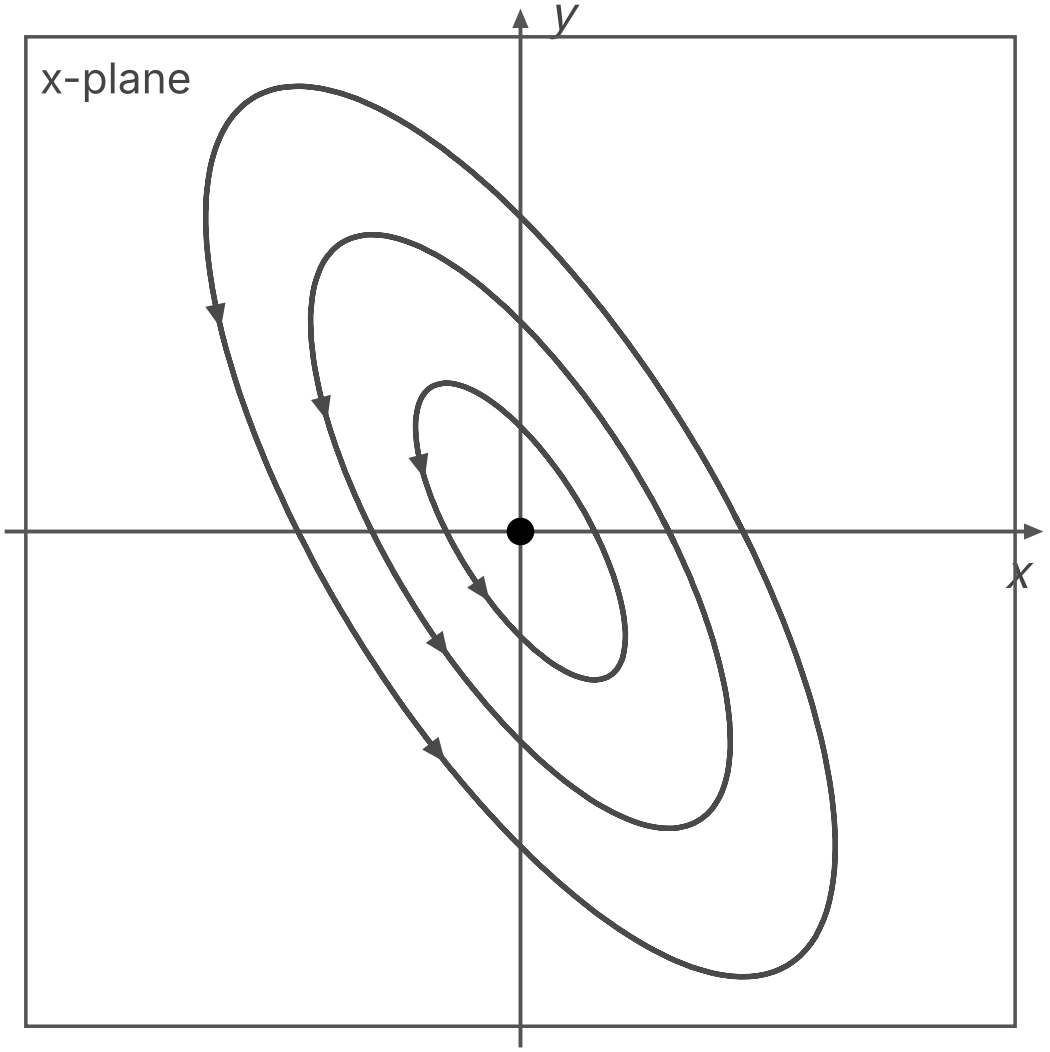
Phase portrait

$\rho = 0$ and $\omega = 3$: the origin is a **centre**. In the $\mathbf{z}$-plane $r = r_0 e^{0t}$ and $\theta = \theta_0 - 3t$: the motion is always **clockwise** ($\dot{\theta} = -\omega < 0$), and the orbits are circles of period $T = 2\pi/\omega$ (closed orbits).

Sense of rotation in the $\mathbf{x}$-plane, tested directly on the vector field (Topic 4 Sec. 7.8): at $(0,1)^T$, $\dot{\mathbf{x}} = A(0,1)^T = \begin{pmatrix}-3\\3\end{pmatrix}$, so $\dot{x} = -3$ is < 0: to the left at the top of the plane: the rotation is **counter-clockwise**. Cross-check: $\det Q = -5$ is < 0, so Q reverses orientation and the clockwise motion of the $\mathbf{z}$-plane becomes counter-clockwise in the $\mathbf{x}$-plane, in agreement.



Canonical $\mathbf{z}$-plane (centre): clockwise, $\dot{\theta} = -\omega$.



Original (x, y)-plane (centre), counter-clockwise.

(iii) $\lambda_{\pm} = -2 \pm i$ with $\boldsymbol{\eta}_{\pm} = (1 \mp i,\ 3 \pm i)$, $\dot{x} > 0$

Split the eigenvector of $\lambda_{+} = -2 + i$ into real and imaginary parts, $\boldsymbol{\eta}_{+} = \begin{pmatrix} 1 - i \\ 3 + i \end{pmatrix} = \mathbf{u} + i\mathbf{w}$, so $\rho = -2, \omega = 1$ and

$$\mathbf{u} = \begin{pmatrix} 1 \\ 3 \end{pmatrix}, \qquad \mathbf{w} = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$$

The matrix is not given; it is rebuilt from $A\mathbf{u} = \rho\mathbf{u} - \omega\mathbf{w}$, $A\mathbf{w} = \omega\mathbf{u} + \rho\mathbf{w}$ (Topic 4 Eq. 7.4), which say $AQ = QJ$ with $Q = [\mathbf{u} \mid \mathbf{w}]$, so $A = QJQ^{-1}$:

$$Q = \begin{pmatrix} 1 & -1 \\ 3 & 1 \end{pmatrix}, \quad J = \begin{pmatrix} -2 & 1 \\ -1 & -2 \end{pmatrix}, \quad \det Q = 4, \quad Q^{-1} = \begin{pmatrix} \frac{1}{4} & \frac{1}{4} \\ -\frac{3}{4} & \frac{1}{4} \end{pmatrix}$$

$$
\begin{aligned}
(QJ)_{11} &= 1\cdot(-2) + (-1)\cdot(-1) = -1\\
(QJ)_{12} &= 1\cdot 1 + (-1)\cdot(-2) = 3\\
(QJ)_{21} &= 3\cdot(-2) + 1\cdot(-1) = -7\\
(QJ)_{22} &= 3\cdot 1 + 1\cdot(-2) = 1
\end{aligned}
$$

$$
QJ = \begin{pmatrix} -1 & 3 \\ -7 & 1 \end{pmatrix}
$$

$$
\begin{aligned}
(A)_{11} &= (-1)\cdot\frac{1}{4} + 3\cdot(-\frac{3}{4}) = -\frac{5}{2}\\
(A)_{12} &= (-1)\cdot\frac{1}{4} + 3\cdot\frac{1}{4} = \frac{1}{2}\\
(A)_{21} &= (-7)\cdot\frac{1}{4} + 1\cdot(-\frac{3}{4}) = -\frac{5}{2}\\
(A)_{22} &= (-7)\cdot\frac{1}{4} + 1\cdot\frac{1}{4} = -\frac{3}{2}
\end{aligned}
$$

$$
A = QJQ^{-1} = \begin{pmatrix} -\frac{5}{2} & \frac{1}{2} \\ -\frac{5}{2} & -\frac{3}{2} \end{pmatrix}
$$

Check of the given sign: $\dot{x}$ at $(0,1)$ equals $a_{12} = \frac{1}{2}$, which is positive, in agreement with the statement ($\dot{x} > 0$).

How A acts on $\mathbf{u}$ and $\mathbf{w}$

Separating the real and imaginary parts of $A\mathbf{v} = \lambda\mathbf{v}$ gives (Topic 4, Eq. 7.4) $A\mathbf{u} = \rho\mathbf{u} - \omega\mathbf{w}$ and $A\mathbf{w} = \omega\mathbf{u} + \rho\mathbf{w}$. Verification with the numbers:

$$
A\mathbf{u} = \begin{pmatrix} -1 \\ -7 \end{pmatrix}, \qquad \rho\mathbf{u} - \omega\mathbf{w} = (-2)\begin{pmatrix} 1 \\ 3 \end{pmatrix} - 1\begin{pmatrix} -1 \\ 1 \end{pmatrix} = \begin{pmatrix} -1 \\ -7 \end{pmatrix}
$$

$$
A\mathbf{w} = \begin{pmatrix} 3 \\ 1 \end{pmatrix}, \qquad \omega\mathbf{u} + \rho\mathbf{w} = 1\begin{pmatrix} 1 \\ 3 \end{pmatrix} + (-2)\begin{pmatrix} -1 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 \\ 1 \end{pmatrix}
$$

Change of coordinates

$\mathbf{u}$ and $\mathbf{w}$ are linearly independent (they are not parallel when $\omega \neq 0$; Topic 4, Lemma of Sec. 7.3). With the columns in the order $\mathbf{u}$, $\mathbf{w}$,

$$
Q = [\mathbf{u} \mid \mathbf{w}] = \begin{pmatrix} 1 & -1 \\ 3 & 1 \end{pmatrix}
$$

$\det Q = (1)(1) - (-1)(3) = 4$, which is not zero, so Q is invertible. For a 2×2 matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad-bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$ (Robinson, Appendix B):

$$Q^{-1} = \frac{1}{4}\begin{pmatrix} 1 & 1 \\ -3 & 1 \end{pmatrix} = \begin{pmatrix} \frac{1}{4} & \frac{1}{4} \\ -\frac{3}{4} & \frac{1}{4} \end{pmatrix}$$

Compute $AQ = [A\mathbf{u} \mid A\mathbf{w}]$ and multiply by Q^{-1} on the left, entry by entry:

$$AQ = \begin{pmatrix} -1 & 3 \\ -7 & 1 \end{pmatrix}$$

$$\begin{aligned} (Q^{-1}AQ)_{11} &= \frac{1}{4}\cdot(-1) + \frac{1}{4}\cdot(-7) = -2 \\ (Q^{-1}AQ)_{12} &= \frac{1}{4}\cdot 3 + \frac{1}{4}\cdot 1 = 1 \\ (Q^{-1}AQ)_{21} &= (-\frac{3}{4})\cdot(-1) + \frac{1}{4}\cdot(-7) = -1 \\ (Q^{-1}AQ)_{22} &= (-\frac{3}{4})\cdot 3 + \frac{1}{4}\cdot 1 = -2 \end{aligned}$$

$$Q^{-1}AQ = \begin{pmatrix} -2 & 1 \\ -1 & -2 \end{pmatrix} = J = \begin{pmatrix} \rho & \omega \\ -\omega & \rho \end{pmatrix}$$

Solution in the canonical plane

$$\dot{z}_1 = -2z_1 + z_2, \qquad \dot{z}_2 = -z_1 - 2z_2$$

The two equations are coupled, but they say "scale and rotate", which polar coordinates express directly: $z_1 = r\cos\theta$, $z_2 = r\sin\theta$ (Topic 4, Sec. 7.5).

The radius. From $r^2 = z_1^2 + z_2^2$ the chain rule gives $2r\dot{r} = 2z_1\dot{z}_1 + 2z_2\dot{z}_2$. Substituting $\dot{z}_1 = \rho z_1 + \omega z_2$ and $\dot{z}_2 = -\omega z_1 + \rho z_2$:

$$r\dot{r} = z_1(\rho z_1 + \omega z_2) + z_2(-\omega z_1 + \rho z_2) = \rho z_1^2 + \omega z_1 z_2 - \omega z_1 z_2 + \rho z_2^2 = \rho r^2$$

The terms in ω cancel. Dividing by $r \neq 0$ (the origin is the equilibrium): $\dot{r} = \rho r$.

The angle. Differentiate $\tan\theta = z_2/z_1$ with the chain rule on the left and the quotient rule on the right: $\sec^2\theta\,\dot{\theta} = (z_1\dot{z}_2 - z_2\dot{z}_1)/z_1^2$. Since $\sec^2\theta = r^2/z_1^2$, $\dot{\theta} = (z_1\dot{z}_2 - z_2\dot{z}_1)/r^2$, and

$$z_1\dot{z}_2 - z_2\dot{z}_1 = z_1(-\omega z_1 + \rho z_2) - z_2(\rho z_1 + \omega z_2) = -\omega z_1^2 + \rho z_1 z_2 - \rho z_1 z_2 - \omega z_2^2 = -\omega r^2 \implies \dot{\theta} = -\omega$$

Integration. $\frac{d}{dt}\left(e^{-\rho t} r\right) = e^{-\rho t}(\dot{r} - \rho r) = 0$, and Barrow's rule between 0 and t, $\left[e^{-\rho s} r(s)\right]_0^t = 0$, gives $r(t) = r_0 e^{\rho t}$. For the angle, $\theta(t) - \theta_0 = \int_0^t (-\omega)\, ds = -\omega t$. Back to Cartesian coordinates, with $c_1 = r_0 \cos\theta_0 = z_1(0)$, $c_2 = r_0 \sin\theta_0 = z_2(0)$ and the subtraction formulas $\cos(\theta_0 - \omega t) = \cos\theta_0 \cos\omega t + \sin\theta_0 \sin\omega t$, $\sin(\theta_0 - \omega t) = \sin\theta_0 \cos\omega t - \cos\theta_0 \sin\omega t$:

$$z_1(t) = e^{-2t}\left(c_1 \cos 1t + c_2 \sin 1t\right), \qquad z_2(t) = e^{-2t}\left(c_2 \cos 1t - c_1 \sin 1t\right)$$

Check by substitution (product rule and chain rule): $\dot{z}_1 = \rho z_1 + e^{\rho t}\omega\left(-c_1 \sin\omega t + c_2 \cos\omega t\right) = \rho z_1 + \omega z_2$ and $\dot{z}_2 = \rho z_2 + e^{\rho t}\omega\left(-c_2 \sin\omega t - c_1 \cos\omega t\right) = \rho z_2 - \omega z_1$, as required.

Back to the original coordinates

$\mathbf{x} = Q\mathbf{z} = z_1\mathbf{u} + z_2\mathbf{w}$, so

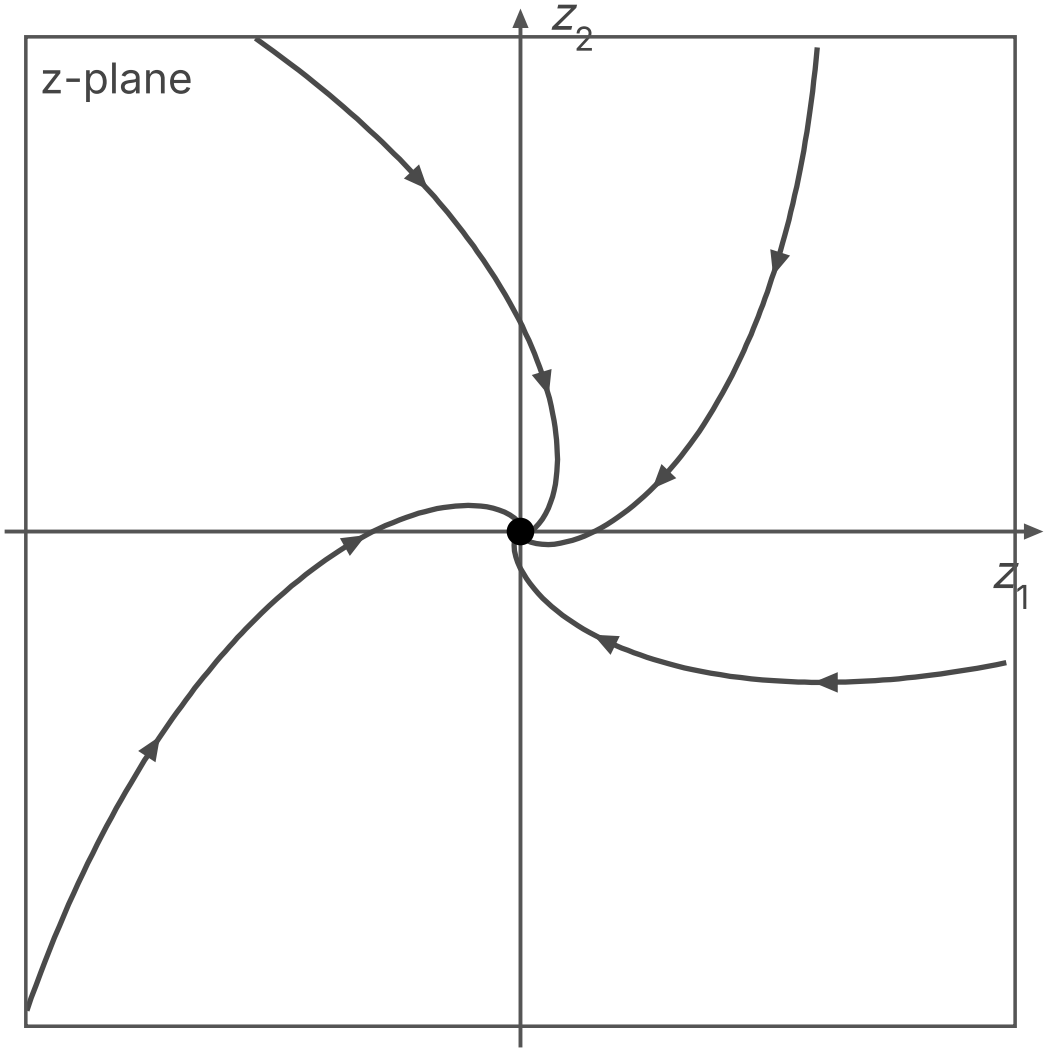
$$\mathbf{x}(t) = e^{-2t}\left[\left(c_1 \cos 1t + c_2 \sin 1t\right)\begin{pmatrix}1\\3\end{pmatrix} + \left(c_2 \cos 1t - c_1 \sin 1t\right)\begin{pmatrix}-1\\1\end{pmatrix}\right]$$

$$x(t) = c_1 e^{-2t}\sin t + c_1 e^{-2t}\cos t + c_2 e^{-2t}\sin t - c_2 e^{-2t}\cos t, \qquad y(t) = -c_1 e^{-2t}\sin t + 3c_1 e^{-2t}\cos t + 3c_2 e^{-2t}\sin t + c_2 e^{-2t}\cos t$$

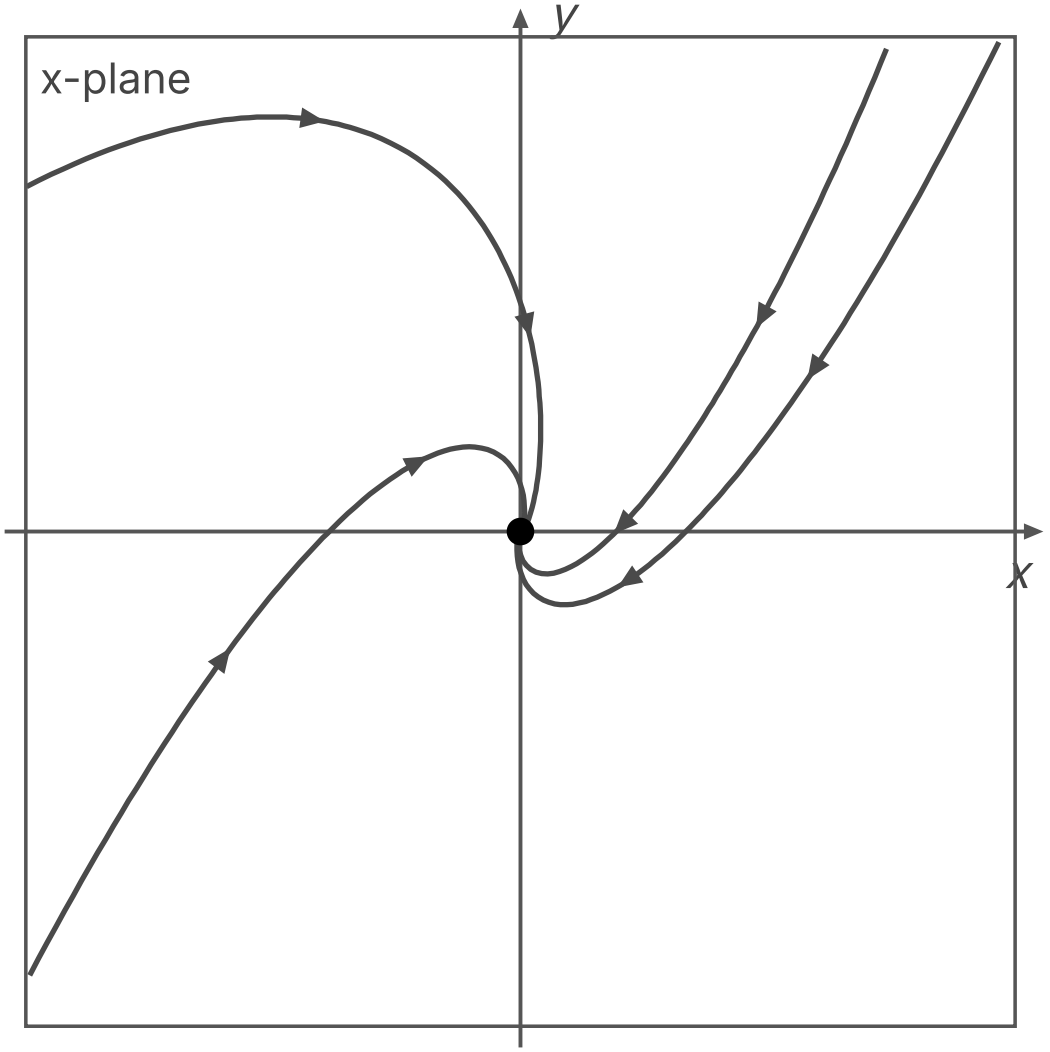
Phase portrait

$\rho = -2$ and $\omega = 1$: the origin is a **stable spiral**. In the $\mathbf{z}$-plane $r = r_0 e^{-2t}$ and $\theta = \theta_0 - 1t$: the motion is always **clockwise** ($\dot{\theta} = -\omega < 0$), and one full turn takes $T = 2\pi/\omega$ while the radius is multiplied by $e^{2\pi\rho/\omega} = e^{-4\pi}$ (shrinking).

Sense of rotation in the $\mathbf{x}$-plane, tested directly on the vector field (Topic 4 Sec. 7.8): at $(0,1)^T$, $\dot{\mathbf{x}} = A(0,1)^T = \begin{pmatrix}\frac{1}{2}\\ -\frac{3}{2}\end{pmatrix}$, so $\dot{x} = \frac{1}{2}$ is > 0: to the right at the top of the plane: the rotation is **clockwise**. Cross-check: $\det Q = 4$ is > 0, so Q preserves orientation and the clockwise motion of the $\mathbf{z}$-plane becomes clockwise in the $\mathbf{x}$-plane, in agreement.



Canonical $\mathbf{z}$-plane (stable spiral): clockwise, $\dot{\theta} = -\omega$.



Original (x, y)-plane (stable spiral), clockwise.

(iv) $\lambda_{\pm} = -1 \pm i$ with $\boldsymbol{\eta}_{\pm} = (1,\ \pm i),\ \dot{x} > 0$

Split the eigenvector of $\lambda_{+} = -1 + i$ into real and imaginary parts, $\boldsymbol{\eta}_{+} = \begin{pmatrix} 1 \\ i \end{pmatrix} = \mathbf{u} + i\mathbf{w}$, so $\rho = -1, \omega = 1$ and

$$\mathbf{u} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad \mathbf{w} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

The matrix is not given; it is rebuilt from $A\mathbf{u} = \rho\mathbf{u} - \omega\mathbf{w}$, $A\mathbf{w} = \omega\mathbf{u} + \rho\mathbf{w}$ (Topic 4 Eq. 7.4), which say $AQ = QJ$ with $Q = [\mathbf{u} \mid \mathbf{w}]$, so $A = QJQ^{-1}$:

$$Q = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad J = \begin{pmatrix} -1 & 1 \\ -1 & -1 \end{pmatrix}, \quad \det Q = 1, \quad Q^{-1} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$
\begin{aligned}
(QJ)_{11} &= 1\cdot(-1) + 0\cdot(-1) = -1\\
(QJ)_{12} &= 1\cdot 1 + 0\cdot(-1) = 1\\
(QJ)_{21} &= 0\cdot(-1) + 1\cdot(-1) = -1\\
(QJ)_{22} &= 0\cdot 1 + 1\cdot(-1) = -1
\end{aligned}
$$

$$
QJ = \begin{pmatrix} -1 & 1 \\ -1 & -1 \end{pmatrix}
$$

$$
\begin{aligned}
(A)_{11} &= (-1)\cdot 1 + 1\cdot 0 = -1\\
(A)_{12} &= (-1)\cdot 0 + 1\cdot 1 = 1\\
(A)_{21} &= (-1)\cdot 1 + (-1)\cdot 0 = -1\\
(A)_{22} &= (-1)\cdot 0 + (-1)\cdot 1 = -1
\end{aligned}
$$

$$
A = QJQ^{-1} = \begin{pmatrix} -1 & 1 \\ -1 & -1 \end{pmatrix}
$$

Check of the given sign: $\dot{x}$ at $(0,1)$ equals $a_{12} = 1$, which is positive, in agreement with the statement ($\dot{x} > 0$).

How A acts on $\mathbf{u}$ and $\mathbf{w}$

Separating the real and imaginary parts of $A\mathbf{v} = \lambda\mathbf{v}$ gives (Topic 4, Eq. 7.4) $A\mathbf{u} = \rho\mathbf{u} - \omega\mathbf{w}$ and $A\mathbf{w} = \omega\mathbf{u} + \rho\mathbf{w}$. Verification with the numbers:

$$
A\mathbf{u} = \begin{pmatrix} -1 \\ -1 \end{pmatrix}, \qquad \rho\mathbf{u} - \omega\mathbf{w} = (-1)\begin{pmatrix} 1 \\ 0 \end{pmatrix} - 1\begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} -1 \\ -1 \end{pmatrix}
$$

$$
A\mathbf{w} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}, \qquad \omega\mathbf{u} + \rho\mathbf{w} = 1\begin{pmatrix} 1 \\ 0 \end{pmatrix} + (-1)\begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}
$$

Change of coordinates

$\mathbf{u}$ and $\mathbf{w}$ are linearly independent (they are not parallel when $\omega \neq 0$; Topic 4, Lemma of Sec. 7.3). With the columns in the order $\mathbf{u}, \mathbf{w}$,

$$
Q = [\mathbf{u} \mid \mathbf{w}] = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}
$$

$\det Q = (1)(1) - (0)(0) = 1$, which is not zero, so Q is invertible. For a 2×2 matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad-bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$ (Robinson, Appendix B):

$$Q^{-1} = \frac{1}{1}\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

Compute $AQ = [A\mathbf{u} \mid A\mathbf{w}]$ and multiply by Q^{-1} on the left, entry by entry:

$$AQ = \begin{pmatrix} -1 & 1 \\ -1 & -1 \end{pmatrix}$$

$$\begin{aligned} (Q^{-1}AQ)_{11} &= 1 \cdot (-1) + 0 \cdot (-1) = -1 \\ (Q^{-1}AQ)_{12} &= 1 \cdot 1 + 0 \cdot (-1) = 1 \\ (Q^{-1}AQ)_{21} &= 0 \cdot (-1) + 1 \cdot (-1) = -1 \\ (Q^{-1}AQ)_{22} &= 0 \cdot 1 + 1 \cdot (-1) = -1 \end{aligned}$$

$$Q^{-1}AQ = \begin{pmatrix} -1 & 1 \\ -1 & -1 \end{pmatrix} = J = \begin{pmatrix} \rho & \omega \\ -\omega & \rho \end{pmatrix}$$

Solution in the canonical plane

$$\dot{z}_1 = -z_1 + z_2, \qquad \dot{z}_2 = -z_1 - z_2$$

The two equations are coupled, but they say "scale and rotate", which polar coordinates express directly: $z_1 = r\cos\theta$, $z_2 = r\sin\theta$ (Topic 4, Sec. 7.5).

The radius. From $r^2 = z_1^2 + z_2^2$ the chain rule gives $2r\dot{r} = 2z_1\dot{z}_1 + 2z_2\dot{z}_2$. Substituting $\dot{z}_1 = \rho z_1 + \omega z_2$ and $\dot{z}_2 = -\omega z_1 + \rho z_2$:

$$r\dot{r} = z_1(\rho z_1 + \omega z_2) + z_2(-\omega z_1 + \rho z_2) = \rho z_1^2 + \omega z_1 z_2 - \omega z_1 z_2 + \rho z_2^2 = \rho r^2$$

The terms in ω cancel. Dividing by $r \neq 0$ (the origin is the equilibrium): $\dot{r} = \rho r$.

The angle. Differentiate $\tan\theta = z_2/z_1$ with the chain rule on the left and the quotient rule on the right: $\sec^2\theta\,\dot{\theta} = (z_1\dot{z}_2 - z_2\dot{z}_1)/z_1^2$. Since $\sec^2\theta = r^2/z_1^2$, $\dot{\theta} = (z_1\dot{z}_2 - z_2\dot{z}_1)/r^2$, and

$$z_1\dot{z}_2 - z_2\dot{z}_1 = z_1(-\omega z_1 + \rho z_2) - z_2(\rho z_1 + \omega z_2) = -\omega z_1^2 + \rho z_1 z_2 - \rho z_1 z_2 - \omega z_2^2 = -\omega r^2 \implies \dot{\theta} = -\omega$$

Integration. $\frac{d}{dt}\left(e^{-\rho t}r\right) = e^{-\rho t}(\dot r - \rho r) = 0$, and Barrow's rule between 0 and t, $\left[e^{-\rho s}r(s)\right]_0^t = 0$, gives $r(t) = r_0 e^{\rho t}$. For the angle, $\theta(t) - \theta_0 = \int_0^t (-\omega)\, ds = -\omega t$. Back to Cartesian coordinates, with $c_1 = r_0\cos\theta_0 = z_1(0), c_2 = r_0\sin\theta_0 = z_2(0)$ and the subtraction formulas $\cos(\theta_0 - \omega t) = \cos\theta_0\cos\omega t + \sin\theta_0\sin\omega t$, $\sin(\theta_0 - \omega t) = \sin\theta_0\cos\omega t - \cos\theta_0\sin\omega t$:

$$z_1(t) = e^{-1t}\left(c_1\cos 1t + c_2\sin 1t\right), \qquad z_2(t) = e^{-1t}\left(c_2\cos 1t - c_1\sin 1t\right)$$

Check by substitution (product rule and chain rule): $\dot z_1 = \rho z_1 + e^{\rho t}\omega\left(-c_1\sin\omega t + c_2\cos\omega t\right) = \rho z_1 + \omega z_2$ and $\dot z_2 = \rho z_2 + e^{\rho t}\omega\left(-c_2\sin\omega t - c_1\cos\omega t\right) = \rho z_2 - \omega z_1$, as required.

Back to the original coordinates

$\mathbf{x} = Q\mathbf{z} = z_1\mathbf{u} + z_2\mathbf{w}$, so

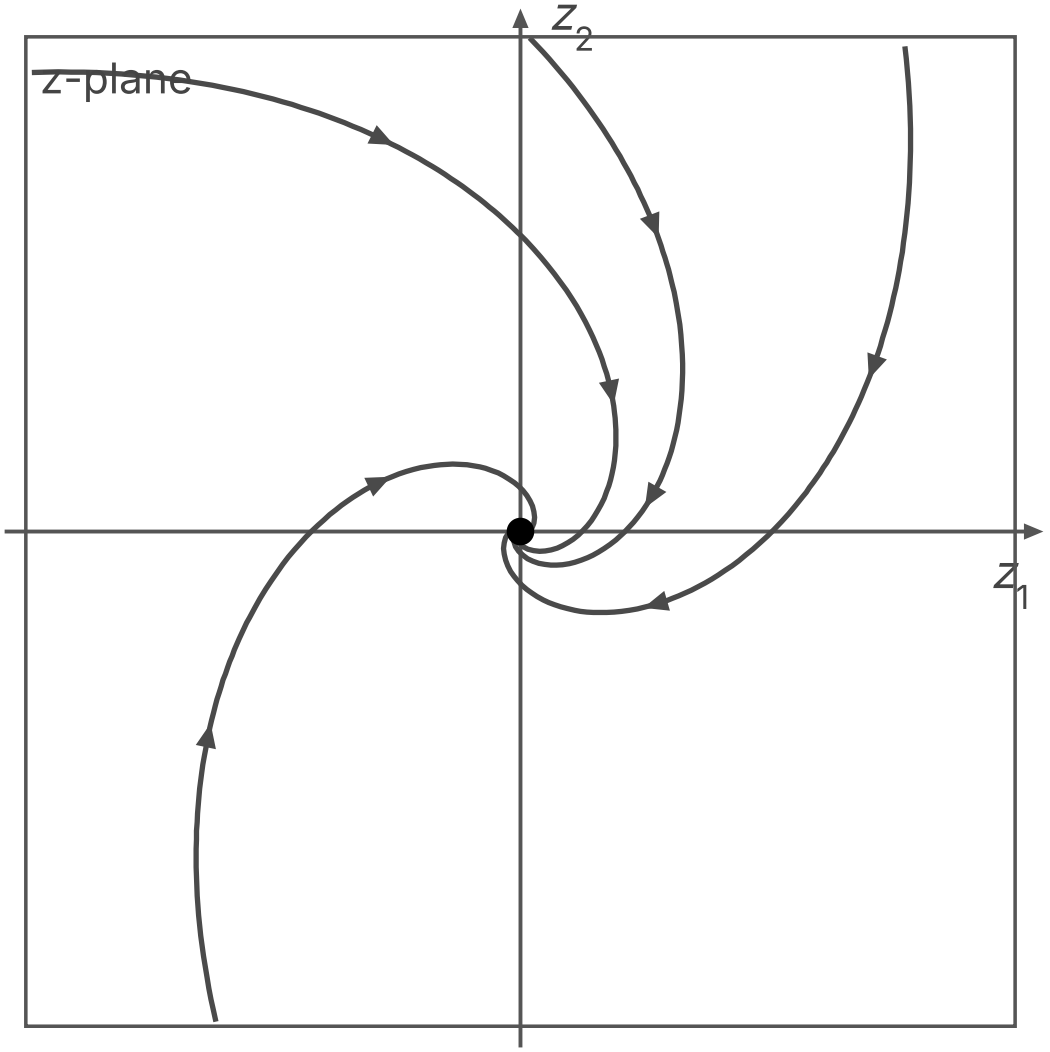
$$\mathbf{x}(t) = e^{-1t}\left[\left(c_1\cos 1t + c_2\sin 1t\right)\begin{pmatrix}1\\0\end{pmatrix} + \left(c_2\cos 1t - c_1\sin 1t\right)\begin{pmatrix}0\\1\end{pmatrix}\right]$$

$$x(t) = c_1 e^{-t}\cos t + c_2 e^{-t}\sin t, \qquad y(t) = -c_1 e^{-t}\sin t + c_2 e^{-t}\cos t$$

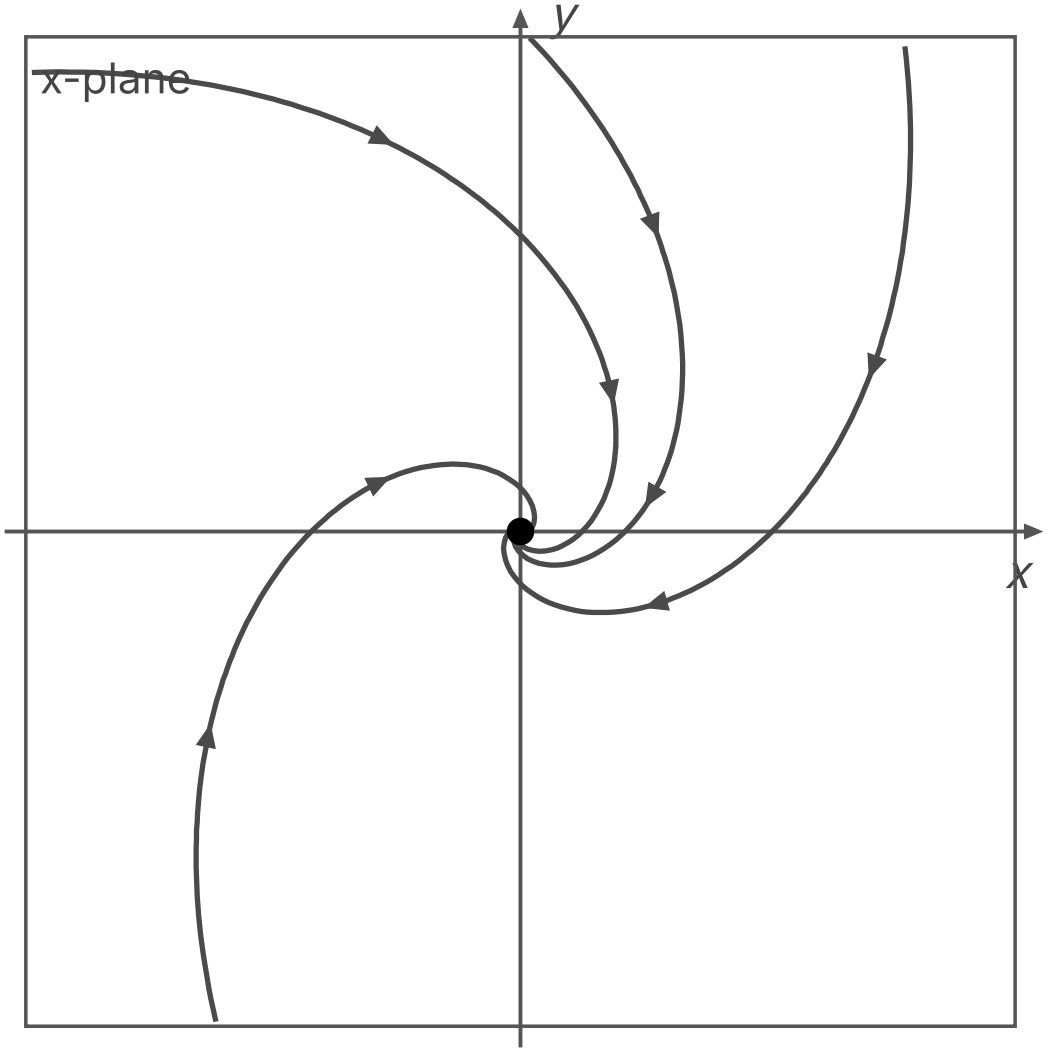
Phase portrait

$\rho = -1$ and $\omega = 1$: the origin is a **stable spiral**. In the $\mathbf{z}$-plane $r = r_0 e^{-1t}$ and $\theta = \theta_0 - 1t$: the motion is always **clockwise** ($\dot\theta = -\omega < 0$), and one full turn takes $T = 2\pi/\omega$ while the radius is multiplied by $e^{2\pi\rho/\omega} = e^{-2\pi}$ (shrinking).

Sense of rotation in the $\mathbf{x}$-plane, tested directly on the vector field (Topic 4 Sec. 7.8): at $(0,1)^T$, $\dot{\mathbf{x}} = A(0,1)^T = \begin{pmatrix}1\\-1\end{pmatrix}$, so $\dot x = 1$ is > 0: to the right at the top of the plane: the rotation is **clockwise**. Cross-check: $\det Q = 1$ is > 0, so Q preserves orientation and the clockwise motion of the $\mathbf{z}$-plane becomes clockwise in the $\mathbf{x}$-plane, in agreement.



Canonical $\mathbf{z}$-plane (stable spiral): clockwise, $\dot{\theta} = -\omega$.



Original (x, y)-plane (stable spiral), clockwise.

(v) $\lambda_{\pm} = 2 \pm 2i$ with $\boldsymbol{\eta}_{\pm} = (\pm 3i,\ 5 \mp 4i)$, $\dot{x} < 0$

Split the eigenvector of $\lambda_{+} = 2 + 2i$ into real and imaginary parts, $\boldsymbol{\eta}_{+} = \begin{pmatrix} 3i \\ 5 - 4i \end{pmatrix} = \mathbf{u} + i\mathbf{w}$, so $\rho = 2, \omega = 2$ and

$$\mathbf{u} = \begin{pmatrix} 0 \\ 5 \end{pmatrix}, \qquad \mathbf{w} = \begin{pmatrix} 3 \\ -4 \end{pmatrix}$$

The matrix is not given; it is rebuilt from $A\mathbf{u} = \rho\mathbf{u} - \omega\mathbf{w}$, $A\mathbf{w} = \omega\mathbf{u} + \rho\mathbf{w}$ (Topic 4 Eq. 7.4), which say $AQ = QJ$ with $Q = [\mathbf{u} \mid \mathbf{w}]$, so $A = QJQ^{-1}$:

$$Q = \begin{pmatrix} 0 & 3 \\ 5 & -4 \end{pmatrix}, \quad J = \begin{pmatrix} 2 & 2 \\ -2 & 2 \end{pmatrix}, \quad \det Q = -15, \quad Q^{-1} = \begin{pmatrix} \frac{4}{15} & \frac{1}{5} \\ \frac{1}{3} & 0 \end{pmatrix}$$

$$
\begin{aligned}
(QJ)_{11} &= 0\cdot 2 + 3\cdot(-2) = -6\\
(QJ)_{12} &= 0\cdot 2 + 3\cdot 2 = 6\\
(QJ)_{21} &= 5\cdot 2 + (-4)\cdot(-2) = 18\\
(QJ)_{22} &= 5\cdot 2 + (-4)\cdot 2 = 2
\end{aligned}
$$

$$
QJ = \begin{pmatrix} -6 & 6 \\ 18 & 2 \end{pmatrix}
$$

$$
\begin{aligned}
(A)_{11} &= (-6)\cdot\frac{4}{15} + 6\cdot\frac{1}{3} = \frac{2}{5}\\
(A)_{12} &= (-6)\cdot\frac{1}{5} + 6\cdot 0 = -\frac{6}{5}\\
(A)_{21} &= 18\cdot\frac{4}{15} + 2\cdot\frac{1}{3} = \frac{82}{15}\\
(A)_{22} &= 18\cdot\frac{1}{5} + 2\cdot 0 = \frac{18}{5}
\end{aligned}
$$

$$
A = QJQ^{-1} = \begin{pmatrix} \frac{2}{5} & -\frac{6}{5} \\ \frac{82}{15} & \frac{18}{5} \end{pmatrix}
$$

Check of the given sign: $\dot{x}$ at $(0,1)$ equals $a_{12} = -\frac{6}{5}$, which is negative, in agreement with the statement ($\dot{x} < 0$).

How A acts on $\mathbf{u}$ and $\mathbf{w}$

Separating the real and imaginary parts of $A\mathbf{v} = \lambda\mathbf{v}$ gives (Topic 4, Eq. 7.4) $A\mathbf{u} = \rho\mathbf{u} - \omega\mathbf{w}$ and $A\mathbf{w} = \omega\mathbf{u} + \rho\mathbf{w}$. Verification with the numbers:

$$
A\mathbf{u} = \begin{pmatrix} -6 \\ 18 \end{pmatrix}, \qquad \rho\mathbf{u} - \omega\mathbf{w} = 2\begin{pmatrix} 0 \\ 5 \end{pmatrix} - 2\begin{pmatrix} 3 \\ -4 \end{pmatrix} = \begin{pmatrix} -6 \\ 18 \end{pmatrix}
$$

$$
A\mathbf{w} = \begin{pmatrix} 6 \\ 2 \end{pmatrix}, \qquad \omega\mathbf{u} + \rho\mathbf{w} = 2\begin{pmatrix} 0 \\ 5 \end{pmatrix} + 2\begin{pmatrix} 3 \\ -4 \end{pmatrix} = \begin{pmatrix} 6 \\ 2 \end{pmatrix}
$$

Change of coordinates

$\mathbf{u}$ and $\mathbf{w}$ are linearly independent (they are not parallel when $\omega \neq 0$; Topic 4, Lemma of Sec. 7.3). With the columns in the order $\mathbf{u}$, $\mathbf{w}$,

$$
Q = [\mathbf{u} \mid \mathbf{w}] = \begin{pmatrix} 0 & 3 \\ 5 & -4 \end{pmatrix}
$$

$\det Q = (0)(-4) - (3)(5) = -15$, which is not zero, so Q is invertible. For a 2×2 matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad-bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$ (Robinson, Appendix B):

$$Q^{-1} = \frac{1}{-15}\begin{pmatrix} -4 & -3 \\ -5 & 0 \end{pmatrix} = \begin{pmatrix} \frac{4}{15} & \frac{1}{5} \\ \frac{1}{3} & 0 \end{pmatrix}$$

Compute $AQ = [A\mathbf{u} \mid A\mathbf{w}]$ and multiply by Q^{-1} on the left, entry by entry:

$$AQ = \begin{pmatrix} -6 & 6 \\ 18 & 2 \end{pmatrix}$$

$$\begin{aligned} (Q^{-1}AQ)_{11} &= \frac{4}{15}\cdot(-6) + \frac{1}{5}\cdot 18 = 2 \\ (Q^{-1}AQ)_{12} &= \frac{4}{15}\cdot 6 + \frac{1}{5}\cdot 2 = 2 \\ (Q^{-1}AQ)_{21} &= \frac{1}{3}\cdot(-6) + 0\cdot 18 = -2 \\ (Q^{-1}AQ)_{22} &= \frac{1}{3}\cdot 6 + 0\cdot 2 = 2 \end{aligned}$$

$$Q^{-1}AQ = \begin{pmatrix} 2 & 2 \\ -2 & 2 \end{pmatrix} = J = \begin{pmatrix} \rho & \omega \\ -\omega & \rho \end{pmatrix}$$

Solution in the canonical plane

$$\dot{z}_1 = 2z_1 + 2z_2, \qquad \dot{z}_2 = -2z_1 + 2z_2$$

The two equations are coupled, but they say "scale and rotate", which polar coordinates express directly: $z_1 = r\cos\theta$, $z_2 = r\sin\theta$ (Topic 4, Sec. 7.5).

The radius. From $r^2 = z_1^2 + z_2^2$ the chain rule gives $2r\dot{r} = 2z_1\dot{z}_1 + 2z_2\dot{z}_2$. Substituting $\dot{z}_1 = \rho z_1 + \omega z_2$ and $\dot{z}_2 = -\omega z_1 + \rho z_2$:

$$r\dot{r} = z_1(\rho z_1 + \omega z_2) + z_2(-\omega z_1 + \rho z_2) = \rho z_1^2 + \omega z_1 z_2 - \omega z_1 z_2 + \rho z_2^2 = \rho r^2$$

The terms in ω cancel. Dividing by $r \neq 0$ (the origin is the equilibrium): $\dot{r} = \rho r$.

The angle. Differentiate $\tan\theta = z_2/z_1$ with the chain rule on the left and the quotient rule on the right: $\sec^2\theta\,\dot{\theta} = (z_1\dot{z}_2 - z_2\dot{z}_1)/z_1^2$. Since $\sec^2\theta = r^2/z_1^2$, $\dot{\theta} = (z_1\dot{z}_2 - z_2\dot{z}_1)/r^2$, and

$$z_1\dot z_2 - z_2\dot z_1 = z_1(-\omega z_1 + \rho z_2) - z_2(\rho z_1 + \omega z_2) = -\omega z_1^2 + \rho z_1 z_2 - \rho z_1 z_2 - \omega z_2^2 = -\omega r^2 \implies \dot\theta = -\omega$$

Integration. $\frac{d}{dt}\left(e^{-\rho t} r\right) = e^{-\rho t}(\dot r - \rho r) = 0$, and Barrow's rule between 0 and t, $\left[e^{-\rho s} r(s)\right]_0^t = 0$, gives $r(t) = r_0 e^{\rho t}$. For the angle, $\theta(t) - \theta_0 = \int_0^t (-\omega)\, ds = -\omega t$. Back to Cartesian coordinates, with $c_1 = r_0\cos\theta_0 = z_1(0)$, $c_2 = r_0 \sin\theta_0 = z_2(0)$ and the subtraction formulas $\cos(\theta_0 - \omega t) = \cos\theta_0\cos\omega t + \sin\theta_0\sin\omega t$, $\sin(\theta_0 - \omega t) = \sin\theta_0\cos\omega t - \cos\theta_0\sin\omega t$:

$$z_1(t) = e^{2t}\left(c_1\cos 2t + c_2 \sin 2t\right), \qquad z_2(t) = e^{2t}\left(c_2\cos 2t - c_1\sin 2t\right)$$

Check by substitution (product rule and chain rule): $\dot z_1 = \rho z_1 + e^{\rho t}\omega\left(-c_1\sin\omega t + c_2\cos\omega t\right) = \rho z_1 + \omega z_2$ and $\dot z_2 = \rho z_2 + e^{\rho t}\omega\left(-c_2\sin\omega t - c_1\cos\omega t\right) = \rho z_2 - \omega z_1$, as required.

Back to the original coordinates

$\mathbf{x} = Q\mathbf{z} = z_1\mathbf{u} + z_2\mathbf{w}$, so

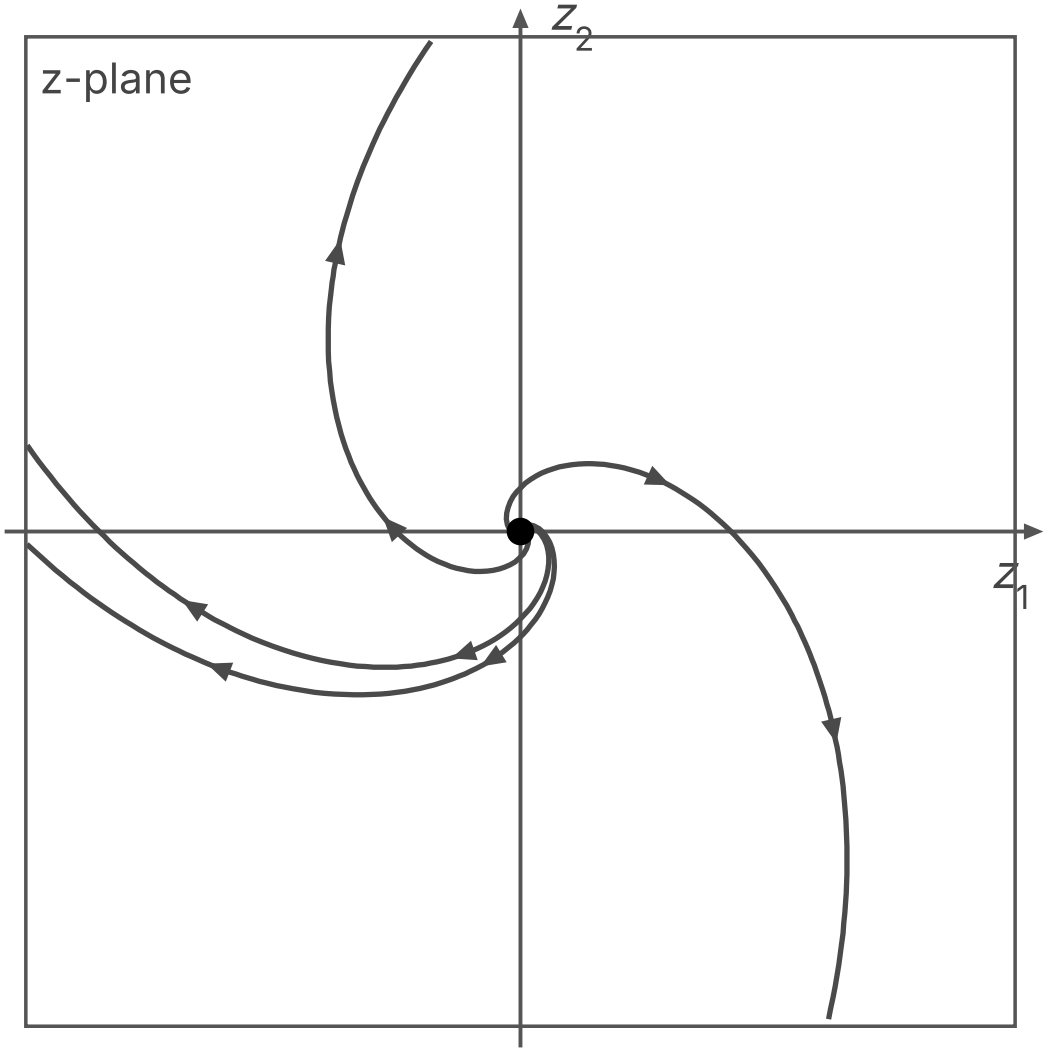
$$\mathbf{x}(t) = e^{2t}\left[(c_1\cos 2t + c_2\sin 2t)\begin{pmatrix}0\\5\end{pmatrix} + (c_2\cos 2t - c_1\sin 2t)\begin{pmatrix}3\\-4\end{pmatrix}\right]$$

$$x(t) = -3c_1 e^{2t}\sin 2t + 3c_2 e^{2t}\cos 2t, \qquad y(t) = 4c_1 e^{2t}\sin 2t + 5c_1 e^{2t}\cos 2t + 5c_2 e^{2t}\sin 2t - 4c_2 e^{2t}\cos 2t$$

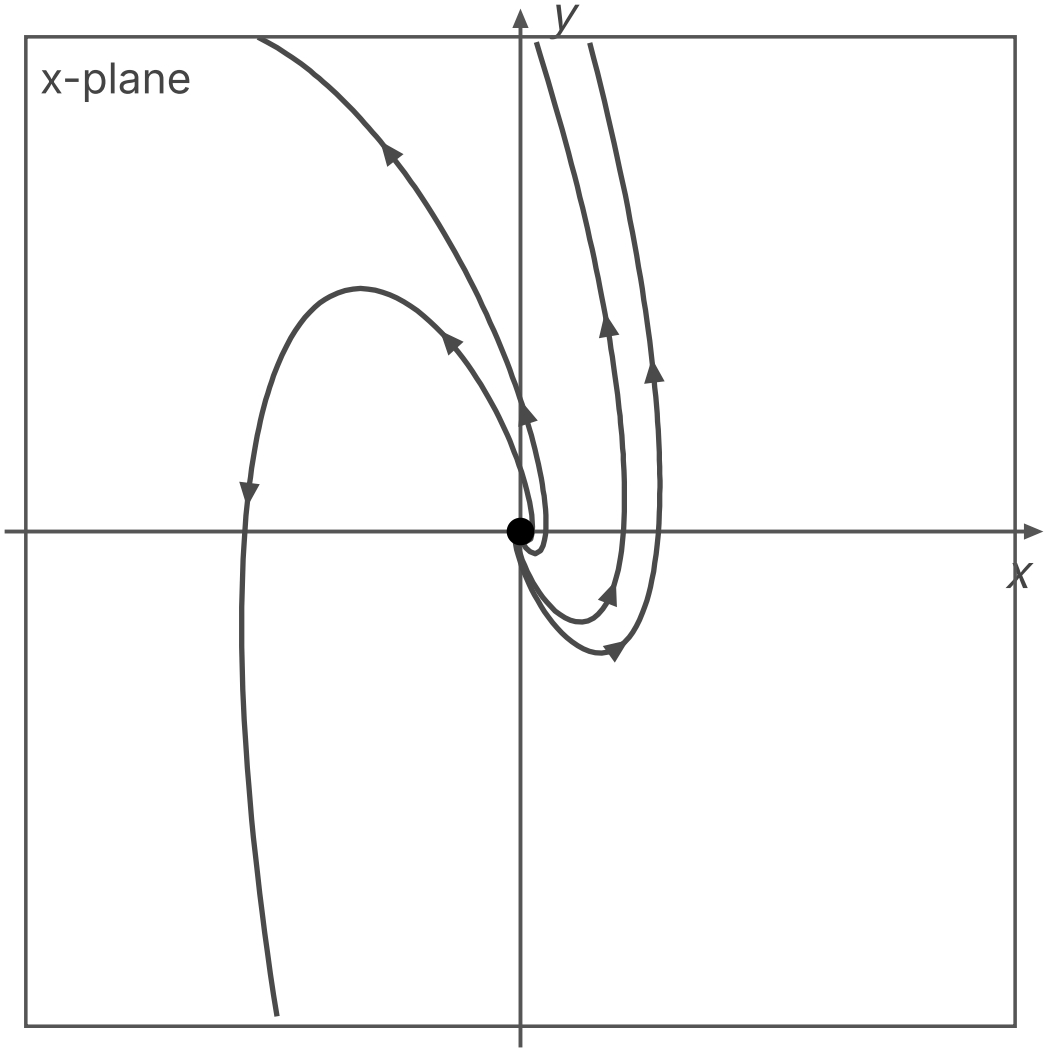
Phase portrait

$\rho = 2$ and $\omega = 2$: the origin is a **unstable spiral**. In the $\mathbf{z}$-plane $r = r_0 e^{2t}$ and $\theta = \theta_0 - 2t$: the motion is always **clockwise** ($\dot\theta = -\omega < 0$), and one full turn takes $T = 2\pi/\omega$ while the radius is multiplied by $e^{2\pi\rho/\omega} = e^{2\pi}$ (growing).

Sense of rotation in the $\mathbf{x}$-plane, tested directly on the vector field (Topic 4 Sec. 7.8): at $(0,1)^T$, $\dot{\mathbf{x}} = A(0,1)^T = \begin{pmatrix}-\frac{6}{5}\\ \frac{18}{5}\end{pmatrix}$, so $\dot x = -\frac{6}{5}$ is < 0: to the left at the top of the plane: the rotation is **counter-clockwise**. Cross-check: $\det Q = -15$ is < 0, so Q reverses orientation and the clockwise motion of the $\mathbf{z}$-plane becomes counter-clockwise in the $\mathbf{x}$-plane, in agreement.



Canonical $\mathbf{z}$-plane (unstable spiral): clockwise, $\dot{\theta} = -\omega$.



Original (x, y)-plane (unstable spiral), counter-clockwise.

(vi) $\lambda_{\pm} = 5 \pm 3i$ with $\boldsymbol{\eta}_{\pm} = (2 \pm 5i,\ \mp i),\ \dot{x} < 0$

Split the eigenvector of $\lambda_{+} = 5 + 3i$ into real and imaginary parts, $\boldsymbol{\eta}_{+} = \begin{pmatrix} 2 + 5i \\ -i \end{pmatrix} = \mathbf{u} + i\mathbf{w}$, so $\rho = 5, \omega = 3$ and

$$\mathbf{u} = \begin{pmatrix} 2 \\ 0 \end{pmatrix}, \qquad \mathbf{w} = \begin{pmatrix} 5 \\ -1 \end{pmatrix}$$

The matrix is not given; it is rebuilt from $A\mathbf{u} = \rho\mathbf{u} - \omega\mathbf{w}$, $A\mathbf{w} = \omega\mathbf{u} + \rho\mathbf{w}$ (Topic 4 Eq. 7.4), which say $AQ = QJ$ with $Q = [\mathbf{u} \mid \mathbf{w}]$, so $A = QJQ^{-1}$:

$$Q = \begin{pmatrix} 2 & 5 \\ 0 & -1 \end{pmatrix}, \quad J = \begin{pmatrix} 5 & 3 \\ -3 & 5 \end{pmatrix}, \quad \det Q = -2, \quad Q^{-1} = \begin{pmatrix} \frac{1}{2} & \frac{5}{2} \\ 0 & -1 \end{pmatrix}$$

$$
\begin{aligned}
(QJ)_{11} &= 2\cdot 5 + 5\cdot(-3) = -5\\
(QJ)_{12} &= 2\cdot 3 + 5\cdot 5 = 31\\
(QJ)_{21} &= 0\cdot 5 + (-1)\cdot(-3) = 3\\
(QJ)_{22} &= 0\cdot 3 + (-1)\cdot 5 = -5
\end{aligned}
$$

$$
QJ = \begin{pmatrix} -5 & 31 \\ 3 & -5 \end{pmatrix}
$$

$$
\begin{aligned}
(A)_{11} &= (-5)\cdot\frac{1}{2} + 31\cdot 0 = -\frac{5}{2}\\
(A)_{12} &= (-5)\cdot\frac{5}{2} + 31\cdot(-1) = -\frac{87}{2}\\
(A)_{21} &= 3\cdot\frac{1}{2} + (-5)\cdot 0 = \frac{3}{2}\\
(A)_{22} &= 3\cdot\frac{5}{2} + (-5)\cdot(-1) = \frac{25}{2}
\end{aligned}
$$

$$
A = QJQ^{-1} = \begin{pmatrix} -\frac{5}{2} & -\frac{87}{2} \\ \frac{3}{2} & \frac{25}{2} \end{pmatrix}
$$

Check of the given sign: $\dot{x}$ at $(0,1)$ equals $a_{12} = -\frac{87}{2}$, which is negative, in agreement with the statement ($\dot{x} < 0$).

How A acts on $\mathbf{u}$ and $\mathbf{w}$

Separating the real and imaginary parts of $A\mathbf{v} = \lambda\mathbf{v}$ gives (Topic 4, Eq. 7.4) $A\mathbf{u} = \rho\mathbf{u} - \omega\mathbf{w}$ and $A\mathbf{w} = \omega\mathbf{u} + \rho\mathbf{w}$. Verification with the numbers:

$$
A\mathbf{u} = \begin{pmatrix} -5 \\ 3 \end{pmatrix}, \qquad \rho\mathbf{u} - \omega\mathbf{w} = 5\begin{pmatrix} 2 \\ 0 \end{pmatrix} - 3\begin{pmatrix} 5 \\ -1 \end{pmatrix} = \begin{pmatrix} -5 \\ 3 \end{pmatrix}
$$

$$
A\mathbf{w} = \begin{pmatrix} 31 \\ -5 \end{pmatrix}, \qquad \omega\mathbf{u} + \rho\mathbf{w} = 3\begin{pmatrix} 2 \\ 0 \end{pmatrix} + 5\begin{pmatrix} 5 \\ -1 \end{pmatrix} = \begin{pmatrix} 31 \\ -5 \end{pmatrix}
$$

Change of coordinates

$\mathbf{u}$ and $\mathbf{w}$ are linearly independent (they are not parallel when $\omega \neq 0$; Topic 4, Lemma of Sec. 7.3). With the columns in the order $\mathbf{u}$, $\mathbf{w}$,

$$
Q = [\mathbf{u} \mid \mathbf{w}] = \begin{pmatrix} 2 & 5 \\ 0 & -1 \end{pmatrix}
$$

$\det Q = (2)(-1) - (5)(0) = -2$, which is not zero, so Q is invertible. For a 2×2 matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad-bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$ (Robinson, Appendix B):

$$Q^{-1} = \frac{1}{-2}\begin{pmatrix} -1 & -5 \\ 0 & 2 \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & \frac{5}{2} \\ 0 & -1 \end{pmatrix}$$

Compute $AQ = [A\mathbf{u} \mid A\mathbf{w}]$ and multiply by Q^{-1} on the left, entry by entry:

$$AQ = \begin{pmatrix} -5 & 31 \\ 3 & -5 \end{pmatrix}$$

$$\begin{aligned} (Q^{-1}AQ)_{11} &= \frac{1}{2}\cdot(-5) + \frac{5}{2}\cdot 3 = 5 \\ (Q^{-1}AQ)_{12} &= \frac{1}{2}\cdot 31 + \frac{5}{2}\cdot(-5) = 3 \\ (Q^{-1}AQ)_{21} &= 0\cdot(-5) + (-1)\cdot 3 = -3 \\ (Q^{-1}AQ)_{22} &= 0\cdot 31 + (-1)\cdot(-5) = 5 \end{aligned}$$

$$Q^{-1}AQ = \begin{pmatrix} 5 & 3 \\ -3 & 5 \end{pmatrix} = J = \begin{pmatrix} \rho & \omega \\ -\omega & \rho \end{pmatrix}$$

Solution in the canonical plane

$$\dot{z}_1 = 5z_1 + 3z_2, \qquad \dot{z}_2 = -3z_1 + 5z_2$$

The two equations are coupled, but they say "scale and rotate", which polar coordinates express directly: $z_1 = r\cos\theta$, $z_2 = r\sin\theta$ (Topic 4, Sec. 7.5).

The radius. From $r^2 = z_1^2 + z_2^2$ the chain rule gives $2r\dot{r} = 2z_1\dot{z}_1 + 2z_2\dot{z}_2$. Substituting $\dot{z}_1 = \rho z_1 + \omega z_2$ and $\dot{z}_2 = -\omega z_1 + \rho z_2$:

$$r\dot{r} = z_1(\rho z_1 + \omega z_2) + z_2(-\omega z_1 + \rho z_2) = \rho z_1^2 + \omega z_1 z_2 - \omega z_1 z_2 + \rho z_2^2 = \rho r^2$$

The terms in ω cancel. Dividing by $r \neq 0$ (the origin is the equilibrium): $\dot{r} = \rho r$.

The angle. Differentiate $\tan\theta = z_2/z_1$ with the chain rule on the left and the quotient rule on the right: $\sec^2\theta\,\dot{\theta} = (z_1\dot{z}_2 - z_2\dot{z}_1)/z_1^2$. Since $\sec^2\theta = r^2/z_1^2$, $\dot{\theta} = (z_1\dot{z}_2 - z_2\dot{z}_1)/r^2$, and

$$z_1\dot{z}_2 - z_2\dot{z}_1 = z_1(-\omega z_1 + \rho z_2) - z_2(\rho z_1 + \omega z_2) = -\omega z_1^2 + \rho z_1 z_2 - \rho z_1 z_2 - \omega z_2^2 = -\omega r^2 \implies \dot{\theta} = -\omega$$

Integration. $\frac{d}{dt}\left(e^{-\rho t} r\right) = e^{-\rho t}(\dot{r} - \rho r) = 0$, and Barrow's rule between 0 and t, $\left[e^{-\rho s} r(s)\right]_0^t = 0$, gives $r(t) = r_0 e^{\rho t}$. For the angle, $\theta(t) - \theta_0 = \int_0^t (-\omega)\, ds = -\omega t$. Back to Cartesian coordinates, with $c_1 = r_0 \cos\theta_0 = z_1(0)$, $c_2 = r_0 \sin\theta_0 = z_2(0)$ and the subtraction formulas $\cos(\theta_0 - \omega t) = \cos\theta_0 \cos\omega t + \sin\theta_0 \sin\omega t$, $\sin(\theta_0 - \omega t) = \sin\theta_0 \cos\omega t - \cos\theta_0 \sin\omega t$:

$$z_1(t) = e^{5t}\left(c_1 \cos 3t + c_2 \sin 3t\right), \qquad z_2(t) = e^{5t}\left(c_2 \cos 3t - c_1 \sin 3t\right)$$

Check by substitution (product rule and chain rule): $\dot{z}_1 = \rho z_1 + e^{\rho t}\omega\left(-c_1 \sin\omega t + c_2 \cos\omega t\right) = \rho z_1 + \omega z_2$ and $\dot{z}_2 = \rho z_2 + e^{\rho t}\omega\left(-c_2 \sin\omega t - c_1 \cos\omega t\right) = \rho z_2 - \omega z_1$, as required.

Back to the original coordinates

$\mathbf{x} = Q\mathbf{z} = z_1\mathbf{u} + z_2\mathbf{w}$, so

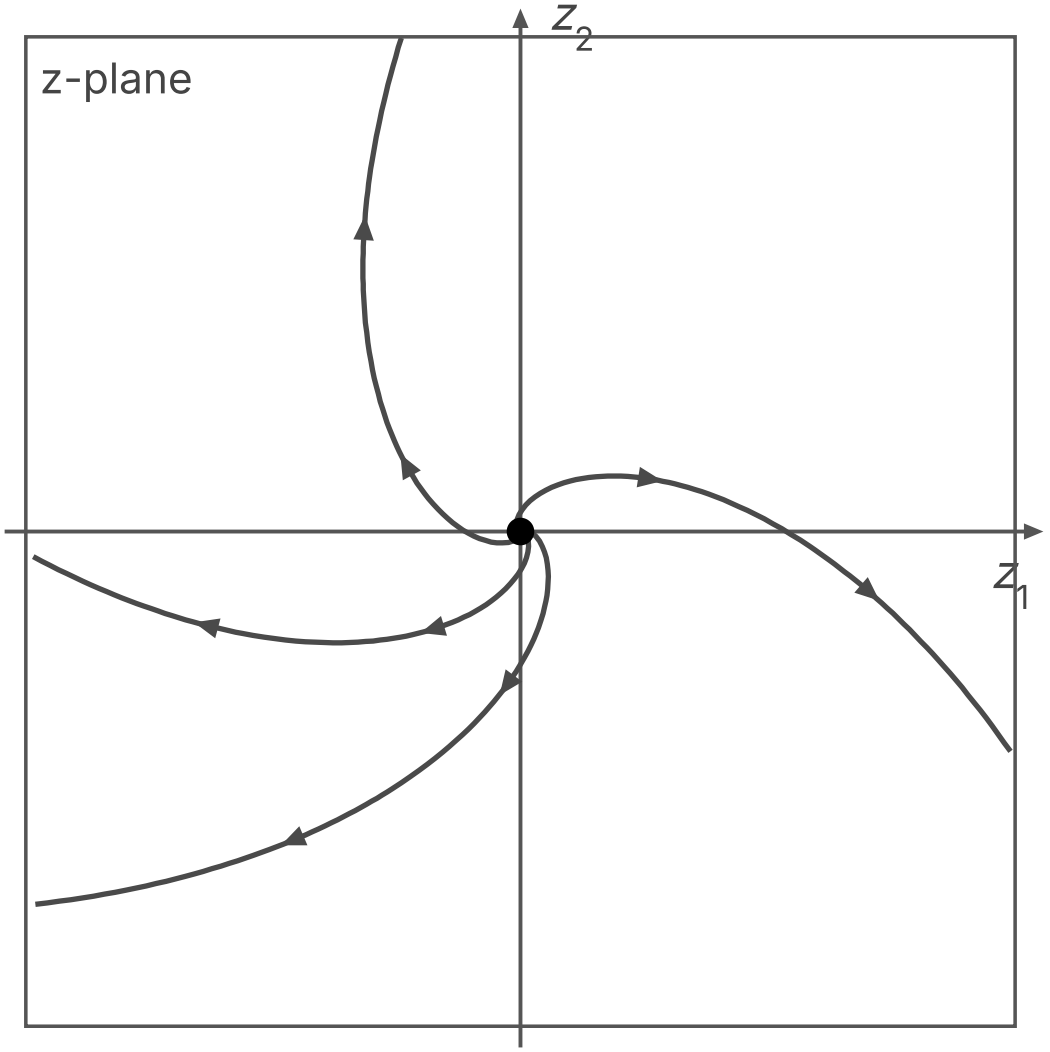
$$\mathbf{x}(t) = e^{5t}\left[\left(c_1 \cos 3t + c_2 \sin 3t\right)\begin{pmatrix}2\\0\end{pmatrix} + \left(c_2 \cos 3t - c_1 \sin 3t\right)\begin{pmatrix}5\\-1\end{pmatrix}\right]$$

$$x(t) = -5c_1 e^{5t}\sin 3t + 2c_1 e^{5t}\cos 3t + 2c_2 e^{5t}\sin 3t + 5c_2 e^{5t}\cos 3t, \qquad y(t) = c_1 e^{5t}\sin 3t - c_2 e^{5t}\cos 3t$$

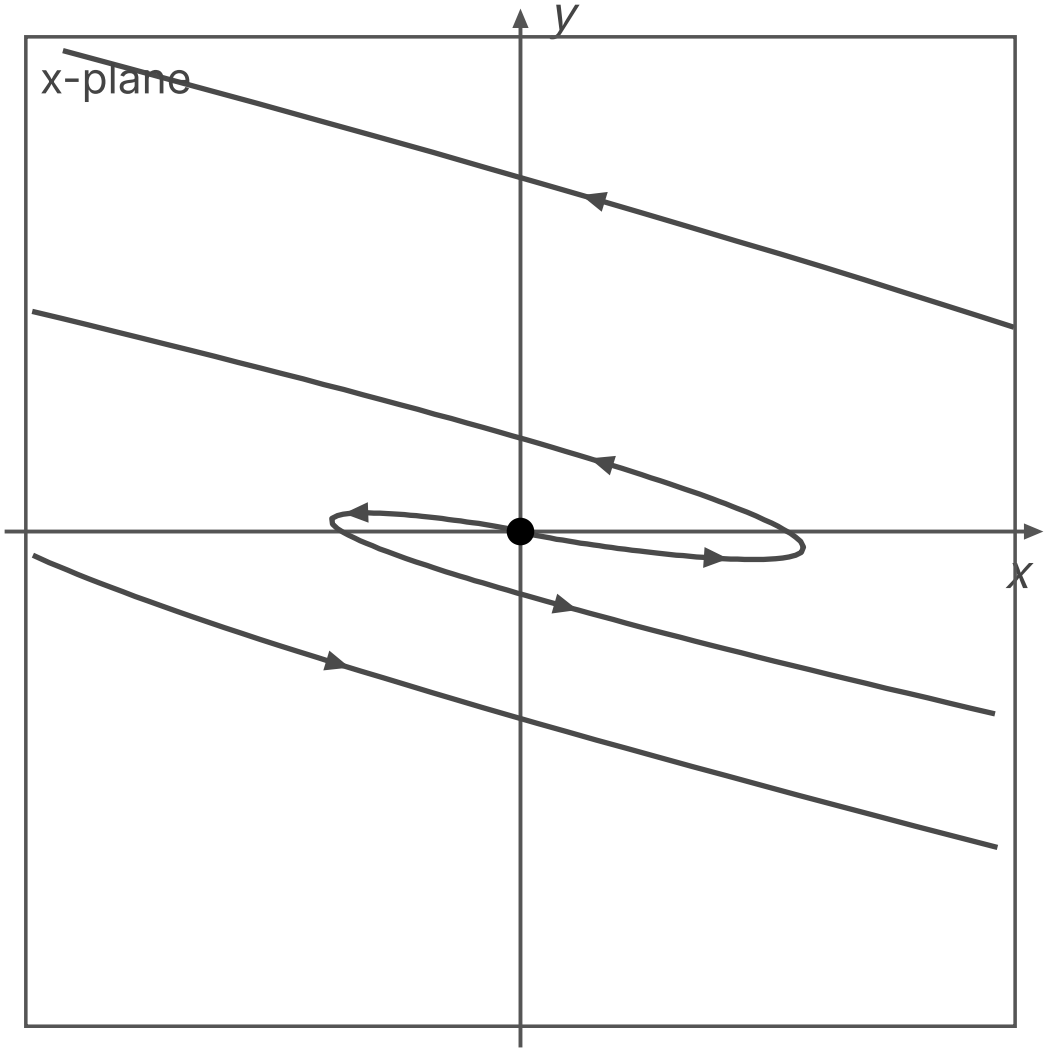
Phase portrait

$\rho = 5$ and $\omega = 3$: the origin is a **unstable spiral**. In the $\mathbf{z}$-plane $r = r_0 e^{5t}$ and $\theta = \theta_0 - 3t$: the motion is always **clockwise** ($\dot{\theta} = -\omega < 0$), and one full turn takes $T = 2\pi/\omega$ while the radius is multiplied by $e^{2\pi\rho/\omega} = e^{\frac{10\pi}{3}}$ (growing).

Sense of rotation in the $\mathbf{x}$-plane, tested directly on the vector field (Topic 4 Sec. 7.8): at $(0,1)^T$, $\dot{\mathbf{x}} = A(0,1)^T = \begin{pmatrix}-\frac{87}{2}\\ \frac{25}{2}\end{pmatrix}$, so $\dot{x} = -\frac{87}{2}$ is < 0: to the left at the top of the plane: the rotation is **counter-clockwise**. Cross-check: $\det Q = -2$ is < 0, so Q reverses orientation and the clockwise motion of the $\mathbf{z}$-plane becomes counter-clockwise in the $\mathbf{x}$-plane, in agreement.



Canonical $\mathbf{z}$-plane (unstable spiral): clockwise, $\dot{\theta} = -\omega$.



Original (x, y)-plane (unstable spiral), counter-clockwise.

(vii) $\lambda_{\pm} = \pm 7i$ with $\boldsymbol{\eta}_{\pm} = (1 \pm i,\ -1 \pm 2i)$, $\dot{x} > 0$

Split the eigenvector of $\lambda_{+} = 7i$ into real and imaginary parts, $\boldsymbol{\eta}_{+} = \begin{pmatrix} 1+i \\ -1+2i \end{pmatrix} = \mathbf{u} + i\mathbf{w}$, so $\rho = 0, \omega = 7$ and

$$\mathbf{u} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}, \qquad \mathbf{w} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

The matrix is not given; it is rebuilt from $A\mathbf{u} = \rho\mathbf{u} - \omega\mathbf{w}$, $A\mathbf{w} = \omega\mathbf{u} + \rho\mathbf{w}$ (Topic 4 Eq. 7.4), which say $AQ = QJ$ with $Q = [\mathbf{u} \mid \mathbf{w}]$, so $A = QJQ^{-1}$:

$$Q = \begin{pmatrix} 1 & 1 \\ -1 & 2 \end{pmatrix}, \quad J = \begin{pmatrix} 0 & 7 \\ -7 & 0 \end{pmatrix}, \quad \det Q = 3, \quad Q^{-1} = \begin{pmatrix} \frac{2}{3} & -\frac{1}{3} \\ \frac{1}{3} & \frac{1}{3} \end{pmatrix}$$

$$
\begin{aligned}
(QJ)_{11} &= 1\cdot 0 + 1\cdot(-7) = -7\\
(QJ)_{12} &= 1\cdot 7 + 1\cdot 0 = 7\\
(QJ)_{21} &= (-1)\cdot 0 + 2\cdot(-7) = -14\\
(QJ)_{22} &= (-1)\cdot 7 + 2\cdot 0 = -7
\end{aligned}
$$

$$
QJ = \begin{pmatrix} -7 & 7 \\ -14 & -7 \end{pmatrix}
$$

$$
\begin{aligned}
(A)_{11} &= (-7)\cdot\frac{2}{3} + 7\cdot\frac{1}{3} = -\frac{7}{3}\\
(A)_{12} &= (-7)\cdot\left(-\frac{1}{3}\right) + 7\cdot\frac{1}{3} = \frac{14}{3}\\
(A)_{21} &= (-14)\cdot\frac{2}{3} + (-7)\cdot\frac{1}{3} = -\frac{35}{3}\\
(A)_{22} &= (-14)\cdot\left(-\frac{1}{3}\right) + (-7)\cdot\frac{1}{3} = \frac{7}{3}
\end{aligned}
$$

$$
A = QJQ^{-1} = \begin{pmatrix} -\frac{7}{3} & \frac{14}{3} \\ -\frac{35}{3} & \frac{7}{3} \end{pmatrix}
$$

Check of the given sign: $\dot{x}$ at $(0,1)$ equals $a_{12} = \frac{14}{3}$, which is positive, in agreement with the statement ($\dot{x} > 0$).

How A acts on $\mathbf{u}$ and $\mathbf{w}$

Separating the real and imaginary parts of $A\mathbf{v} = \lambda\mathbf{v}$ gives (Topic 4, Eq. 7.4) $A\mathbf{u} = \rho\mathbf{u} - \omega\mathbf{w}$ and $A\mathbf{w} = \omega\mathbf{u} + \rho\mathbf{w}$. Verification with the numbers:

$$
A\mathbf{u} = \begin{pmatrix} -7 \\ -14 \end{pmatrix}, \qquad \rho\mathbf{u} - \omega\mathbf{w} = 0\begin{pmatrix} 1 \\ -1 \end{pmatrix} - 7\begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} -7 \\ -14 \end{pmatrix}
$$

$$
A\mathbf{w} = \begin{pmatrix} 7 \\ -7 \end{pmatrix}, \qquad \omega\mathbf{u} + \rho\mathbf{w} = 7\begin{pmatrix} 1 \\ -1 \end{pmatrix} + 0\begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 7 \\ -7 \end{pmatrix}
$$

Change of coordinates

$\mathbf{u}$ and $\mathbf{w}$ are linearly independent (they are not parallel when $\omega \neq 0$; Topic 4, Lemma of Sec. 7.3). With the columns in the order $\mathbf{u}$, $\mathbf{w}$,

$$
Q = [\mathbf{u} \mid \mathbf{w}] = \begin{pmatrix} 1 & 1 \\ -1 & 2 \end{pmatrix}
$$

$\det Q = (1)(2) - (1)(-1) = 3$, which is not zero, so Q is invertible. For a 2×2 matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad-bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$ (Robinson, Appendix B):

$$Q^{-1} = \frac{1}{3}\begin{pmatrix} 2 & -1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} \frac{2}{3} & -\frac{1}{3} \\ \frac{1}{3} & \frac{1}{3} \end{pmatrix}$$

Compute $AQ = [A\mathbf{u} \mid A\mathbf{w}]$ and multiply by Q^{-1} on the left, entry by entry:

$$AQ = \begin{pmatrix} -7 & 7 \\ -14 & -7 \end{pmatrix}$$

$$\begin{aligned} (Q^{-1}AQ)_{11} &= \frac{2}{3}\cdot(-7) + (-\frac{1}{3})\cdot(-14) = 0 \\ (Q^{-1}AQ)_{12} &= \frac{2}{3}\cdot 7 + (-\frac{1}{3})\cdot(-7) = 7 \\ (Q^{-1}AQ)_{21} &= \frac{1}{3}\cdot(-7) + \frac{1}{3}\cdot(-14) = -7 \\ (Q^{-1}AQ)_{22} &= \frac{1}{3}\cdot 7 + \frac{1}{3}\cdot(-7) = 0 \end{aligned}$$

$$Q^{-1}AQ = \begin{pmatrix} 0 & 7 \\ -7 & 0 \end{pmatrix} = J = \begin{pmatrix} \rho & \omega \\ -\omega & \rho \end{pmatrix}$$

Solution in the canonical plane

$$\dot{z}_1 = 7z_2, \qquad \dot{z}_2 = -7z_1$$

The two equations are coupled, but they say "scale and rotate", which polar coordinates express directly: $z_1 = r\cos\theta$, $z_2 = r\sin\theta$ (Topic 4, Sec. 7.5).

The radius. From $r^2 = z_1^2 + z_2^2$ the chain rule gives $2r\dot{r} = 2z_1\dot{z}_1 + 2z_2\dot{z}_2$. Substituting $\dot{z}_1 = \rho z_1 + \omega z_2$ and $\dot{z}_2 = -\omega z_1 + \rho z_2$:

$$r\dot{r} = z_1(\rho z_1 + \omega z_2) + z_2(-\omega z_1 + \rho z_2) = \rho z_1^2 + \omega z_1 z_2 - \omega z_1 z_2 + \rho z_2^2 = \rho r^2$$

The terms in ω cancel. Dividing by $r \neq 0$ (the origin is the equilibrium): $\dot{r} = \rho r$.

The angle. Differentiate $\tan\theta = z_2/z_1$ with the chain rule on the left and the quotient rule on the right: $\sec^2\theta\,\dot{\theta} = (z_1\dot{z}_2 - z_2\dot{z}_1)/z_1^2$. Since $\sec^2\theta = r^2/z_1^2$, $\dot{\theta} = (z_1\dot{z}_2 - z_2\dot{z}_1)/r^2$, and

$$z_1\dot{z}_2 - z_2\dot{z}_1 = z_1(-\omega z_1 + \rho z_2) - z_2(\rho z_1 + \omega z_2) = -\omega z_1^2 + \rho z_1 z_2 - \rho z_1 z_2 - \omega z_2^2 = -\omega r^2 \implies \dot{\theta} = -\omega$$

Integration. $\frac{d}{dt}\left(e^{-\rho t} r\right) = e^{-\rho t}(\dot{r} - \rho r) = 0$, and Barrow's rule between 0 and t, $\left[e^{-\rho s} r(s)\right]_0^t = 0$, gives $r(t) = r_0 e^{\rho t}$. For the angle, $\theta(t) - \theta_0 = \int_0^t (-\omega)\, ds = -\omega t$. Back to Cartesian coordinates, with $c_1 = r_0 \cos\theta_0 = z_1(0)$, $c_2 = r_0 \sin\theta_0 = z_2(0)$ and the subtraction formulas $\cos(\theta_0 - \omega t) = \cos\theta_0 \cos\omega t + \sin\theta_0 \sin\omega t$, $\sin(\theta_0 - \omega t) = \sin\theta_0 \cos\omega t - \cos\theta_0 \sin\omega t$:

$$z_1(t) = e^{0t}\left(c_1 \cos 7t + c_2 \sin 7t\right), \qquad z_2(t) = e^{0t}\left(c_2 \cos 7t - c_1 \sin 7t\right)$$

Check by substitution (product rule and chain rule): $\dot{z}_1 = \rho z_1 + e^{\rho t}\omega\left(-c_1 \sin\omega t + c_2 \cos\omega t\right) = \rho z_1 + \omega z_2$ and $\dot{z}_2 = \rho z_2 + e^{\rho t}\omega\left(-c_2 \sin\omega t - c_1 \cos\omega t\right) = \rho z_2 - \omega z_1$, as required.

Back to the original coordinates

$\mathbf{x} = Q\mathbf{z} = z_1\mathbf{u} + z_2\mathbf{w}$, so

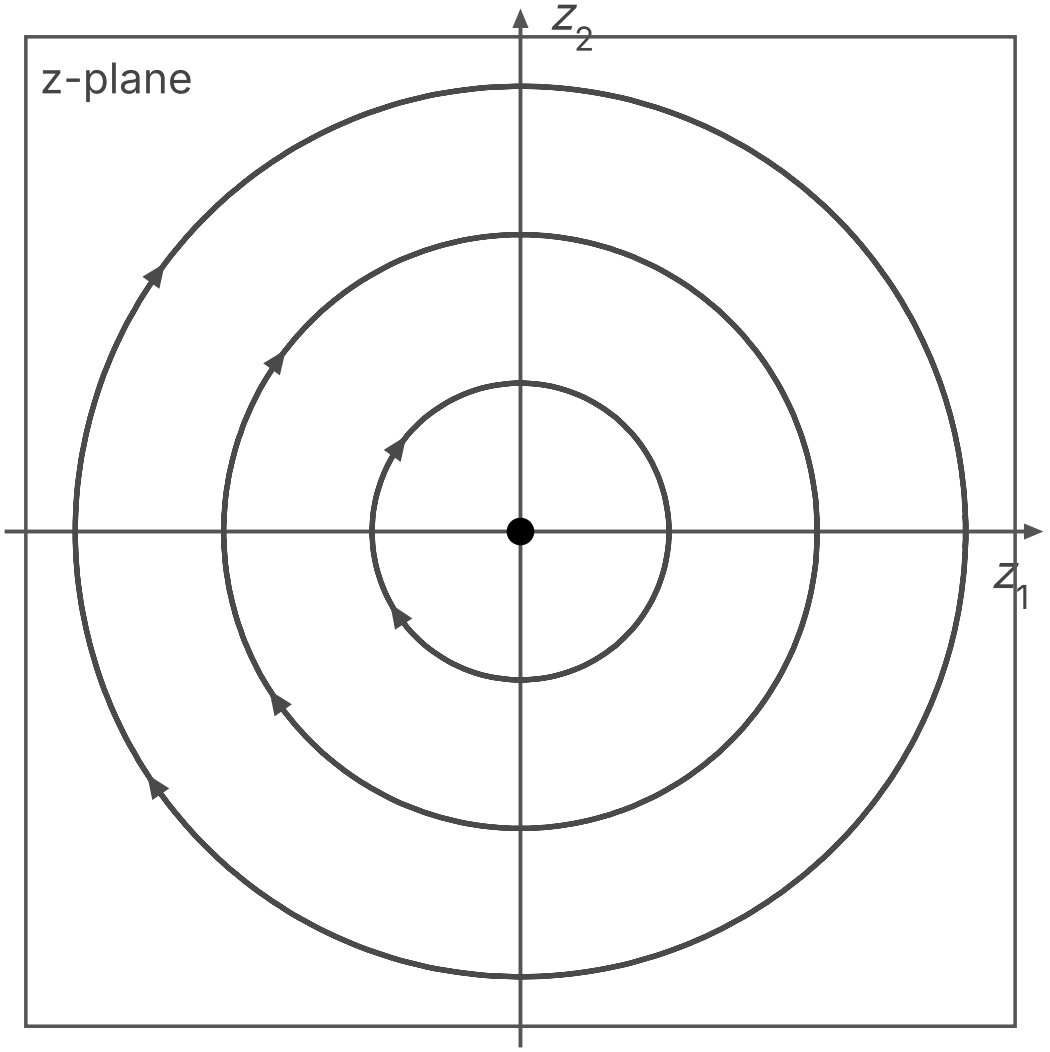
$$\mathbf{x}(t) = e^{0t}\left[\left(c_1 \cos 7t + c_2 \sin 7t\right)\begin{pmatrix}1\\-1\end{pmatrix} + \left(c_2 \cos 7t - c_1 \sin 7t\right)\begin{pmatrix}1\\2\end{pmatrix}\right]$$

$$x(t) = -c_1 \sin 7t + c_1 \cos 7t + c_2 \sin 7t + c_2 \cos 7t, \qquad y(t) = -2c_1 \sin 7t - c_1 \cos 7t - c_2 \sin 7t + 2c_2 \cos 7t$$

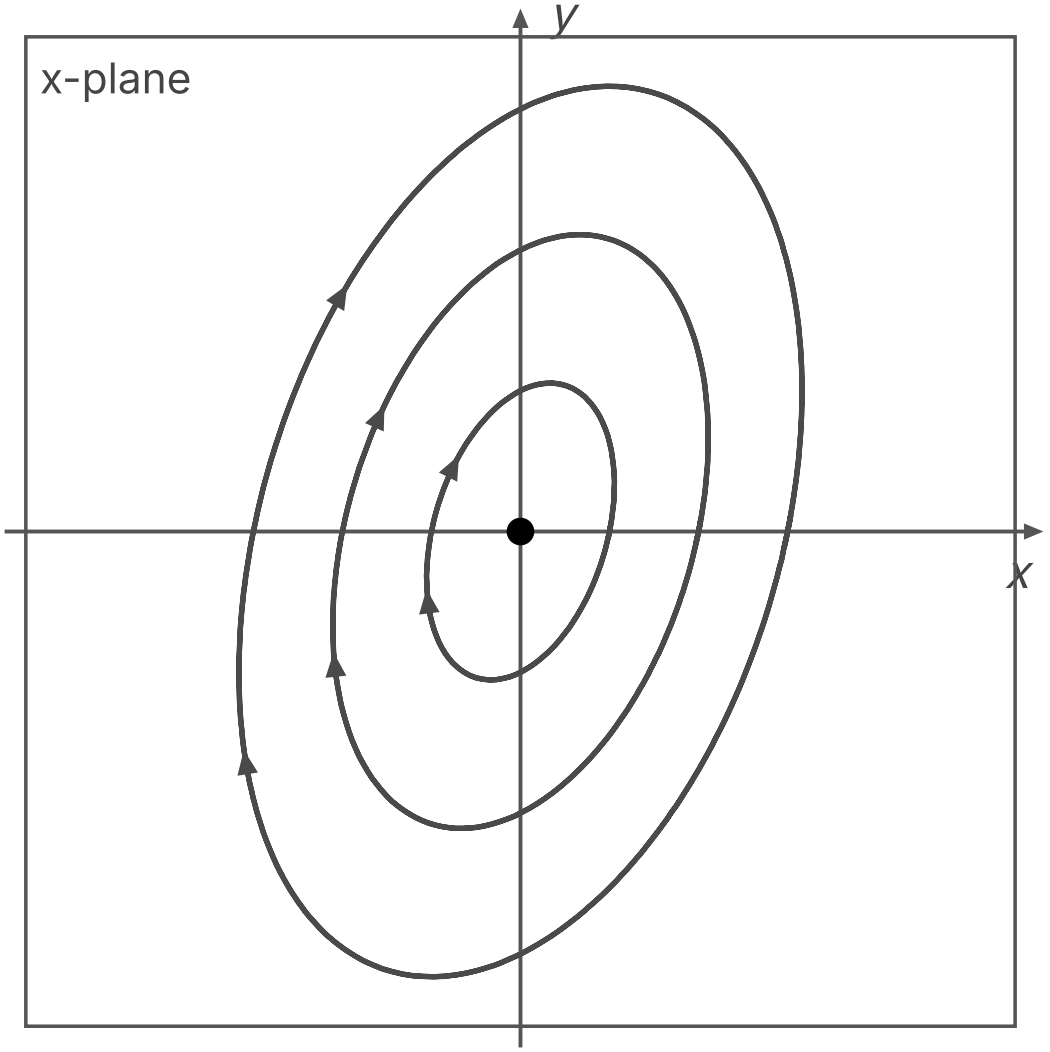
Phase portrait

$\rho = 0$ and $\omega = 7$: the origin is a **centre**. In the $\mathbf{z}$-plane $r = r_0 e^{0t}$ and $\theta = \theta_0 - 7t$: the motion is always **clockwise** ($\dot{\theta} = -\omega < 0$), and the orbits are circles of period $T = 2\pi/\omega$ (closed orbits).

Sense of rotation in the $\mathbf{x}$-plane, tested directly on the vector field (Topic 4 Sec. 7.8): at $(0,1)^T$, $\dot{\mathbf{x}} = A(0,1)^T = \begin{pmatrix}\frac{14}{3}\\ \frac{7}{3}\end{pmatrix}$, so $\dot{x} = \frac{14}{3}$ is > 0: to the right at the top of the plane: the rotation is **clockwise**. Cross-check: $\det Q = 3$ is > 0, so Q preserves orientation and the clockwise motion of the $\mathbf{z}$-plane becomes clockwise in the $\mathbf{x}$-plane, in agreement.



Canonical $\mathbf{z}$-plane (centre): clockwise, $\dot{\theta} = -\omega$.



Original (x, y)-plane (centre), clockwise.

(viii) $\lambda_{\pm} = -13 \pm 17i$ with $\boldsymbol{\eta}_{\pm} = (\pm 6i - 8,\ 4 \mp 5i)$, $\dot{x} > 0$

Split the eigenvector of $\lambda_{+} = -13 + 17i$ into real and imaginary parts, $\boldsymbol{\eta}_{+} = \begin{pmatrix} -8 + 6i \\ 4 - 5i \end{pmatrix} = \mathbf{u} + i\mathbf{w}$, so $\rho = -13, \omega = 17$ and

$$\mathbf{u} = \begin{pmatrix} -8 \\ 4 \end{pmatrix}, \qquad \mathbf{w} = \begin{pmatrix} 6 \\ -5 \end{pmatrix}$$

The matrix is not given; it is rebuilt from $A\mathbf{u} = \rho\mathbf{u} - \omega\mathbf{w}$, $A\mathbf{w} = \omega\mathbf{u} + \rho\mathbf{w}$ (Topic 4 Eq. 7.4), which say $AQ = QJ$ with $Q = [\mathbf{u} \mid \mathbf{w}]$, so $A = QJQ^{-1}$:

$$Q = \begin{pmatrix} -8 & 6 \\ 4 & -5 \end{pmatrix}, \quad J = \begin{pmatrix} -13 & 17 \\ -17 & -13 \end{pmatrix}, \quad \det Q = 16, \quad Q^{-1} = \begin{pmatrix} -\frac{5}{16} & -\frac{3}{8} \\ -\frac{1}{4} & -\frac{1}{2} \end{pmatrix}$$

$$
\begin{aligned}
(QJ)_{11} &= (-8)\cdot(-13) + 6\cdot(-17) = 2\\
(QJ)_{12} &= (-8)\cdot 17 + 6\cdot(-13) = -214\\
(QJ)_{21} &= 4\cdot(-13) + (-5)\cdot(-17) = 33\\
(QJ)_{22} &= 4\cdot 17 + (-5)\cdot(-13) = 133
\end{aligned}
$$

$$
QJ = \begin{pmatrix} 2 & -214 \\ 33 & 133 \end{pmatrix}
$$

$$
\begin{aligned}
(A)_{11} &= 2\cdot(-\frac{5}{16}) + (-214)\cdot(-\frac{1}{4}) = \frac{423}{8}\\
(A)_{12} &= 2\cdot(-\frac{3}{8}) + (-214)\cdot(-\frac{1}{2}) = \frac{425}{4}\\
(A)_{21} &= 33\cdot(-\frac{5}{16}) + 133\cdot(-\frac{1}{4}) = -\frac{697}{16}\\
(A)_{22} &= 33\cdot(-\frac{3}{8}) + 133\cdot(-\frac{1}{2}) = -\frac{631}{8}
\end{aligned}
$$

$$
A = QJQ^{-1} = \begin{pmatrix} \frac{423}{8} & \frac{425}{4} \\ -\frac{697}{16} & -\frac{631}{8} \end{pmatrix}
$$

Check of the given sign: $\dot{x}$ at $(0,1)$ equals $a_{12} = \frac{425}{4}$, which is positive, in agreement with the statement ($\dot{x} > 0$).

How A acts on $\mathbf{u}$ and $\mathbf{w}$

Separating the real and imaginary parts of $A\mathbf{v} = \lambda\mathbf{v}$ gives (Topic 4, Eq. 7.4) $A\mathbf{u} = \rho\mathbf{u} - \omega\mathbf{w}$ and $A\mathbf{w} = \omega\mathbf{u} + \rho\mathbf{w}$. Verification with the numbers:

$$
A\mathbf{u} = \begin{pmatrix} 2 \\ 33 \end{pmatrix}, \qquad \rho\mathbf{u} - \omega\mathbf{w} = (-13)\begin{pmatrix} -8 \\ 4 \end{pmatrix} - 17\begin{pmatrix} 6 \\ -5 \end{pmatrix} = \begin{pmatrix} 2 \\ 33 \end{pmatrix}
$$

$$
A\mathbf{w} = \begin{pmatrix} -214 \\ 133 \end{pmatrix}, \qquad \omega\mathbf{u} + \rho\mathbf{w} = 17\begin{pmatrix} -8 \\ 4 \end{pmatrix} + (-13)\begin{pmatrix} 6 \\ -5 \end{pmatrix} = \begin{pmatrix} -214 \\ 133 \end{pmatrix}
$$

Change of coordinates

$\mathbf{u}$ and $\mathbf{w}$ are linearly independent (they are not parallel when $\omega \neq 0$; Topic 4, Lemma of Sec. 7.3). With the columns in the order $\mathbf{u}$, $\mathbf{w}$,

$$
Q = [\,\mathbf{u} \mid \mathbf{w}\,] = \begin{pmatrix} -8 & 6 \\ 4 & -5 \end{pmatrix}
$$

$\det Q = (-8)(-5) - (6)(4) = 16$, which is not zero, so Q is invertible. For a 2×2 matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad-bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$ (Robinson, Appendix B):

$$Q^{-1} = \frac{1}{16}\begin{pmatrix} -5 & -6 \\ -4 & -8 \end{pmatrix} = \begin{pmatrix} -\frac{5}{16} & -\frac{3}{8} \\ -\frac{1}{4} & -\frac{1}{2} \end{pmatrix}$$

Compute $AQ = [A\mathbf{u} \mid A\mathbf{w}]$ and multiply by Q^{-1} on the left, entry by entry:

$$AQ = \begin{pmatrix} 2 & -214 \\ 33 & 133 \end{pmatrix}$$

$$\begin{aligned} (Q^{-1}AQ)_{11} &= (-\frac{5}{16}) \cdot 2 + (-\frac{3}{8}) \cdot 33 = -13 \\ (Q^{-1}AQ)_{12} &= (-\frac{5}{16}) \cdot (-214) + (-\frac{3}{8}) \cdot 133 = 17 \\ (Q^{-1}AQ)_{21} &= (-\frac{1}{4}) \cdot 2 + (-\frac{1}{2}) \cdot 33 = -17 \\ (Q^{-1}AQ)_{22} &= (-\frac{1}{4}) \cdot (-214) + (-\frac{1}{2}) \cdot 133 = -13 \end{aligned}$$

$$Q^{-1}AQ = \begin{pmatrix} -13 & 17 \\ -17 & -13 \end{pmatrix} = J = \begin{pmatrix} \rho & \omega \\ -\omega & \rho \end{pmatrix}$$

Solution in the canonical plane

$$\dot{z}_1 = -13z_1 + 17z_2, \qquad \dot{z}_2 = -17z_1 - 13z_2$$

The two equations are coupled, but they say "scale and rotate", which polar coordinates express directly: $z_1 = r\cos\theta$, $z_2 = r\sin\theta$ (Topic 4, Sec. 7.5).

The radius. From $r^2 = z_1^2 + z_2^2$ the chain rule gives $2r\dot{r} = 2z_1\dot{z}_1 + 2z_2\dot{z}_2$. Substituting $\dot{z}_1 = \rho z_1 + \omega z_2$ and $\dot{z}_2 = -\omega z_1 + \rho z_2$:

$$r\dot{r} = z_1(\rho z_1 + \omega z_2) + z_2(-\omega z_1 + \rho z_2) = \rho z_1^2 + \omega z_1 z_2 - \omega z_1 z_2 + \rho z_2^2 = \rho r^2$$

The terms in ω cancel. Dividing by $r \neq 0$ (the origin is the equilibrium): $\dot{r} = \rho r$.

The angle. Differentiate $\tan\theta = z_2/z_1$ with the chain rule on the left and the quotient rule on the right: $\sec^2\theta\,\dot{\theta} = (z_1\dot{z}_2 - z_2\dot{z}_1)/z_1^2$. Since $\sec^2\theta = r^2/z_1^2$, $\dot{\theta} = (z_1\dot{z}_2 - z_2\dot{z}_1)/r^2$, and

$$z_1\dot z_2 - z_2\dot z_1 = z_1(-\omega z_1 + \rho z_2) - z_2(\rho z_1 + \omega z_2) = -\omega z_1^2 + \rho z_1 z_2 - \rho z_1 z_2 - \omega z_2^2 = -\omega r^2 \implies \dot\theta = -\omega$$

Integration. $\frac{d}{dt}\left(e^{-\rho t} r\right) = e^{-\rho t}(\dot r - \rho r) = 0$, and Barrow's rule between 0 and t, $\left[e^{-\rho s} r(s)\right]_0^t = 0$, gives $r(t) = r_0 e^{\rho t}$. For the angle, $\theta(t) - \theta_0 = \int_0^t (-\omega)\, ds = -\omega t$. Back to Cartesian coordinates, with $c_1 = r_0\cos\theta_0 = z_1(0)$, $c_2 = r_0 \sin\theta_0 = z_2(0)$ and the subtraction formulas $\cos(\theta_0 - \omega t) = \cos\theta_0\cos\omega t + \sin\theta_0\sin\omega t$, $\sin(\theta_0 - \omega t) = \sin\theta_0\cos\omega t - \cos\theta_0\sin\omega t$:

$$z_1(t) = e^{-13t}\left(c_1\cos 17t + c_2 \sin 17t\right), \qquad z_2(t) = e^{-13t}\left(c_2\cos 17t - c_1\sin 17t\right)$$

Check by substitution (product rule and chain rule): $\dot z_1 = \rho z_1 + e^{\rho t}\omega\left(-c_1\sin\omega t + c_2\cos\omega t\right) = \rho z_1 + \omega z_2$ and $\dot z_2 = \rho z_2 + e^{\rho t}\omega\left(-c_2\sin\omega t - c_1\cos\omega t\right) = \rho z_2 - \omega z_1$, as required.

Back to the original coordinates

$\mathbf{x} = Q\mathbf{z} = z_1\mathbf{u} + z_2\mathbf{w}$, so

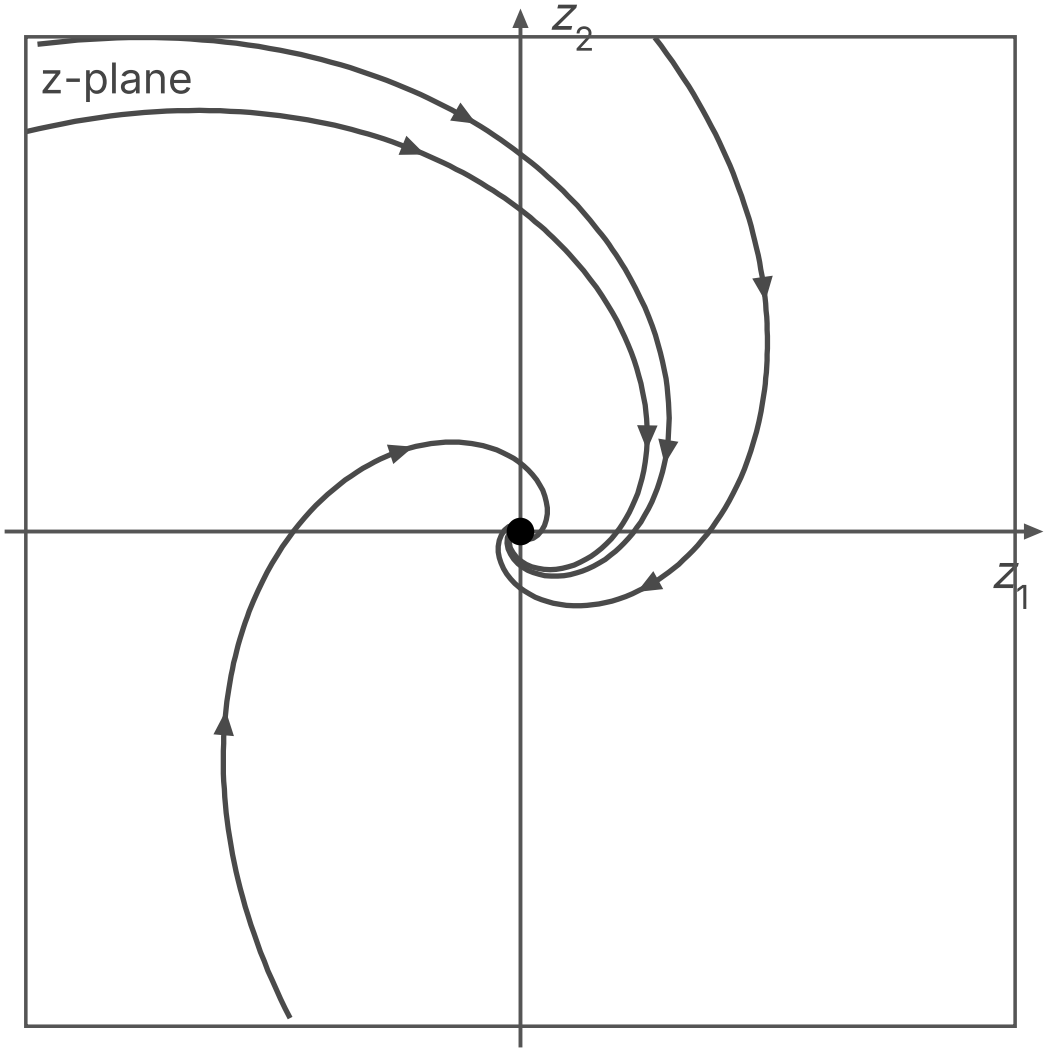
$$\mathbf{x}(t) = e^{-13t}\left[(c_1\cos 17t + c_2\sin 17t)\begin{pmatrix}-8\\4\end{pmatrix} + (c_2\cos 17t - c_1\sin 17t)\begin{pmatrix}6\\-5\end{pmatrix}\right]$$

$$x(t) = -6c_1 e^{-13t}\sin 17t - 8c_1 e^{-13t}\cos 17t - 8c_2 e^{-13t}\sin 17t + 6c_2 e^{-13t}\cos 17t, \qquad y(t) = 5c_1 e^{-13t}\sin 17t + 4c_1 e^{-13t}\cos 17t + 4c_2 e^{-13t}\sin 17t - 5c_2 e^{-13t}\cos 17t$$

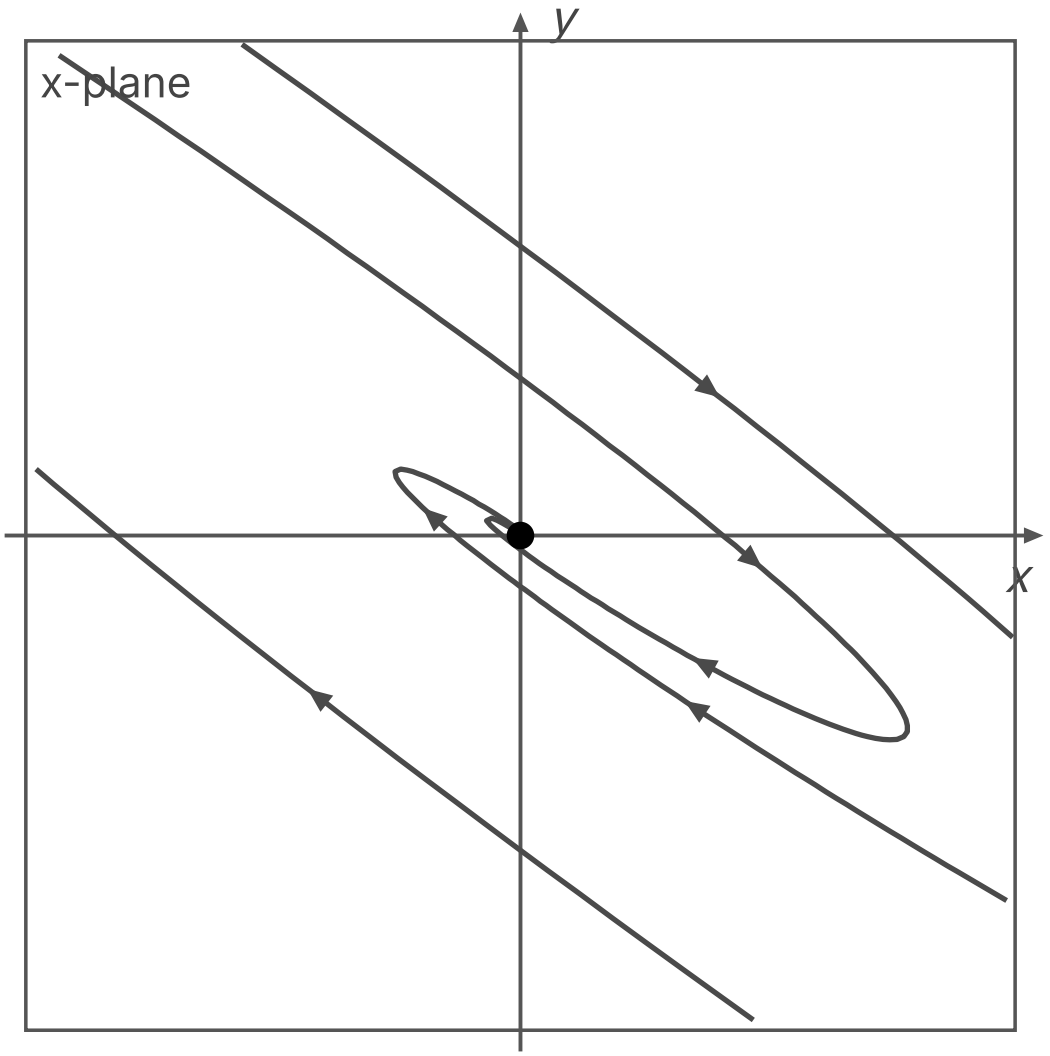
Phase portrait

$\rho = -13$ and $\omega = 17$: the origin is a **stable spiral**. In the $\mathbf{z}$-plane $r = r_0 e^{-13t}$ and $\theta = \theta_0 - 17t$: the motion is always **clockwise** ($\dot\theta = -\omega < 0$), and one full turn takes $T = 2\pi/\omega$ while the radius is multiplied by $e^{2\pi\rho/\omega} = e^{-\frac{26\pi}{17}}$ (shrinking).

Sense of rotation in the $\mathbf{x}$-plane, tested directly on the vector field (Topic 4 Sec. 7.8): at $(0,1)^T$, $\dot{\mathbf{x}} = A(0,1)^T = \begin{pmatrix}\frac{425}{4}\\ -\frac{631}{8}\end{pmatrix}$, so $\dot x = \frac{425}{4}$ is > 0: to the right at the top of the plane: the rotation is **clockwise**. Cross-check: $\det Q = 16$ is > 0, so Q preserves orientation and the clockwise motion of the $\mathbf{z}$-plane becomes clockwise in the $\mathbf{x}$-plane, in agreement.



Canonical $\mathbf{z}$-plane (stable spiral): clockwise, $\dot{\theta} = -\omega$.



Original (x, y)-plane (stable spiral), clockwise.

PHASE 4: INTERPRETATION AND CHECKS

Item	A	Type	Sense of rotation
(i)	$\begin{pmatrix} 7 & -3 \\ 15 & -5 \end{pmatrix}$	unstable spiral	counter-clockwise
(ii)	$\begin{pmatrix} -3 & -3 \\ 6 & 3 \end{pmatrix}$	centre	counter-clockwise
(iii)	$\begin{pmatrix} -\frac{5}{2} & \frac{1}{2} \\ -\frac{5}{2} & -\frac{3}{2} \end{pmatrix}$	stable spiral	clockwise
(iv)	$\begin{pmatrix} -1 & 1 \\ -1 & -1 \end{pmatrix}$	stable spiral	clockwise
(v)	$\begin{pmatrix} \frac{2}{5} & -\frac{6}{5} \\ \frac{82}{15} & \frac{18}{5} \end{pmatrix}$	unstable spiral	counter-clockwise
(vi)	$\begin{pmatrix} -\frac{5}{2} & -\frac{87}{2} \\ \frac{3}{2} & \frac{25}{2} \end{pmatrix}$	unstable spiral	counter-clockwise

Item	A	Type	Sense of rotation
(vii)	$\begin{pmatrix} -\frac{7}{3} & \frac{14}{3} \\ -\frac{35}{3} & \frac{7}{3} \end{pmatrix}$	centre	clockwise
(viii)	$\begin{pmatrix} \frac{423}{8} & \frac{425}{4} \\ -\frac{697}{16} & -\frac{631}{8} \end{pmatrix}$	stable spiral	clockwise

- In every item $\det Q$ and the sign of a_{12} agree (the code asserts $\det Q > 0 \Leftrightarrow a_{12} > 0$), which is the Topic 4 rule: the $\mathbf{z}$-plane turns clockwise, and Q reverses the sense exactly when $\det Q < 0$.
- Centres (ii) and (vii) ($\rho = 0$): $\operatorname{tr} A = 0$ and $\det A = \omega^2 > 0$; the orbits are closed ellipses with period $2\pi/\omega$ ($2\pi/3$ and $2\pi/7$).
- Spirals: (i), (v), (vi) unstable; (iii), (iv), (viii) stable. The growth/decay factor per turn is $e^{2\pi\rho/\omega}$, stated in each item; (viii) decays extremely fast ($e^{-26\pi/17} \approx 0.008$ per turn), so its spiral is nearly invisible after one turn.
- The eigen-data are consistent: $A\boldsymbol{\eta}_+ = \lambda_+\boldsymbol{\eta}_+$ holds in each item by construction of A (Eq. 7.4 verified entry by entry).

EXERCISE 2

General Solutions for the Eight Complex Eigen-data

Original statement · probls_ch4-ii_2627.pdf

Real General Solution • Euler's Formula

STATEMENT

> Write down the general solution of the equation $d\mathbf{x}/dt = A\mathbf{x}$, when the eigenvalues ($\lambda_\pm$) and eigenvectors ($\boldsymbol{\eta}_\pm$) of A are those in the previous exercise.

PHASE 1: SETUP

The eight eigen-data are those of Exercise 1 (listed there). For each, $\boldsymbol{\eta}_+ = \mathbf{u} + i\mathbf{w}$ is the eigenvector of $\lambda_+ = \rho + i\omega$.

PHASE 2: FORMULATION: THE REAL GENERAL SOLUTION

Complex solutions $e^{\lambda_+ t}\boldsymbol{\eta}_+$ and $e^{\lambda_- t}\bar{\boldsymbol{\eta}}_+$ combine into real ones: choose conjugate coefficients, $\mathbf{x} = 2\,\mathrm{Re}\left[Ce^{(\rho+i\omega)t}(\mathbf{u}+i\mathbf{w})\right]$ with $C = \alpha + i\beta$. Expanding $e^{i\omega t} = \cos\omega t + i\sin\omega t$ (Euler) and keeping the real part, with $c_1 = 2\alpha$, $c_2 = -2\beta$ (Topic 4 Eq. 7.3, Robinson Eq. 29.2):

$$\mathbf{x}(t) = e^{\rho t}\Big[\,(c_1\cos\omega t + c_2\sin\omega t)\,\mathbf{u} + (c_2\cos\omega t - c_1\sin\omega t)\,\mathbf{w}\Big]$$

Check of the derivation of the real part: $(a+ib)(\mathbf{u}+i\mathbf{w})$ has real part $a\mathbf{u} - b\mathbf{w}$ because $i\cdot i = -1$; with $a = \alpha\cos\omega t - \beta\sin\omega t$ and $b = \beta\cos\omega t + \alpha\sin\omega t$ this is $\frac{1}{2}[(c_1\cos\omega t + c_2\sin\omega t)\mathbf{u} + (c_2\cos\omega t - c_1\sin\omega t)\mathbf{w}]$ after substituting $\alpha = c_1/2$, $\beta = -c_2/2$.

> METHOD OF THE CLASS NOTES [NOTES, P. 10]
>
> The notes write the real general solution as $\mathbf{x} = C_1\,\mathrm{Re}\{\mathbf{v}e^{\lambda t}\} + C_2\,\mathrm{Im}\{\mathbf{v}e^{\lambda t}\}$ with $\mathbf{v} = \mathbf{v}_1 + i\mathbf{v}_2$ and $\lambda = \rho + i\omega$. Since $\mathbf{v}e^{\lambda t} = e^{\rho t}(\cos\omega t + i\sin\omega t)(\mathbf{v}_1 + i\mathbf{v}_2)$,
>
> $$\mathrm{Re}\{\mathbf{v}e^{\lambda t}\} = e^{\rho t}(\cos\omega t\,\mathbf{v}_1 - \sin\omega t\,\mathbf{v}_2), \qquad \mathrm{Im}\{\mathbf{v}e^{\lambda t}\} = e^{\rho t}(\sin\omega t\,\mathbf{v}_1 + \cos\omega t\,\mathbf{v}_2),$$
>
> so it is the formula above with $c_1 = C_1$ and $c_2 = C_2$.

PHASE 3: DERIVATION: THE EIGHT SOLUTIONS

(i) $\lambda_\pm = 1 \pm 3i$ with $\boldsymbol{\eta}_\pm = (1,\ 2 \mp i),\ \dot{x} < 0$

$\rho = 1, \omega = 3, \mathbf{u} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \mathbf{w} = \begin{pmatrix} 0 \\ -1 \end{pmatrix}$. Substituting in the general formula,

$$\mathbf{x}(t) = e^{1t}\left[(c_1 \cos 3t + c_2 \sin 3t)\begin{pmatrix} 1 \\ 2 \end{pmatrix} + (c_2 \cos 3t - c_1 \sin 3t)\begin{pmatrix} 0 \\ -1 \end{pmatrix}\right]$$

Component by component:

$$x(t) = c_1 e^{t} \cos 3t + c_2 e^{t} \sin 3t, \qquad y(t) = c_1 e^{t} \sin 3t + 2c_1 e^{t} \cos 3t + 2c_2 e^{t} \sin 3t - c_2 e^{t} \cos 3t$$

Check: with $A = \begin{pmatrix} 7 & -3 \\ 15 & -5 \end{pmatrix}$ (Exercise 1), $\dot{\mathbf{x}} - A\mathbf{x} = \mathbf{0}$ identically (verified symbolically).

(ii) $\lambda_\pm = \pm 3i$ with $\boldsymbol{\eta}_\pm = (1 \pm 2i,\ 1 \mp 3i),\ \dot{x} < 0$

$\rho = 0, \omega = 3, \mathbf{u} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \mathbf{w} = \begin{pmatrix} 2 \\ -3 \end{pmatrix}$. Substituting in the general formula,

$$\mathbf{x}(t) = e^{0t}\left[(c_1 \cos 3t + c_2 \sin 3t)\begin{pmatrix} 1 \\ 1 \end{pmatrix} + (c_2 \cos 3t - c_1 \sin 3t)\begin{pmatrix} 2 \\ -3 \end{pmatrix}\right]$$

Component by component:

$$x(t) = -2c_1 \sin 3t + c_1 \cos 3t + c_2 \sin 3t + 2c_2 \cos 3t, \qquad y(t) = 3c_1 \sin 3t + c_1 \cos 3t + c_2 \sin 3t - 3c_2 \cos 3t$$

Check: with $A = \begin{pmatrix} -3 & -3 \\ 6 & 3 \end{pmatrix}$ (Exercise 1), $\dot{\mathbf{x}} - A\mathbf{x} = \mathbf{0}$ identically (verified symbolically).

(iii) $\lambda_\pm = -2 \pm i$ with $\boldsymbol{\eta}_\pm = (1 \mp i,\ 3 \pm i),\ \dot{x} > 0$

$\rho = -2, \omega = 1, \mathbf{u} = \begin{pmatrix} 1 \\ 3 \end{pmatrix}, \mathbf{w} = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$. Substituting in the general formula,

$$\mathbf{x}(t) = e^{-2t}\left[(c_1 \cos 1t + c_2 \sin 1t)\begin{pmatrix} 1 \\ 3 \end{pmatrix} + (c_2 \cos 1t - c_1 \sin 1t)\begin{pmatrix} -1 \\ 1 \end{pmatrix}\right]$$

Component by component:

$$x(t) = c_1 e^{-2t}\sin t + c_1 e^{-2t}\cos t + c_2 e^{-2t}\sin t - c_2 e^{-2t}\cos t, \qquad y(t) = -c_1 e^{-2t}\sin t + 3c_1 e^{-2t}\cos t + 3c_2 e^{-2t}\sin t + c_2 e^{-2t}\cos t$$

Check: with $A = \begin{pmatrix} -\frac{5}{2} & \frac{1}{2} \\ -\frac{5}{2} & -\frac{3}{2} \end{pmatrix}$ (Exercise 1), $\dot{\mathbf{x}} - A\mathbf{x} = \mathbf{0}$ identically (verified symbolically).

(iv) $\lambda_\pm = -1 \pm i$ with $\boldsymbol{\eta}_\pm = (1,\ \pm i),\ \dot{x} > 0$

$\rho = -1, \omega = 1, \mathbf{u} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \mathbf{w} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$. Substituting in the general formula,

$$\mathbf{x}(t) = e^{-1t}\left[(c_1\cos 1t + c_2\sin 1t)\begin{pmatrix} 1 \\ 0 \end{pmatrix} + (c_2\cos 1t - c_1\sin 1t)\begin{pmatrix} 0 \\ 1 \end{pmatrix}\right]$$

Component by component:

$$x(t) = c_1 e^{-t}\cos t + c_2 e^{-t}\sin t, \qquad y(t) = -c_1 e^{-t}\sin t + c_2 e^{-t}\cos t$$

Check: with $A = \begin{pmatrix} -1 & 1 \\ -1 & -1 \end{pmatrix}$ (Exercise 1), $\dot{\mathbf{x}} - A\mathbf{x} = \mathbf{0}$ identically (verified symbolically).

(v) $\lambda_\pm = 2 \pm 2i$ with $\boldsymbol{\eta}_\pm = (\pm 3i,\ 5 \mp 4i),\ \dot{x} < 0$

$\rho = 2, \omega = 2, \mathbf{u} = \begin{pmatrix} 0 \\ 5 \end{pmatrix}, \mathbf{w} = \begin{pmatrix} 3 \\ -4 \end{pmatrix}$. Substituting in the general formula,

$$\mathbf{x}(t) = e^{2t}\left[(c_1\cos 2t + c_2\sin 2t)\begin{pmatrix} 0 \\ 5 \end{pmatrix} + (c_2\cos 2t - c_1\sin 2t)\begin{pmatrix} 3 \\ -4 \end{pmatrix}\right]$$

Component by component:

$$x(t) = -3c_1 e^{2t}\sin 2t + 3c_2 e^{2t}\cos 2t, \qquad y(t) = 4c_1 e^{2t}\sin 2t + 5c_1 e^{2t}\cos 2t + 5c_2 e^{2t}\sin 2t - 4c_2 e^{2t}\cos 2t$$

Check: with $A = \begin{pmatrix} \frac{2}{5} & -\frac{6}{5} \\ \frac{82}{15} & \frac{18}{5} \end{pmatrix}$ (Exercise 1), $\dot{\mathbf{x}} - A\mathbf{x} = \mathbf{0}$ identically (verified symbolically).

(vi) $\lambda_\pm = 5 \pm 3i$ with $\boldsymbol{\eta}_\pm = (2 \pm 5i,\ \mp i),\ \dot{x} < 0$

$\rho = 5, \omega = 3, \mathbf{u} = \begin{pmatrix} 2 \\ 0 \end{pmatrix}, \mathbf{w} = \begin{pmatrix} 5 \\ -1 \end{pmatrix}$. Substituting in the general formula,

$$\mathbf{x}(t) = e^{5t}\left[(c_1\cos 3t + c_2\sin 3t)\begin{pmatrix}2\\0\end{pmatrix} + (c_2\cos 3t - c_1\sin 3t)\begin{pmatrix}5\\-1\end{pmatrix}\right]$$

Component by component:

$$x(t) = -5c_1e^{5t}\sin 3t + 2c_1e^{5t}\cos 3t + 2c_2e^{5t}\sin 3t + 5c_2e^{5t}\cos 3t, \qquad y(t) = c_1e^{5t}\sin 3t - c_2e^{5t}\cos 3t$$

Check: with $A = \begin{pmatrix}-\frac{5}{2} & -\frac{87}{2}\\ \frac{3}{2} & \frac{25}{2}\end{pmatrix}$ (Exercise 1), $\dot{\mathbf{x}} - A\mathbf{x} = \mathbf{0}$ identically (verified symbolically).

(vii) $\lambda_\pm = \pm 7i$ with $\boldsymbol{\eta}_\pm = (1 \pm i,\ -1 \pm 2i)$, $\dot{x} > 0$

$\rho = 0, \omega = 7, \mathbf{u} = \begin{pmatrix}1\\-1\end{pmatrix}, \mathbf{w} = \begin{pmatrix}1\\2\end{pmatrix}$. Substituting in the general formula,

$$\mathbf{x}(t) = e^{0t}\left[(c_1\cos 7t + c_2\sin 7t)\begin{pmatrix}1\\-1\end{pmatrix} + (c_2\cos 7t - c_1\sin 7t)\begin{pmatrix}1\\2\end{pmatrix}\right]$$

Component by component:

$$x(t) = -c_1\sin 7t + c_1\cos 7t + c_2\sin 7t + c_2\cos 7t, \qquad y(t) = -2c_1\sin 7t - c_1\cos 7t - c_2\sin 7t + 2c_2\cos 7t$$

Check: with $A = \begin{pmatrix}-\frac{7}{3} & \frac{14}{3}\\ -\frac{35}{3} & \frac{7}{3}\end{pmatrix}$ (Exercise 1), $\dot{\mathbf{x}} - A\mathbf{x} = \mathbf{0}$ identically (verified symbolically).

(viii) $\lambda_\pm = -13 \pm 17i$ with $\boldsymbol{\eta}_\pm = (\pm 6i - 8,\ 4 \mp 5i)$, $\dot{x} > 0$

$\rho = -13, \omega = 17, \mathbf{u} = \begin{pmatrix}-8\\4\end{pmatrix}, \mathbf{w} = \begin{pmatrix}6\\-5\end{pmatrix}$. Substituting in the general formula,

$$\mathbf{x}(t) = e^{-13t}\left[(c_1\cos 17t + c_2\sin 17t)\begin{pmatrix}-8\\4\end{pmatrix} + (c_2\cos 17t - c_1\sin 17t)\begin{pmatrix}6\\-5\end{pmatrix}\right]$$

Component by component:

$$x(t) = -6c_1e^{-13t}\sin 17t - 8c_1e^{-13t}\cos 17t - 8c_2e^{-13t}\sin 17t + 6c_2e^{-13t}\cos 17t, \qquad y(t) = 5c_1e^{-13t}\sin 17t + 4c_1e^{-13t}\cos 17t + 4c_2e^{-13t}\sin 17t - 5c_2e^{-13t}\cos 17t$$

Check: with $A = \begin{pmatrix}\frac{423}{8} & \frac{425}{4}\\ -\frac{697}{16} & -\frac{631}{8}\end{pmatrix}$ (Exercise 1), $\dot{\mathbf{x}} - A\mathbf{x} = \mathbf{0}$ identically (verified symbolically).

PHASE 4: INTERPRETATION AND CHECKS

- The two constants are real and are the initial coordinates in the $(\mathbf{u}, \mathbf{w})$ basis: $(c_1, c_2)^T = Q^{-1}\mathbf{x}_0$.
- Centres ($\rho = 0$): (ii) and (vii) give purely periodic $\mathbf{x}(t)$ with period $2\pi/3$ and $2\pi/7$; spirals have an envelope $e^{\rho t}$.
- Linear independence: $e^{\rho t}(\cos \omega t\, \mathbf{u} - \sin \omega t\, \mathbf{w})$ and $e^{\rho t}(\sin \omega t\, \mathbf{u} + \cos \omega t\, \mathbf{w})$ have Wronskian $e^{2\rho t} \det[\mathbf{u}\ \mathbf{w}] \neq 0$ since $\mathbf{u}, \mathbf{w}$ are independent (Exercise 4).
- Dimensional check: the exponent ρt and the phase ωt are dimensionless when ρ and ω are rates in s^{-1} and t in s.

EXERCISE 3

Canonical Form of Four Systems with Complex Eigenvalues

Original statement · probls_ch4-ii_2627.pdf

Canonical Form • Q = [u | w] • Four Matrices

STATEMENT

For the following equations find the eigenvalues and eigenvectors of the matrix on the right-hand side, and hence find the coordinate transformation that will put the equations into their standard simple (canonical) form. Show that this transformation has the desired effect.

- (i) $\dfrac{d\mathbf{x}}{dt} = \begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix} \mathbf{x}$;
- (ii) $\dfrac{d\mathbf{x}}{dt} = \begin{pmatrix} -2 & 3 \\ -6 & 4 \end{pmatrix} \mathbf{x}$;
- (iii) $\dfrac{d\mathbf{x}}{dt} = \begin{pmatrix} -11 & -2 \\ 13 & -9 \end{pmatrix} \mathbf{x}$;
- (iv) $\dfrac{d\mathbf{x}}{dt} = \begin{pmatrix} 7 & -5 \\ 10 & -3 \end{pmatrix} \mathbf{x}$.

PHASE 1: SETUP

The standard (canonical) form for a complex pair $\rho \pm i\omega$ is $\dot{\mathbf{z}} = J\mathbf{z}$ with $J = \begin{pmatrix} \rho & \omega \\ -\omega & \rho \end{pmatrix}$, reached by $\mathbf{z} = Q^{-1}\mathbf{x}$ with $Q = [\mathbf{u} \mid \mathbf{w}]$, $\mathbf{u} + i\mathbf{w}$ the eigenvector of $\rho + i\omega$ (Topic 4 Sec. 7.4). For every matrix $\Delta = (\operatorname{tr} A)^2 - 4 \det A < 0$ (checked in each item), so the eigenvalues are complex conjugate.

PHASE 2: FORMULATION

Steps for each matrix: characteristic equation, eigenvalue $\rho + i\omega$, eigenvector via the first row, $\mathbf{u}$, $\mathbf{w}$, relations (7.4), Q and Q^{-1}, and the explicit product $Q^{-1}AQ = J$.

METHOD OF THE CLASS NOTES [NOTES, P. 10]

The construction of the notes: complex diagonalisation $A = PDP^{-1}$ with $P = (\mathbf{v}_+ \mid \mathbf{v}_-)$, then $Q = (\mathbf{v}_1 \mid \mathbf{v}_2) = PM$ with $M = \begin{pmatrix} 1/2 & -i/2 \\ 1/2 & i/2 \end{pmatrix}$, so that $Q^{-1}AQ = M^{-1}P^{-1}APM = M^{-1}DM = \begin{pmatrix} \rho & \omega \\ -\omega & \rho \end{pmatrix}$. The explicit product $Q^{-1}AQ$ computed for each matrix below checks this canonical form.

PHASE 3: DERIVATION

(i) $A = \begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix}$

Characteristic equation $\det(A - \lambda I) = 0$ (Topic 4, Eq. 27.8):

$$\det(A - \lambda I) = \begin{vmatrix} 0 - \lambda & -1 \\ 1 & -1 - \lambda \end{vmatrix} = (0 - \lambda)(-1 - \lambda) - (-1)(1)$$

$$= \lambda^2 - (0 + (-1))\lambda + (0 \cdot (-1) - (-1) \cdot 1) = \lambda^2 - (-1)\lambda + (1) = \lambda^2 + \lambda + 1$$

$\operatorname{tr} A = -1$, $\det A = 1$, discriminant $\Delta = (\operatorname{tr} A)^2 - 4 \det A = 1 - 4 = -3$.

$\Delta < 0$: a complex-conjugate pair. $\lambda = \dfrac{-1 \pm \sqrt{-3}}{2} = \dfrac{-1 \pm \sqrt{3}i}{2}$, so

$$\lambda_{\pm} = -\frac{1}{2} \pm \frac{\sqrt{3}}{2}\, i \qquad (\rho = -\frac{1}{2},\ \omega = \frac{\sqrt{3}}{2} > 0)$$

Eigenvector of the eigenvalue with positive imaginary part

For $\lambda = \rho + i\omega = -\frac{1}{2} + \frac{\sqrt{3}i}{2}$ the first row of $(A - \lambda I)\mathbf{v} = \mathbf{0}$ reads $(\frac{1}{2} - \frac{\sqrt{3}i}{2})\, v_1 + (-1)\, v_2 = 0$ (the coefficient of v_2 is non-zero because $\Delta < 0$ forces $bc < 0$). Choose $v_1 = -1$; then $v_2 = -(\frac{1}{2} - \frac{\sqrt{3}i}{2}) = -\frac{1}{2} + \frac{\sqrt{3}i}{2}$:

$$\mathbf{v} = \begin{pmatrix} -1 \\ -\frac{1}{2} + \frac{\sqrt{3}i}{2} \end{pmatrix} = \underbrace{\begin{pmatrix} -1 \\ -\frac{1}{2} \end{pmatrix}}_{\mathbf{u}} + i \underbrace{\begin{pmatrix} 0 \\ \frac{\sqrt{3}}{2} \end{pmatrix}}_{\mathbf{w}}$$

Check with the second row: $1 \cdot (-1) + (-\frac{1}{2} - \frac{\sqrt{3}i}{2})(-\frac{1}{2} + \frac{\sqrt{3}i}{2}) = -1 + 1 = 0$ (using $i^2 = -1$).

How A acts on $\mathbf{u}$ and $\mathbf{w}$

Separating the real and imaginary parts of $A\mathbf{v} = \lambda\mathbf{v}$ gives (Topic 4, Eq. 7.4) $A\mathbf{u} = \rho\mathbf{u} - \omega\mathbf{w}$ and $A\mathbf{w} = \omega\mathbf{u} + \rho\mathbf{w}$. Verification with the numbers:

$$A\mathbf{u} = \begin{pmatrix} \frac{1}{2} \\ -\frac{1}{2} \end{pmatrix}, \qquad \rho\mathbf{u} - \omega\mathbf{w} = (-\frac{1}{2}) \begin{pmatrix} -1 \\ -\frac{1}{2} \end{pmatrix} - \frac{\sqrt{3}}{2} \begin{pmatrix} 0 \\ \frac{\sqrt{3}}{2} \end{pmatrix} = \begin{pmatrix} \frac{1}{2} \\ -\frac{1}{2} \end{pmatrix}$$

$$A\mathbf{w} = \begin{pmatrix} -\frac{\sqrt{3}}{2} \\ -\frac{\sqrt{3}}{2} \end{pmatrix}, \qquad \omega\mathbf{u} + \rho\mathbf{w} = \frac{\sqrt{3}}{2} \begin{pmatrix} -1 \\ -\frac{1}{2} \end{pmatrix} + (-\frac{1}{2}) \begin{pmatrix} 0 \\ \frac{\sqrt{3}}{2} \end{pmatrix} = \begin{pmatrix} -\frac{\sqrt{3}}{2} \\ -\frac{\sqrt{3}}{2} \end{pmatrix}$$

Change of coordinates

$\mathbf{u}$ and $\mathbf{w}$ are linearly independent (they are not parallel when $\omega \neq 0$; Topic 4, Lemma of Sec. 7.3). With the columns in the order $\mathbf{u}, \mathbf{w}$,

$$Q = [\mathbf{u} \mid \mathbf{w}] = \begin{pmatrix} -1 & 0 \\ -\frac{1}{2} & \frac{\sqrt{3}}{2} \end{pmatrix}$$

$\det Q = (-1)(\frac{\sqrt{3}}{2}) - (0)(-\frac{1}{2}) = -\frac{\sqrt{3}}{2}$, which is not zero, so Q is invertible. For a 2×2 matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad-bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$ (Robinson, Appendix B):

$$Q^{-1} = \frac{1}{-\frac{\sqrt{3}}{2}} \begin{pmatrix} \frac{\sqrt{3}}{2} & 0 \\ \frac{1}{2} & -1 \end{pmatrix} = \begin{pmatrix} -1 & 0 \\ -\frac{\sqrt{3}}{3} & \frac{2\sqrt{3}}{3} \end{pmatrix}$$

Compute $AQ = [A\mathbf{u} \mid A\mathbf{w}]$ and multiply by Q^{-1} on the left, entry by entry:

$$AQ = \begin{pmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} \\ -\frac{1}{2} & -\frac{\sqrt{3}}{2} \end{pmatrix}$$

$$\begin{aligned}
(Q^{-1}AQ)_{11} &= (-1)\cdot\frac{1}{2} + 0\cdot(-\frac{1}{2}) = -\frac{1}{2} \\
(Q^{-1}AQ)_{12} &= (-1)\cdot(-\frac{\sqrt{3}}{2}) + 0\cdot(-\frac{\sqrt{3}}{2}) = \frac{\sqrt{3}}{2} \\
(Q^{-1}AQ)_{21} &= (-\frac{\sqrt{3}}{3})\cdot\frac{1}{2} + \frac{2\sqrt{3}}{3}\cdot(-\frac{1}{2}) = -\frac{\sqrt{3}}{2} \\
(Q^{-1}AQ)_{22} &= (-\frac{\sqrt{3}}{3})\cdot(-\frac{\sqrt{3}}{2}) + \frac{2\sqrt{3}}{3}\cdot(-\frac{\sqrt{3}}{2}) = -\frac{1}{2}
\end{aligned}$$

$$Q^{-1}AQ = \begin{pmatrix} -\frac{1}{2} & \frac{\sqrt{3}}{2} \\ -\frac{\sqrt{3}}{2} & -\frac{1}{2} \end{pmatrix} = J = \begin{pmatrix} \rho & \omega \\ -\omega & \rho \end{pmatrix}$$

Solution in the canonical plane

$$\dot{z}_1 = -\frac{1}{2}z_1 + \frac{\sqrt{3}}{2}z_2, \qquad \dot{z}_2 = -\frac{\sqrt{3}}{2}z_1 - \frac{1}{2}z_2$$

The two equations are coupled, but they say "scale and rotate", which polar coordinates express directly: $z_1 = r\cos\theta$, $z_2 = r\sin\theta$ (Topic 4, Sec. 7.5).

The radius. From $r^2 = z_1^2 + z_2^2$ the chain rule gives $2r\dot{r} = 2z_1\dot{z}_1 + 2z_2\dot{z}_2$. Substituting $\dot{z}_1 = \rho z_1 + \omega z_2$ and $\dot{z}_2 = -\omega z_1 + \rho z_2$:

$$r\dot r = z_1(\rho z_1 + \omega z_2) + z_2(-\omega z_1 + \rho z_2) = \rho z_1^2 + \omega z_1 z_2 - \omega z_1 z_2 + \rho z_2^2 = \rho r^2$$

The terms in ω cancel. Dividing by $r \neq 0$ (the origin is the equilibrium): $\dot r = \rho r$.

The angle. Differentiate $\tan\theta = z_2/z_1$ with the chain rule on the left and the quotient rule on the right: $\sec^2\theta\,\dot\theta = (z_1\dot z_2 - z_2\dot z_1)/z_1^2$. Since $\sec^2\theta = r^2/z_1^2$, $\dot\theta = (z_1\dot z_2 - z_2\dot z_1)/r^2$, and

$$z_1\dot z_2 - z_2\dot z_1 = z_1(-\omega z_1 + \rho z_2) - z_2(\rho z_1 + \omega z_2) = -\omega z_1^2 + \rho z_1 z_2 - \rho z_1 z_2 - \omega z_2^2 = -\omega r^2 \implies \dot\theta = -\omega$$

Integration. $\frac{d}{dt}\left(e^{-\rho t}r\right) = e^{-\rho t}(\dot r - \rho r) = 0$, and Barrow's rule between 0 and t, $\left[e^{-\rho s}r(s)\right]_0^t = 0$, gives $r(t) = r_0 e^{\rho t}$. For the angle, $\theta(t) - \theta_0 = \int_0^t (-\omega)\,ds = -\omega t$. Back to Cartesian coordinates, with $c_1 = r_0\cos\theta_0 = z_1(0)$, $c_2 = r_0\sin\theta_0 = z_2(0)$ and the subtraction formulas $\cos(\theta_0 - \omega t) = \cos\theta_0\cos\omega t + \sin\theta_0\sin\omega t$, $\sin(\theta_0 - \omega t) = \sin\theta_0\cos\omega t - \cos\theta_0\sin\omega t$:

$$z_1(t) = e^{-\frac{1}{2}t}\left(c_1\cos\frac{\sqrt3}{2}t + c_2\sin\frac{\sqrt3}{2}t\right), \qquad z_2(t) = e^{-\frac{1}{2}t}\left(c_2\cos\frac{\sqrt3}{2}t - c_1\sin\frac{\sqrt3}{2}t\right)$$

Check by substitution (product rule and chain rule): $\dot z_1 = \rho z_1 + e^{\rho t}\omega\left(-c_1\sin\omega t + c_2\cos\omega t\right) = \rho z_1 + \omega z_2$ and $\dot z_2 = \rho z_2 + e^{\rho t}\omega\left(-c_2\sin\omega t - c_1\cos\omega t\right) = \rho z_2 - \omega z_1$, as required.

Back to the original coordinates
$\mathbf{x} = Q\mathbf{z} = z_1\mathbf{u} + z_2\mathbf{w}$, so

$$\mathbf{x}(t) = e^{-\frac{1}{2}t}\left[\left(c_1\cos\frac{\sqrt3}{2}t + c_2\sin\frac{\sqrt3}{2}t\right)\begin{pmatrix}-1\\-\frac{1}{2}\end{pmatrix} + \left(c_2\cos\frac{\sqrt3}{2}t - c_1\sin\frac{\sqrt3}{2}t\right)\begin{pmatrix}0\\\frac{\sqrt3}{2}\end{pmatrix}\right]$$

$$x(t) = -c_1 e^{-\frac{t}{2}}\cos\frac{\sqrt3 t}{2} - c_2 e^{-\frac{t}{2}}\sin\frac{\sqrt3 t}{2}, \qquad y(t) = -\frac{\sqrt3 c_1 e^{-\frac{t}{2}}\sin\frac{\sqrt3 t}{2}}{2} - \frac{c_1 e^{-\frac{t}{2}}\cos\frac{\sqrt3 t}{2}}{2} - \frac{c_2 e^{-\frac{t}{2}}\sin\frac{\sqrt3 t}{2}}{2} + \frac{\sqrt3 c_2 e^{-\frac{t}{2}}\cos\frac{\sqrt3 t}{2}}{2}$$

The canonical system is $\dot z_1 = -\frac{1}{2}z_1 + (\frac{\sqrt3}{2})z_2$, $\dot z_2 = -(\frac{\sqrt3}{2})z_1 + -\frac{1}{2}z_2$: a rotation clockwise with angular speed $\omega = \frac{\sqrt3}{2}$ combined with a radial factor $e^{-\frac{1}{2}t}$.

(ii) $A = \begin{pmatrix}-2 & 3\\ -6 & 4\end{pmatrix}$

Characteristic equation $\det(A - \lambda I) = 0$ (Topic 4, Eq. 27.8):

$$\det(A - \lambda I) = \begin{vmatrix}-2-\lambda & 3\\ -6 & 4-\lambda\end{vmatrix} = (-2-\lambda)(4-\lambda) - (3)(-6)$$

$$= \lambda^2 - (-2+4)\lambda + (-2\cdot 4 - 3\cdot(-6)) = \lambda^2 - (2)\lambda + (10) = \lambda^2 - 2\lambda + 10$$

$\operatorname{tr} A = 2$, $\det A = 10$, discriminant $\Delta = (\operatorname{tr} A)^2 - 4\det A = 4 - 40 = -36$.

$\Delta < 0$: a complex-conjugate pair. $\lambda = \dfrac{2 \pm \sqrt{-36}}{2} = \dfrac{2 \pm 6i}{2}$, so

$$\lambda_{\pm} = 1 \pm 3\,i \qquad (\rho = 1,\ \omega = 3 > 0)$$

Eigenvector of the eigenvalue with positive imaginary part

For $\lambda = \rho + i\omega = 1 + 3i$ the first row of $(A - \lambda I)\mathbf{v} = \mathbf{0}$ reads $(-3 - 3i)\,v_1 + 3\,v_2 = 0$ (the coefficient of v_2 is non-zero because $\Delta < 0$ forces $bc < 0$). Choose $v_1 = 3$; then $v_2 = -(-3 - 3i) = 3 + 3i$:

$$\mathbf{v} = \begin{pmatrix} 3 \\ 3 + 3i \end{pmatrix} = \underbrace{\begin{pmatrix} 3 \\ 3 \end{pmatrix}}_{\mathbf{u}} + i\,\underbrace{\begin{pmatrix} 0 \\ 3 \end{pmatrix}}_{\mathbf{w}}$$

Check with the second row: $-6 \cdot 3 + (3 - 3i)(3 + 3i) = -18 + 18 = 0$ (using $i^2 = -1$).

How A acts on $\mathbf{u}$ and $\mathbf{w}$

Separating the real and imaginary parts of $A\mathbf{v} = \lambda\mathbf{v}$ gives (Topic 4, Eq. 7.4) $A\mathbf{u} = \rho\mathbf{u} - \omega\mathbf{w}$ and $A\mathbf{w} = \omega\mathbf{u} + \rho\mathbf{w}$. Verification with the numbers:

$$A\mathbf{u} = \begin{pmatrix} 3 \\ -6 \end{pmatrix}, \qquad \rho\mathbf{u} - \omega\mathbf{w} = 1\begin{pmatrix} 3 \\ 3 \end{pmatrix} - 3\begin{pmatrix} 0 \\ 3 \end{pmatrix} = \begin{pmatrix} 3 \\ -6 \end{pmatrix}$$

$$A\mathbf{w} = \begin{pmatrix} 9 \\ 12 \end{pmatrix}, \qquad \omega\mathbf{u} + \rho\mathbf{w} = 3\begin{pmatrix} 3 \\ 3 \end{pmatrix} + 1\begin{pmatrix} 0 \\ 3 \end{pmatrix} = \begin{pmatrix} 9 \\ 12 \end{pmatrix}$$

Change of coordinates

$\mathbf{u}$ and $\mathbf{w}$ are linearly independent (they are not parallel when $\omega \neq 0$; Topic 4, Lemma of Sec. 7.3). With the columns in the order $\mathbf{u}$, $\mathbf{w}$,

$$Q = [\mathbf{u} \mid \mathbf{w}] = \begin{pmatrix} 3 & 0 \\ 3 & 3 \end{pmatrix}$$

$\det Q = (3)(3) - (0)(3) = 9$, which is not zero, so Q is invertible. For a 2×2 matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad-bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$ (Robinson, Appendix B):

$$Q^{-1} = \frac{1}{9}\begin{pmatrix} 3 & 0 \\ -3 & 3 \end{pmatrix} = \begin{pmatrix} \frac{1}{3} & 0 \\ -\frac{1}{3} & \frac{1}{3} \end{pmatrix}$$

Compute $AQ = [A\mathbf{u} \mid A\mathbf{w}]$ and multiply by Q^{-1} on the left, entry by entry:

$$AQ = \begin{pmatrix} 3 & 9 \\ -6 & 12 \end{pmatrix}$$

$$\begin{aligned}
(Q^{-1}AQ)_{11} &= \frac{1}{3}\cdot 3 + 0 \cdot (-6) = 1 \\
(Q^{-1}AQ)_{12} &= \frac{1}{3}\cdot 9 + 0 \cdot 12 = 3 \\
(Q^{-1}AQ)_{21} &= (-\frac{1}{3})\cdot 3 + \frac{1}{3}\cdot (-6) = -3 \\
(Q^{-1}AQ)_{22} &= (-\frac{1}{3})\cdot 9 + \frac{1}{3}\cdot 12 = 1
\end{aligned}$$

$$Q^{-1}AQ = \begin{pmatrix} 1 & 3 \\ -3 & 1 \end{pmatrix} = J = \begin{pmatrix} \rho & \omega \\ -\omega & \rho \end{pmatrix}$$

Solution in the canonical plane

$$\dot{z}_1 = z_1 + 3z_2, \qquad \dot{z}_2 = -3z_1 + z_2$$

The two equations are coupled, but they say "scale and rotate", which polar coordinates express directly: $z_1 = r\cos\theta$, $z_2 = r\sin\theta$ (Topic 4, Sec. 7.5).

The radius. From $r^2 = z_1^2 + z_2^2$ the chain rule gives $2r\dot{r} = 2z_1\dot{z}_1 + 2z_2\dot{z}_2$. Substituting $\dot{z}_1 = \rho z_1 + \omega z_2$ and $\dot{z}_2 = -\omega z_1 + \rho z_2$:

$$r\dot{r} = z_1(\rho z_1 + \omega z_2) + z_2(-\omega z_1 + \rho z_2) = \rho z_1^2 + \omega z_1 z_2 - \omega z_1 z_2 + \rho z_2^2 = \rho r^2$$

The terms in ω cancel. Dividing by $r \neq 0$ (the origin is the equilibrium): $\dot{r} = \rho r$.

The angle. Differentiate $\tan\theta = z_2/z_1$ with the chain rule on the left and the quotient rule on the right: $\sec^2\theta\,\dot{\theta} = (z_1\dot{z}_2 - z_2\dot{z}_1)/z_1^2$. Since $\sec^2\theta = r^2/z_1^2$, $\dot{\theta} = (z_1\dot{z}_2 - z_2\dot{z}_1)/r^2$, and

$$z_1\dot{z}_2 - z_2\dot{z}_1 = z_1(-\omega z_1 + \rho z_2) - z_2(\rho z_1 + \omega z_2) = -\omega z_1^2 + \rho z_1 z_2 - \rho z_1 z_2 - \omega z_2^2 = -\omega r^2 \implies \dot{\theta} = -\omega$$

Integration. $\frac{d}{dt}\left(e^{-\rho t}r\right) = e^{-\rho t}(\dot r - \rho r) = 0$, and Barrow's rule between 0 and t, $\left[e^{-\rho s}r(s)\right]_0^t = 0$, gives $r(t) = r_0 e^{\rho t}$. For the angle, $\theta(t) - \theta_0 = \int_0^t (-\omega)\, ds = -\omega t$. Back to Cartesian coordinates, with $c_1 = r_0\cos\theta_0 = z_1(0)$, $c_2 = r_0\sin\theta_0 = z_2(0)$ and the subtraction formulas $\cos(\theta_0 - \omega t) = \cos\theta_0\cos\omega t + \sin\theta_0\sin\omega t$, $\sin(\theta_0 - \omega t) = \sin\theta_0\cos\omega t - \cos\theta_0\sin\omega t$:

$$z_1(t) = e^{1t}\left(c_1\cos 3t + c_2\sin 3t\right), \qquad z_2(t) = e^{1t}\left(c_2\cos 3t - c_1\sin 3t\right)$$

Check by substitution (product rule and chain rule): $\dot z_1 = \rho z_1 + e^{\rho t}\omega\left(-c_1\sin\omega t + c_2\cos\omega t\right) = \rho z_1 + \omega z_2$ and $\dot z_2 = \rho z_2 + e^{\rho t}\omega\left(-c_2\sin\omega t - c_1\cos\omega t\right) = \rho z_2 - \omega z_1$, as required.

Back to the original coordinates
$\mathbf{x} = Q\mathbf{z} = z_1\mathbf{u} + z_2\mathbf{w}$, so

$$\mathbf{x}(t) = e^{1t}\left[\left(c_1\cos 3t + c_2\sin 3t\right)\begin{pmatrix}3\\3\end{pmatrix} + \left(c_2\cos 3t - c_1\sin 3t\right)\begin{pmatrix}0\\3\end{pmatrix}\right]$$

$$x(t) = 3c_1e^t\cos 3t + 3c_2e^t\sin 3t, \qquad y(t) = -3c_1e^t\sin 3t + 3c_1e^t\cos 3t + 3c_2e^t\sin 3t + 3c_2e^t\cos 3t$$

The canonical system is $\dot z_1 = 1z_1 + (3)z_2$, $\dot z_2 = -(3)z_1 + 1z_2$: a rotation clockwise with angular speed $\omega = 3$ combined with a radial factor e^{1t}.

(iii) $A = \begin{pmatrix}-11 & -2\\ 13 & -9\end{pmatrix}$

Characteristic equation $\det(A - \lambda I) = 0$ (Topic 4, Eq. 27.8):

$$\det(A - \lambda I) = \begin{vmatrix}-11-\lambda & -2\\ 13 & -9-\lambda\end{vmatrix} = (-11-\lambda)(-9-\lambda) - (-2)(13)$$

$$= \lambda^2 - (-11 + (-9))\lambda + (-11\cdot(-9) - (-2)\cdot 13) = \lambda^2 - (-20)\lambda + (125) = \lambda^2 + 20\lambda + 125$$

$\operatorname{tr} A = -20$, $\det A = 125$, discriminant $\Delta = (\operatorname{tr} A)^2 - 4\det A = 400 - 500 = -100$.

$\Delta < 0$: a complex-conjugate pair. $\lambda = \dfrac{-20 \pm \sqrt{-100}}{2} = \dfrac{-20 \pm 10i}{2}$, so

$$\lambda_\pm = -10 \pm 5\,i \qquad (\rho = -10,\ \omega = 5 > 0)$$

Eigenvector of the eigenvalue with positive imaginary part

For $\lambda = \rho + i\omega = -10 + 5i$ the first row of $(A - \lambda I)\mathbf{v} = \mathbf{0}$ reads $(-1 - 5i)\,v_1 + (-2)\,v_2 = 0$ (the coefficient of v_2 is non-zero because $\Delta < 0$ forces $bc < 0$). Choose $v_1 = -2$; then $v_2 = -(-1 - 5i) = 1 + 5i$:

$$\mathbf{v} = \begin{pmatrix} -2 \\ 1 + 5i \end{pmatrix} = \underbrace{\begin{pmatrix} -2 \\ 1 \end{pmatrix}}_{\mathbf{u}} + i\,\underbrace{\begin{pmatrix} 0 \\ 5 \end{pmatrix}}_{\mathbf{w}}$$

Check with the second row: $13 \cdot (-2) + (1 - 5i)(1 + 5i) = -26 + 26 = 0$ (using $i^2 = -1$).

How A acts on $\mathbf{u}$ and $\mathbf{w}$

Separating the real and imaginary parts of $A\mathbf{v} = \lambda\mathbf{v}$ gives (Topic 4, Eq. 7.4) $A\mathbf{u} = \rho\mathbf{u} - \omega\mathbf{w}$ and $A\mathbf{w} = \omega\mathbf{u} + \rho\mathbf{w}$. Verification with the numbers:

$$A\mathbf{u} = \begin{pmatrix} 20 \\ -35 \end{pmatrix}, \qquad \rho\mathbf{u} - \omega\mathbf{w} = (-10)\begin{pmatrix} -2 \\ 1 \end{pmatrix} - 5\begin{pmatrix} 0 \\ 5 \end{pmatrix} = \begin{pmatrix} 20 \\ -35 \end{pmatrix}$$

$$A\mathbf{w} = \begin{pmatrix} -10 \\ -45 \end{pmatrix}, \qquad \omega\mathbf{u} + \rho\mathbf{w} = 5\begin{pmatrix} -2 \\ 1 \end{pmatrix} + (-10)\begin{pmatrix} 0 \\ 5 \end{pmatrix} = \begin{pmatrix} -10 \\ -45 \end{pmatrix}$$

Change of coordinates

$\mathbf{u}$ and $\mathbf{w}$ are linearly independent (they are not parallel when $\omega \neq 0$; Topic 4, Lemma of Sec. 7.3). With the columns in the order $\mathbf{u}$, $\mathbf{w}$,

$$Q = [\mathbf{u} \mid \mathbf{w}] = \begin{pmatrix} -2 & 0 \\ 1 & 5 \end{pmatrix}$$

$\det Q = (-2)(5) - (0)(1) = -10$, which is not zero, so Q is invertible. For a 2×2 matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad - bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$ (Robinson, Appendix B):

$$Q^{-1} = \frac{1}{-10}\begin{pmatrix} 5 & 0 \\ -1 & -2 \end{pmatrix} = \begin{pmatrix} -\frac{1}{2} & 0 \\ \frac{1}{10} & \frac{1}{5} \end{pmatrix}$$

Compute $AQ = [A\mathbf{u} \mid A\mathbf{w}]$ and multiply by Q^{-1} on the left, entry by entry:

$$AQ = \begin{pmatrix} 20 & -10 \\ -35 & -45 \end{pmatrix}$$

$$
\begin{aligned}
(Q^{-1}AQ)_{11} &= (-\frac{1}{2})\cdot 20 + 0\cdot(-35) = -10\\
(Q^{-1}AQ)_{12} &= (-\frac{1}{2})\cdot(-10) + 0\cdot(-45) = 5\\
(Q^{-1}AQ)_{21} &= \frac{1}{10}\cdot 20 + \frac{1}{5}\cdot(-35) = -5\\
(Q^{-1}AQ)_{22} &= \frac{1}{10}\cdot(-10) + \frac{1}{5}\cdot(-45) = -10
\end{aligned}
$$

$$
Q^{-1}AQ = \begin{pmatrix} -10 & 5 \\ -5 & -10 \end{pmatrix} = J = \begin{pmatrix} \rho & \omega \\ -\omega & \rho \end{pmatrix}
$$

Solution in the canonical plane

$$
\dot{z}_1 = -10z_1 + 5z_2, \qquad \dot{z}_2 = -5z_1 - 10z_2
$$

The two equations are coupled, but they say "scale and rotate", which polar coordinates express directly: $z_1 = r\cos\theta$, $z_2 = r\sin\theta$ (Topic 4, Sec. 7.5).

The radius. From $r^2 = z_1^2 + z_2^2$ the chain rule gives $2r\dot{r} = 2z_1\dot{z}_1 + 2z_2\dot{z}_2$. Substituting $\dot{z}_1 = \rho z_1 + \omega z_2$ and $\dot{z}_2 = -\omega z_1 + \rho z_2$:

$$
r\dot{r} = z_1(\rho z_1 + \omega z_2) + z_2(-\omega z_1 + \rho z_2) = \rho z_1^2 + \omega z_1 z_2 - \omega z_1 z_2 + \rho z_2^2 = \rho r^2
$$

The terms in ω cancel. Dividing by $r \neq 0$ (the origin is the equilibrium): $\dot{r} = \rho r$.

The angle. Differentiate $\tan\theta = z_2/z_1$ with the chain rule on the left and the quotient rule on the right: $\sec^2\theta\,\dot{\theta} = (z_1\dot{z}_2 - z_2\dot{z}_1)/z_1^2$. Since $\sec^2\theta = r^2/z_1^2$, $\dot{\theta} = (z_1\dot{z}_2 - z_2\dot{z}_1)/r^2$, and

$$
z_1\dot{z}_2 - z_2\dot{z}_1 = z_1(-\omega z_1 + \rho z_2) - z_2(\rho z_1 + \omega z_2) = -\omega z_1^2 + \rho z_1 z_2 - \rho z_1 z_2 - \omega z_2^2 = -\omega r^2 \implies \dot{\theta} = -\omega
$$

Integration. $\frac{d}{dt}\left(e^{-\rho t}r\right) = e^{-\rho t}(\dot{r} - \rho r) = 0$, and Barrow's rule between 0 and t, $\left[e^{-\rho s}r(s)\right]_0^t = 0$, gives $r(t) = r_0 e^{\rho t}$. For the angle, $\theta(t) - \theta_0 = \int_0^t (-\omega)\,ds = -\omega t$. Back to Cartesian coordinates, with $c_1 = r_0\cos\theta_0 = z_1(0)$, $c_2 = r_0\sin\theta_0 = z_2(0)$ and the subtraction formulas $\cos(\theta_0 - \omega t) = \cos\theta_0\cos\omega t + \sin\theta_0\sin\omega t$, $\sin(\theta_0 - \omega t) = \sin\theta_0\cos\omega t - \cos\theta_0\sin\omega t$:

$$
z_1(t) = e^{-10t}\left(c_1\cos 5t + c_2\sin 5t\right), \qquad z_2(t) = e^{-10t}\left(c_2\cos 5t - c_1\sin 5t\right)
$$

Check by substitution (product rule and chain rule): $\dot{z}_1 = \rho z_1 + e^{\rho t}\omega\left(-c_1\sin\omega t + c_2\cos\omega t\right) = \rho z_1 + \omega z_2$ and $\dot{z}_2 = \rho z_2 + e^{\rho t}\omega\left(-c_2\sin\omega t - c_1\cos\omega t\right) = \rho z_2 - \omega z_1$, as required.

Back to the original coordinates

$\mathbf{x} = Q\mathbf{z} = z_1\mathbf{u} + z_2\mathbf{w}$, so

$$\mathbf{x}(t) = e^{-10t}\left[(c_1\cos 5t + c_2\sin 5t)\begin{pmatrix}-2\\1\end{pmatrix} + (c_2\cos 5t - c_1\sin 5t)\begin{pmatrix}0\\5\end{pmatrix}\right]$$

$$x(t) = -2c_1e^{-10t}\cos 5t - 2c_2e^{-10t}\sin 5t, \qquad y(t) = -5c_1e^{-10t}\sin 5t + c_1e^{-10t}\cos 5t + c_2e^{-10t}\sin 5t + 5c_2e^{-10t}\cos 5t$$

The canonical system is $\dot{z}_1 = -10z_1 + (5)z_2$, $\dot{z}_2 = -(5)z_1 + -10z_2$: a rotation clockwise with angular speed $\omega = 5$ combined with a radial factor e^{-10t}.

(iv) $A = \begin{pmatrix}7 & -5\\ 10 & -3\end{pmatrix}$

Characteristic equation $\det(A - \lambda I) = 0$ (Topic 4, Eq. 27.8):

$$\det(A - \lambda I) = \begin{vmatrix}7-\lambda & -5\\ 10 & -3-\lambda\end{vmatrix} = (7-\lambda)(-3-\lambda) - (-5)(10)$$

$$= \lambda^2 - (7 + (-3))\lambda + (7\cdot(-3) - (-5)\cdot 10) = \lambda^2 - (4)\lambda + (29) = \lambda^2 - 4\lambda + 29$$

$\operatorname{tr} A = 4$, $\det A = 29$, discriminant $\Delta = (\operatorname{tr} A)^2 - 4\det A = 16 - 116 = -100$.

$\Delta < 0$: a complex-conjugate pair. $\lambda = \dfrac{4 \pm \sqrt{-100}}{2} = \dfrac{4 \pm 10i}{2}$, so

$$\lambda_{\pm} = 2 \pm 5\,i \qquad (\rho = 2,\ \omega = 5 > 0)$$

Eigenvector of the eigenvalue with positive imaginary part

For $\lambda = \rho + i\omega = 2 + 5i$ the first row of $(A - \lambda I)\mathbf{v} = \mathbf{0}$ reads $(5 - 5i)\,v_1 + (-5)\,v_2 = 0$ (the coefficient of v_2 is non-zero because $\Delta < 0$ forces $bc < 0$). Choose $v_1 = -5$; then $v_2 = -(5 - 5i) = -5 + 5i$:

$$\mathbf{v} = \begin{pmatrix}-5\\-5+5i\end{pmatrix} = \underbrace{\begin{pmatrix}-5\\-5\end{pmatrix}}_{\mathbf{u}} + i\,\underbrace{\begin{pmatrix}0\\5\end{pmatrix}}_{\mathbf{w}}$$

Check with the second row: $10\cdot(-5) + (-5-5i)(-5+5i) = -50 + 50 = 0$ (using $i^2 = -1$).

How A acts on $\mathbf{u}$ and $\mathbf{w}$

Separating the real and imaginary parts of $A\mathbf{v} = \lambda\mathbf{v}$ gives (Topic 4, Eq. 7.4) $A\mathbf{u} = \rho\mathbf{u} - \omega\mathbf{w}$ and $A\mathbf{w} = \omega\mathbf{u} + \rho\mathbf{w}$. Verification with the numbers:

$$A\mathbf{u} = \begin{pmatrix} -10 \\ -35 \end{pmatrix}, \qquad \rho\mathbf{u} - \omega\mathbf{w} = 2\begin{pmatrix} -5 \\ -5 \end{pmatrix} - 5\begin{pmatrix} 0 \\ 5 \end{pmatrix} = \begin{pmatrix} -10 \\ -35 \end{pmatrix}$$

$$A\mathbf{w} = \begin{pmatrix} -25 \\ -15 \end{pmatrix}, \qquad \omega\mathbf{u} + \rho\mathbf{w} = 5\begin{pmatrix} -5 \\ -5 \end{pmatrix} + 2\begin{pmatrix} 0 \\ 5 \end{pmatrix} = \begin{pmatrix} -25 \\ -15 \end{pmatrix}$$

Change of coordinates

$\mathbf{u}$ and $\mathbf{w}$ are linearly independent (they are not parallel when $\omega \neq 0$; Topic 4, Lemma of Sec. 7.3). With the columns in the order $\mathbf{u}, \mathbf{w}$,

$$Q = [\mathbf{u} \mid \mathbf{w}] = \begin{pmatrix} -5 & 0 \\ -5 & 5 \end{pmatrix}$$

$\det Q = (-5)(5) - (0)(-5) = -25$, which is not zero, so Q is invertible. For a 2×2 matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad-bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$ (Robinson, Appendix B):

$$Q^{-1} = \frac{1}{-25}\begin{pmatrix} 5 & 0 \\ 5 & -5 \end{pmatrix} = \begin{pmatrix} -\frac{1}{5} & 0 \\ -\frac{1}{5} & \frac{1}{5} \end{pmatrix}$$

Compute $AQ = [A\mathbf{u} \mid A\mathbf{w}]$ and multiply by Q^{-1} on the left, entry by entry:

$$AQ = \begin{pmatrix} -10 & -25 \\ -35 & -15 \end{pmatrix}$$

$$\begin{aligned} (Q^{-1}AQ)_{11} &= (-\frac{1}{5}) \cdot (-10) + 0 \cdot (-35) = 2 \\ (Q^{-1}AQ)_{12} &= (-\frac{1}{5}) \cdot (-25) + 0 \cdot (-15) = 5 \\ (Q^{-1}AQ)_{21} &= (-\frac{1}{5}) \cdot (-10) + \frac{1}{5} \cdot (-35) = -5 \\ (Q^{-1}AQ)_{22} &= (-\frac{1}{5}) \cdot (-25) + \frac{1}{5} \cdot (-15) = 2 \end{aligned}$$

$$Q^{-1}AQ = \begin{pmatrix} 2 & 5 \\ -5 & 2 \end{pmatrix} = J = \begin{pmatrix} \rho & \omega \\ -\omega & \rho \end{pmatrix}$$

Solution in the canonical plane

$$\dot{z}_1 = 2z_1 + 5z_2, \qquad \dot{z}_2 = -5z_1 + 2z_2$$

The two equations are coupled, but they say "scale and rotate", which polar coordinates express directly: $z_1 = r\cos\theta$, $z_2 = r\sin\theta$ (Topic 4, Sec. 7.5).

The radius. From $r^2 = z_1^2 + z_2^2$ the chain rule gives $2r\dot{r} = 2z_1\dot{z}_1 + 2z_2\dot{z}_2$. Substituting $\dot{z}_1 = \rho z_1 + \omega z_2$ and $\dot{z}_2 = -\omega z_1 + \rho z_2$:

$$r\dot{r} = z_1(\rho z_1 + \omega z_2) + z_2(-\omega z_1 + \rho z_2) = \rho z_1^2 + \omega z_1 z_2 - \omega z_1 z_2 + \rho z_2^2 = \rho r^2$$

The terms in ω cancel. Dividing by $r \neq 0$ (the origin is the equilibrium): $\dot{r} = \rho r$.

The angle. Differentiate $\tan\theta = z_2/z_1$ with the chain rule on the left and the quotient rule on the right: $\sec^2\theta\,\dot{\theta} = (z_1\dot{z}_2 - z_2\dot{z}_1)/z_1^2$. Since $\sec^2\theta = r^2/z_1^2$, $\dot{\theta} = (z_1\dot{z}_2 - z_2\dot{z}_1)/r^2$, and

$$z_1\dot{z}_2 - z_2\dot{z}_1 = z_1(-\omega z_1 + \rho z_2) - z_2(\rho z_1 + \omega z_2) = -\omega z_1^2 + \rho z_1 z_2 - \rho z_1 z_2 - \omega z_2^2 = -\omega r^2 \implies \dot{\theta} = -\omega$$

Integration. $\frac{d}{dt}\left(e^{-\rho t}r\right) = e^{-\rho t}(\dot{r} - \rho r) = 0$, and Barrow's rule between 0 and t, $\left[e^{-\rho s}r(s)\right]_0^t = 0$, gives $r(t) = r_0 e^{\rho t}$. For the angle, $\theta(t) - \theta_0 = \int_0^t (-\omega)\,ds = -\omega t$. Back to Cartesian coordinates, with $c_1 = r_0\cos\theta_0 = z_1(0)$, $c_2 = r_0\sin\theta_0 = z_2(0)$ and the subtraction formulas $\cos(\theta_0 - \omega t) = \cos\theta_0\cos\omega t + \sin\theta_0\sin\omega t$, $\sin(\theta_0 - \omega t) = \sin\theta_0\cos\omega t - \cos\theta_0\sin\omega t$:

$$z_1(t) = e^{2t}\left(c_1\cos 5t + c_2\sin 5t\right), \qquad z_2(t) = e^{2t}\left(c_2\cos 5t - c_1\sin 5t\right)$$

Check by substitution (product rule and chain rule): $\dot{z}_1 = \rho z_1 + e^{\rho t}\omega\left(-c_1\sin\omega t + c_2\cos\omega t\right) = \rho z_1 + \omega z_2$ and $\dot{z}_2 = \rho z_2 + e^{\rho t}\omega\left(-c_2\sin\omega t - c_1\cos\omega t\right) = \rho z_2 - \omega z_1$, as required.

Back to the original coordinates

$\mathbf{x} = Q\mathbf{z} = z_1\mathbf{u} + z_2\mathbf{w}$, so

$$\mathbf{x}(t) = e^{2t}\left[\left(c_1\cos 5t + c_2\sin 5t\right)\begin{pmatrix} -5 \\ -5 \end{pmatrix} + \left(c_2\cos 5t - c_1\sin 5t\right)\begin{pmatrix} 0 \\ 5 \end{pmatrix}\right]$$

$$x(t) = -5c_1e^{2t}\cos 5t - 5c_2e^{2t}\sin 5t, \qquad y(t) = -5c_1e^{2t}\sin 5t - 5c_1e^{2t}\cos 5t - 5c_2e^{2t}\sin 5t + 5c_2e^{2t}\cos 5t$$

The canonical system is $\dot{z}_1 = 2z_1 + (5)z_2$, $\dot{z}_2 = -(5)z_1 + 2z_2$: a rotation clockwise with angular speed $\omega = 5$ combined with a radial factor e^{2t}.

PHASE 4: INTERPRETATION AND CHECKS

- In each item the explicit product $Q^{-1}AQ$ reproduces J exactly (entry-by-entry arithmetic above).
- (i) has an irrational $\omega = \sqrt{3}/2$ (eigenvalues $-\frac{1}{2} \pm \frac{\sqrt{3}}{2}i$, with the surds carried exactly): stable spiral. (ii) $1 \pm 3i$, unstable spiral; (iii) $-10 \pm 5i$, stable spiral (fast decay, ratio $|\rho|/\omega = 2$); (iv) $2 \pm 5i$, unstable spiral.
- The orientation of Q (det Q sign) decides whether the clockwise rotation of the $\mathbf{z}$-plane stays clockwise in the $\mathbf{x}$-plane; it is $b\omega$ here (positive iff $b > 0$).
- The eigenvalues of each matrix satisfy $\lambda_+ + \lambda_- = \operatorname{tr} A$ and $\lambda_+\lambda_- = \det A$ (checked in the text).

EXERCISE 4

Real and Imaginary Parts of a Complex Eigenvector Are Independent

Original statement · probls_ch4-ii_2627.pdf

Linear Independence • Complex Eigenvectors

STATEMENT

> In the previous chapter we used the result that eigenvectors which correspond to distinct eigenvalues are linearly independent. Use this result to show that the real and imaginary parts of complex eigenvectors are linearly independent.

PHASE 1: SETUP AND HYPOTHESES

Let A be a real 2×2 matrix with a non-real eigenvalue $\lambda = \rho + i\omega$ ($\omega \neq 0$) and eigenvector $\mathbf{v} = \mathbf{u} + i\mathbf{w} \in \mathbb{C}^2$, $\mathbf{u}, \mathbf{w} \in \mathbb{R}^2$. To show: $\mathbf{u}$ and $\mathbf{w}$ are linearly independent over $\mathbb{R}$, i.e. $\alpha\mathbf{u} + \beta\mathbf{w} = \mathbf{0}$ with $\alpha, \beta \in \mathbb{R}$ implies $\alpha = \beta = 0$.

PHASE 2: FORMULATION

Ingredients: (a) $\bar{\mathbf{v}} = \mathbf{u} - i\mathbf{w}$ is an eigenvector for $\bar{\lambda}$ (conjugate $A\mathbf{v} = \lambda\mathbf{v}$ and use $\bar{A} = A$ since A is real); (b) $\lambda \neq \bar{\lambda}$ because $\omega \neq 0$; (c) eigenvectors of distinct eigenvalues are linearly independent (over $\mathbb{C}$) (Topic 4, Appendix B argument).

> METHOD OF THE CLASS NOTES [NOTES, P. 10]
> Route of the notes: $\mathbf{v}_1 = \frac{1}{2}(\mathbf{v}_+ + \mathbf{v}_-)$ and $\mathbf{v}_2 = -\frac{i}{2}(\mathbf{v}_+ - \mathbf{v}_-)$, that is $Q = (\mathbf{v}_1 \mid \mathbf{v}_2) = PM$ with $P = (\mathbf{v}_+ \mid \mathbf{v}_-)$ and $M = \begin{pmatrix} 1/2 & -i/2 \\ 1/2 & i/2 \end{pmatrix}$. P is invertible because $\mathbf{v}_\pm$ belong to the distinct eigenvalues $\rho \pm i\omega$ ($\omega \neq 0$), and $\det M = \frac{i}{4} + \frac{i}{4} = \frac{i}{2} \neq 0$. So $\det Q = \det P \, \det M \neq 0$ and the real vectors $\mathbf{v}_1, \mathbf{v}_2$ are independent: the argument below, written with matrices.

PHASE 3: DERIVATION

Step 1: $\mathbf{v}$ and $\bar{\mathbf{v}}$ are independent

By (a), (b) and the quoted result, $\mathbf{v}$ and $\bar{\mathbf{v}}$ are linearly independent in $\mathbb{C}^2$: $\gamma\mathbf{v} + \delta\bar{\mathbf{v}} = \mathbf{0}$ with $\gamma, \delta \in \mathbb{C}$ forces $\gamma = \delta = 0$.

Step 2: express $\mathbf{u}, \mathbf{w}$ through $\mathbf{v}, \bar{\mathbf{v}}$

$$\mathbf{u} = \frac{\mathbf{v} + \bar{\mathbf{v}}}{2}, \qquad \mathbf{w} = \frac{\mathbf{v} - \bar{\mathbf{v}}}{2i}$$

Step 3: a real relation must be trivial

Suppose $\alpha\mathbf{u} + \beta\mathbf{w} = \mathbf{0}$ with $\alpha, \beta \in \mathbb{R}$. Substitute Step 2 and collect the coefficients of $\mathbf{v}$ and $\bar{\mathbf{v}}$:

$$\alpha\frac{\mathbf{v} + \bar{\mathbf{v}}}{2} + \beta\frac{\mathbf{v} - \bar{\mathbf{v}}}{2i} = \underbrace{\left(\frac{\alpha}{2} + \frac{\beta}{2i}\right)}_{\gamma}\mathbf{v} + \underbrace{\left(\frac{\alpha}{2} - \frac{\beta}{2i}\right)}_{\delta}\bar{\mathbf{v}} = \mathbf{0}$$

By Step 1, $\gamma = \delta = 0$. Using $\frac{1}{i} = -i$: $\gamma = \frac{\alpha - i\beta}{2}$ and $\delta = \frac{\alpha + i\beta}{2}$. Then $\gamma = 0$ means $\alpha - i\beta = 0$; with α, β real, real and imaginary parts give $\alpha = 0$ and $\beta = 0$. ■

Check against the direct argument

Alternative (Topic 4 Sec. 7.3): if $\mathbf{w} = k\mathbf{u}$ (or $\mathbf{u} = \mathbf{0}$), $\mathbf{v}$ would be a complex multiple of one real vector $\mathbf{r}$, forcing $A\mathbf{r} = \lambda\mathbf{r}$ with $A\mathbf{r}$ real, so λ would be real: contradiction. The two proofs agree; the one above only uses the independence of eigenvectors of distinct eigenvalues, as the exercise requests.

PHASE 4: INTERPRETATION AND CHECKS

- Consequence: $Q = [\mathbf{u} \mid \mathbf{w}]$ is invertible, which is what makes the change of coordinates $\mathbf{z} = Q^{-1}\mathbf{x}$ possible for complex eigenvalues (Exercises 3 and 1).
- Numerical check with the eigenvector of Topic 4 Example 7.1, $\mathbf{v} = (5, -1 + 3i)^T$: $\mathbf{u} = (5, -1)^T$, $\mathbf{w} = (0, 3)^T$, $\det[\mathbf{u}\ \mathbf{w}] = 15 \neq 0$.
- The argument fails for a real eigenvalue ($\lambda = \bar{\lambda}$): then $\mathbf{w} = \mathbf{0}$ can occur, consistent with Step 1 breaking down.
- In dimension n the same idea shows that the real and imaginary parts of eigenvectors of different conjugate pairs together with real eigenvectors remain independent.

EXERCISE 5

Building a Matrix from a Complex Pair of Eigenvalues and Eigenvectors

Original statement · probls_ch4-ii_2627.pdf, p. 2

Inverse Problem • A = Q J Q^{-1} • Three Matrices

STATEMENT

Show how to construct a matrix with a complex conjugate pair of eigenvalues $\lambda_\pm = \rho \pm i\omega$ and corresponding eigenvectors $\boldsymbol{\eta}_\pm = \mathbf{v}_1 \pm i\mathbf{v}_2$. Hence find the matrices with the following eigenvalues and eigenvectors:

- (i) $\lambda_\pm = 3 \pm 3i,\ \boldsymbol{\eta}_\pm = (2 \pm i,\ 1 \mp i)$;
- (ii) $\lambda_\pm = \pm 3i,\ \boldsymbol{\eta}_\pm = (\pm i,\ 3 \pm 2i)$;
- (iii) $\lambda_\pm = -2 \pm i,\ \boldsymbol{\eta}_\pm = (1 \pm i,\ 1 \mp i)$.

PHASE 1: SETUP AND HYPOTHESES

We look for a real matrix A (2×2) with $A\boldsymbol{\eta}_+ = \lambda_+ \boldsymbol{\eta}_+$ where $\boldsymbol{\eta}_+ = \mathbf{v}_1 + i\mathbf{v}_2$ and $\lambda_+ = \rho + i\omega$, with $\mathbf{v}_1, \mathbf{v}_2$ real and independent and $\omega \neq 0$. Because A is real, $A\bar{\boldsymbol{\eta}}_+ = \bar{\lambda}_+ \bar{\boldsymbol{\eta}}_+$ follows automatically. (Note on the statement: the sheet writes $\boldsymbol{\eta}_\pm = \mathbf{v}_1 \pm i\mathbf{v}_2$ for the eigenvectors of $\lambda_\pm = \rho \pm i\omega$, i.e. $\mathbf{v}_1 = \mathbf{u}$, $\mathbf{v}_2 = \mathbf{w}$ of Topic 4.)

PHASE 2: FORMULATION: THE CONSTRUCTION

Expand $A(\mathbf{v}_1 + i\mathbf{v}_2) = (\rho + i\omega)(\mathbf{v}_1 + i\mathbf{v}_2)$ and separate real and imaginary parts (both $A\mathbf{v}_j$ are real vectors, $i^2 = -1$):

$$A\mathbf{v}_1 + iA\mathbf{v}_2 = (\rho\mathbf{v}_1 - \omega\mathbf{v}_2) + i(\omega\mathbf{v}_1 + \rho\mathbf{v}_2) \implies A\mathbf{v}_1 = \rho\mathbf{v}_1 - \omega\mathbf{v}_2, \quad A\mathbf{v}_2 = \omega\mathbf{v}_1 + \rho\mathbf{v}_2$$

Collect the two vector equations in one matrix equation, with $Q = [\mathbf{v}_1 \mid \mathbf{v}_2]$ (columns): the first column of QJ is $\rho\mathbf{v}_1 - \omega\mathbf{v}_2$ for $J = \begin{pmatrix} \rho & \omega \\ -\omega & \rho \end{pmatrix}$ and the second is $\omega\mathbf{v}_1 + \rho\mathbf{v}_2$. Hence $AQ = QJ$, and since $\mathbf{v}_1, \mathbf{v}_2$ are independent (Exercise 4) Q is invertible:

$$\boxed{A = QJQ^{-1}, \qquad Q = [\mathbf{v}_1 \mid \mathbf{v}_2], \qquad J = \begin{pmatrix} \rho & \omega \\ -\omega & \rho \end{pmatrix}}$$

This A is real (a product of real matrices), has the eigenvalues of J, $\rho \pm i\omega$ (similar matrices have the same characteristic polynomial), and $A\boldsymbol{\eta}_+ = \lambda_+ \boldsymbol{\eta}_+$ by construction.

METHOD OF THE CLASS NOTES [NOTES, P. 10]

The identities of the notes read backwards: $A = PDP^{-1}$ and $Q = PM$ give $A = QM^{-1}DMQ^{-1} = QJQ^{-1}$ with $J = M^{-1}DM = \begin{pmatrix} \rho & \omega \\ -\omega & \rho \end{pmatrix}$. It is the matrix of the construction above, which only needs the real and imaginary parts of $A\boldsymbol{\eta}_+ = \lambda_+ \boldsymbol{\eta}_+$.

PHASE 3: DERIVATION OF THE THREE MATRICES

(i) $\lambda_\pm = 3 \pm 3i,\ \boldsymbol{\eta}_\pm = (2 \pm i,\ 1 \mp i)$

$\boldsymbol{\eta}_+ = \begin{pmatrix} 2+i \\ 1-i \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \end{pmatrix} + i \begin{pmatrix} 1 \\ -1 \end{pmatrix}$, so $\mathbf{v}_1 = \begin{pmatrix} 2 \\ 1 \end{pmatrix}, \mathbf{v}_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}, \rho = 3, \omega = 3.$

$$Q = \begin{pmatrix} 2 & 1 \\ 1 & -1 \end{pmatrix}, \quad J = \begin{pmatrix} 3 & 3 \\ -3 & 3 \end{pmatrix}, \quad \det Q = -3, \quad Q^{-1} = \begin{pmatrix} \frac{1}{3} & \frac{1}{3} \\ \frac{1}{3} & -\frac{2}{3} \end{pmatrix}$$

$$\begin{aligned} (QJ)_{11} &= 2 \cdot 3 + 1 \cdot (-3) = 3 \\ (QJ)_{12} &= 2 \cdot 3 + 1 \cdot 3 = 9 \\ (QJ)_{21} &= 1 \cdot 3 + (-1) \cdot (-3) = 6 \\ (QJ)_{22} &= 1 \cdot 3 + (-1) \cdot 3 = 0 \end{aligned}$$

$$\begin{aligned} (A)_{11} &= 3 \cdot \frac{1}{3} + 9 \cdot \frac{1}{3} = 4 \\ (A)_{12} &= 3 \cdot \frac{1}{3} + 9 \cdot (-\frac{2}{3}) = -5 \\ (A)_{21} &= 6 \cdot \frac{1}{3} + 0 \cdot \frac{1}{3} = 2 \\ (A)_{22} &= 6 \cdot \frac{1}{3} + 0 \cdot (-\frac{2}{3}) = 2 \end{aligned}$$

$$A = QJQ^{-1} = \begin{pmatrix} 4 & -5 \\ 2 & 2 \end{pmatrix}$$

Verification

$$A\boldsymbol{\eta}_+ = \begin{pmatrix} 4 & -5 \\ 2 & 2 \end{pmatrix} \begin{pmatrix} 2+i \\ 1-i \end{pmatrix} = \begin{pmatrix} 4\,(2+i) + (-5)\,(1-i) = 3+9i \\ 2\,(2+i) + 2\,(1-i) = 6 \end{pmatrix} = (3+3i) \begin{pmatrix} 2+i \\ 1-i \end{pmatrix}$$

Characteristic polynomial: $\lambda^2 - (6)\lambda + (18) = 0$ has $\operatorname{tr} A = 6 = 2\rho$ and $\det A = 18 = \rho^2 + \omega^2$, hence $\lambda = 3 \pm i\sqrt{9}$, i.e. $\rho \pm i\omega$ with $\rho = 3, \omega = 3$.

(ii) $\lambda_\pm = \pm 3i,\ \boldsymbol{\eta}_\pm = (\pm i,\ 3 \pm 2i)$

$\boldsymbol{\eta}_+ = \begin{pmatrix} i \\ 3+2i \end{pmatrix} = \begin{pmatrix} 0 \\ 3 \end{pmatrix} + i \begin{pmatrix} 1 \\ 2 \end{pmatrix}$, so $\mathbf{v}_1 = \begin{pmatrix} 0 \\ 3 \end{pmatrix}, \mathbf{v}_2 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \rho = 0, \omega = 3.$

$$Q = \begin{pmatrix} 0 & 1 \\ 3 & 2 \end{pmatrix}, \quad J = \begin{pmatrix} 0 & 3 \\ -3 & 0 \end{pmatrix}, \quad \det Q = -3, \quad Q^{-1} = \begin{pmatrix} -\frac{2}{3} & \frac{1}{3} \\ 1 & 0 \end{pmatrix}$$

$$\begin{aligned} (QJ)_{11} &= 0 \cdot 0 + 1 \cdot (-3) = -3 \\ (QJ)_{12} &= 0 \cdot 3 + 1 \cdot 0 = 0 \\ (QJ)_{21} &= 3 \cdot 0 + 2 \cdot (-3) = -6 \\ (QJ)_{22} &= 3 \cdot 3 + 2 \cdot 0 = 9 \end{aligned}$$

$$\begin{aligned} (A)_{11} &= (-3) \cdot (-\frac{2}{3}) + 0 \cdot 1 = 2 \\ (A)_{12} &= (-3) \cdot \frac{1}{3} + 0 \cdot 0 = -1 \\ (A)_{21} &= (-6) \cdot (-\frac{2}{3}) + 9 \cdot 1 = 13 \\ (A)_{22} &= (-6) \cdot \frac{1}{3} + 9 \cdot 0 = -2 \end{aligned}$$

$$A = QJQ^{-1} = \begin{pmatrix} 2 & -1 \\ 13 & -2 \end{pmatrix}$$

Verification

$$A\boldsymbol{\eta}_+ = \begin{pmatrix} 2 & -1 \\ 13 & -2 \end{pmatrix} \begin{pmatrix} i \\ 3+2i \end{pmatrix} = \begin{pmatrix} 2\,(i) + (-1)\,(3+2i) = -3 \\ 13\,(i) + (-2)\,(3+2i) = -6+9i \end{pmatrix} = (3i) \begin{pmatrix} i \\ 3+2i \end{pmatrix}$$

Characteristic polynomial: $\lambda^2 - (0)\lambda + (9) = 0$ has $\operatorname{tr} A = 0 = 2\rho$ and $\det A = 9 = \rho^2 + \omega^2$, hence $\lambda = 0 \pm i\sqrt{9}$, i.e. $\rho \pm i\omega$ with $\rho = 0, \omega = 3$.

(iii) $\lambda_\pm = -2 \pm i,\ \boldsymbol{\eta}_\pm = (1 \pm i,\ 1 \mp i)$

$\boldsymbol{\eta}_+ = \begin{pmatrix} 1+i \\ 1-i \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} + i \begin{pmatrix} 1 \\ -1 \end{pmatrix}$, so $\mathbf{v}_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \mathbf{v}_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}, \rho = -2, \omega = 1.$

$$Q = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad J = \begin{pmatrix} -2 & 1 \\ -1 & -2 \end{pmatrix}, \quad \det Q = -2, \quad Q^{-1} = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} \end{pmatrix}$$

$$\begin{aligned} (QJ)_{11} &= 1 \cdot (-2) + 1 \cdot (-1) = -3 \\ (QJ)_{12} &= 1 \cdot 1 + 1 \cdot (-2) = -1 \\ (QJ)_{21} &= 1 \cdot (-2) + (-1) \cdot (-1) = -1 \\ (QJ)_{22} &= 1 \cdot 1 + (-1) \cdot (-2) = 3 \end{aligned}$$

$$\begin{aligned} (A)_{11} &= (-3) \cdot \frac{1}{2} + (-1) \cdot \frac{1}{2} = -2 \\ (A)_{12} &= (-3) \cdot \frac{1}{2} + (-1) \cdot (-\frac{1}{2}) = -1 \\ (A)_{21} &= (-1) \cdot \frac{1}{2} + 3 \cdot \frac{1}{2} = 1 \\ (A)_{22} &= (-1) \cdot \frac{1}{2} + 3 \cdot (-\frac{1}{2}) = -2 \end{aligned}$$

$$A = QJQ^{-1} = \begin{pmatrix} -2 & -1 \\ 1 & -2 \end{pmatrix}$$

Verification

$$A\boldsymbol{\eta}_+ = \begin{pmatrix} -2 & -1 \\ 1 & -2 \end{pmatrix} \begin{pmatrix} 1+i \\ 1-i \end{pmatrix} = \begin{pmatrix} (-2)(1+i) + (-1)(1-i) = -3-i \\ 1(1+i) + (-2)(1-i) = -1+3i \end{pmatrix} = (-2+i) \begin{pmatrix} 1+i \\ 1-i \end{pmatrix}$$

Characteristic polynomial: $\lambda^2 - (-4)\lambda + (5) = 0$ has $\operatorname{tr} A = -4 = 2\rho$ and $\det A = 5 = \rho^2 + \omega^2$, hence $\lambda = -2 \pm i\sqrt{1}$, i.e. $\rho \pm i\omega$ with $\rho = -2, \omega = 1$.

PHASE 4: INTERPRETATION AND CHECKS

- The matrix is **not unique** if only the eigenvalues are prescribed, but it is unique once $\boldsymbol{\eta}_+$ is given (up to the complex scalar multiple that every eigenvector admits: rescaling $\boldsymbol{\eta}_+$ by $c = c_r + ic_i$ changes $(\mathbf{v}_1, \mathbf{v}_2)$ and leaves A unchanged, since $A\bar{\boldsymbol{\eta}}_+$, $A\boldsymbol{\eta}_+$ still determine A on the basis $\{\boldsymbol{\eta}_+, \bar{\boldsymbol{\eta}}_+\}$ of $\mathbb{C}^2$).
- For the three items $\operatorname{tr} A = 6, 0, -4$ and $\det A = 18, 9, 5$ respectively, which are 2ρ and $\rho^2 + \omega^2$ as the construction predicts.
- (ii) has $\rho = 0$: a centre. $A = \begin{pmatrix} 2 & -1 \\ 13 & -2 \end{pmatrix}$ has trace zero and determinant $9 = \omega^2$.
- This is the inverse of Exercise 3: from A to (Q, J) there, from (Q, J) to A here.

EXERCISE 6

Five Systems with a Repeated Eigenvalue: Stars and Improper Nodes

Original statement · probls_ch4-ii_2627.pdf, p. 2

Repeated Eigenvalue • Jordan Chain • Stars and Improper Nodes

STATEMENT

Find the eigenvalues and eigenvectors of the matrices occurring in the following equations, and draw the phase portrait. Find also the coordinate transformation that will put the equation into canonical form and show that this works. Write down the general solution.

- (i) $\dfrac{d\mathbf{x}}{dt} = \begin{pmatrix} 5 & -4 \\ 1 & 1 \end{pmatrix} \mathbf{x}$;
- (ii) $\dfrac{d\mathbf{x}}{dt} = \begin{pmatrix} -6 & 2 \\ -2 & -2 \end{pmatrix} \mathbf{x}$;
- (iii) $\dfrac{d\mathbf{x}}{dt} = \begin{pmatrix} -3 & -1 \\ 1 & -5 \end{pmatrix} \mathbf{x}$;
- (iv) $\dfrac{d\mathbf{x}}{dt} = \begin{pmatrix} 13 & 0 \\ 0 & 13 \end{pmatrix} \mathbf{x}$;
- (v) $\dfrac{d\mathbf{x}}{dt} = \begin{pmatrix} 7 & -4 \\ 1 & 3 \end{pmatrix} \mathbf{x}$.

PHASE 1: SETUP

For every matrix the discriminant $\Delta = (\operatorname{tr} A)^2 - 4 \det A$ is zero (checked in each item), so there is one repeated eigenvalue $\lambda = \frac{1}{2} \operatorname{tr} A$ with algebraic multiplicity $m_a = 2$. The geometric multiplicity $m_g = \dim \ker(A - \lambda I)$ is 2 iff $A = \lambda I$ (a **star**) and 1 otherwise (an **improper node** with a Jordan chain), Topic 4 Sec. 8.2 to 8.4. The canonical form is $J = \begin{pmatrix} \lambda & 1 \\ 0 & \lambda \end{pmatrix}$ with $Q = [\mathbf{v}_1 \mid \mathbf{v}_2]$, $(A - \lambda I)\mathbf{v}_2 = \mathbf{v}_1$; for a star there is nothing to transform.

PHASE 2: FORMULATION

Algorithm (Topic 4 Sec. 8.10): $\lambda = \frac{1}{2}\operatorname{tr} A$; $N = A - \lambda I$; if $N = O$ star, $\mathbf{x} = e^{\lambda t}\mathbf{x}_0$; otherwise $\mathbf{v}_1 \in \ker N$, solve $N\mathbf{v}_2 = \mathbf{v}_1$, $Q = [\mathbf{v}_1 \mid \mathbf{v}_2]$, $\mathbf{z} = Q^{-1}\mathbf{x}$, $z_1 = (c_1 + c_2 t)e^{\lambda t}$, $z_2 = c_2 e^{\lambda t}$, $\mathbf{x} = Q\mathbf{z}$.

METHOD OF THE CLASS NOTES [NOTES, P. 10]

Case 3 of the notes. 3.1, two independent eigenvectors: $A = P(\lambda I)P^{-1} = \lambda I$, so $\mathbf{x} = \mathbf{x}_0 e^{\lambda t}$, a star (item (iv)). 3.2, one eigenvector: the notes choose $\mathbf{v}_2$ independent of the eigenvector and set $\mathbf{v}_1 = (A - \lambda I)\mathbf{v}_2$, which is an eigenvector; then $A\mathbf{v}_2 = \mathbf{v}_1 + \lambda\mathbf{v}_2$, $A = P\begin{pmatrix}\lambda & 1\\ 0 & \lambda\end{pmatrix}P^{-1}$, $z_2 = Ce^{\lambda t}$, $z_1 = (Ct + D)e^{\lambda t}$ and $\mathbf{x} = (Ct + D)e^{\lambda t}\mathbf{v}_1 + Ce^{\lambda t}\mathbf{v}_2$, an improper node.

The algorithm above uses the same relation $N\mathbf{v}_2 = \mathbf{v}_1$ the other way round: it fixes the eigenvector $\mathbf{v}_1$ first and solves for $\mathbf{v}_2$ ($c_1 = D$, $c_2 = C$).

PHASE 3: DERIVATION

(i) $A = \begin{pmatrix}5 & -4\\ 1 & 1\end{pmatrix}$

Characteristic equation $\det(A - \lambda I) = 0$ (Topic 4, Eq. 27.8):

$$\det(A - \lambda I) = \begin{vmatrix}5 - \lambda & -4\\ 1 & 1 - \lambda\end{vmatrix} = (5 - \lambda)(1 - \lambda) - (-4)(1)$$

$$= \lambda^2 - (5 + 1)\lambda + (5 \cdot 1 - (-4) \cdot 1) = \lambda^2 - (6)\lambda + (9) = \lambda^2 - 6\lambda + 9$$

$\operatorname{tr} A = 6$, $\det A = 9$, discriminant $\Delta = (\operatorname{tr} A)^2 - 4\det A = 36 - 36 = 0$.

$\Delta = 0$: the characteristic polynomial is a perfect square, $(\lambda - 3)^2 = \lambda^2 - 6\lambda + 9$, so

$$\lambda_1 = \lambda_2 = \lambda = \tfrac{1}{2}\operatorname{tr} A = 3 \quad (m_a = 2)$$

Geometric multiplicity: $N = A - \lambda I = \begin{pmatrix}2 & -4\\ 1 & -2\end{pmatrix}$.

$N \neq O$ and $\det N = (2)(-2) - (-4)(1) = 0$, so N has rank 1: $m_g = 1$ and there is a single eigendirection (improper node). By Cayley-Hamilton $N^2 = O$ (Topic 4, Sec. 8.4):

$$
\begin{aligned}
(N^2)_{11} &= 2\cdot 2 + (-4)\cdot 1 = 0 \\
(N^2)_{12} &= 2\cdot (-4) + (-4)\cdot (-2) = 0 \\
(N^2)_{21} &= 1\cdot 2 + (-2)\cdot 1 = 0 \\
(N^2)_{22} &= 1\cdot (-4) + (-2)\cdot (-2) = 0
\end{aligned}
$$

Eigenvector

For $\lambda = 3$: $(A - (3)I)\mathbf{v_1} = \mathbf{0}$, that is

$$
\begin{pmatrix} 2 & -4 \\ 1 & -2 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}
$$

$$
\begin{aligned}
2v_1 - 4v_2 &= 0 \quad \text{(row 1)} \\
1v_1 - 2v_2 &= 0 \quad \text{(row 2)}
\end{aligned}
$$

The coefficient matrix is singular ($\det(A - \lambda I) = 0$), so the rows are proportional (row 2 equals $\frac{1}{2}$ times row 1) and express a single condition. Row 1 reads $2v_1 + (-4)v_2 = 0$, so $v_1 = 2\,v_2$: every eigenvector is a multiple of $(2, 1)^T$. The smallest integer representative is

$$
\mathbf{v_1} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}
$$

Check: $A\mathbf{v_1} = \begin{pmatrix} 6 \\ 3 \end{pmatrix} = 3\mathbf{v_1}$.

Generalised eigenvector

Solve $(A - \lambda I)\mathbf{v}_2 = \mathbf{v}_1$, i.e. $N\mathbf{v}_2 = \mathbf{v}_1$ (Eq. 8.2). The system is solvable because $\operatorname{im} N = \ker N$ (Sec. 8.4):

$$
\begin{pmatrix} 2 & -4 \\ 1 & -2 \end{pmatrix} \begin{pmatrix} p \\ q \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}
$$

$$
\begin{aligned}
2p - 4q &= 2 \\
1p - 2q &= 1
\end{aligned}
$$

The two equations are proportional (the right-hand sides are in the same ratio as the rows, because $\mathbf{v}_1 \in \operatorname{im} N$), so one is enough. The solution is not unique: $\mathbf{v}_2 + s\mathbf{v}_1$ also works for every s, because $N\mathbf{v}_1 = \mathbf{0}$.

Take $q = 0$. Row 1 gives $2p = 2$, so $p = 1$:

$$\mathbf{v}_2 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

Check: $N\mathbf{v}_2 = \begin{pmatrix} 2 \\ 1 \end{pmatrix} = \mathbf{v}_1$, equivalently $A\mathbf{v}_2 = \lambda\mathbf{v}_2 + \mathbf{v}_1 = \begin{pmatrix} 5 \\ 1 \end{pmatrix}$ (Eq. 8.2).

Change of coordinates

$$Q = [\mathbf{v}_1 \mid \mathbf{v}_2] = \begin{pmatrix} 2 & 1 \\ 1 & 0 \end{pmatrix}$$

$\det Q = (2)(0) - (1)(1) = -1$, which is not zero, so Q is invertible. For a 2×2 matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad-bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$ (Robinson, Appendix B):

$$Q^{-1} = \frac{1}{-1}\begin{pmatrix} 0 & -1 \\ -1 & 2 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 1 & -2 \end{pmatrix}$$

$$A\mathbf{v}_1 = \begin{pmatrix} 6 \\ 3 \end{pmatrix} = 3\mathbf{v}_1, \qquad A\mathbf{v}_2 = \begin{pmatrix} 5 \\ 1 \end{pmatrix} = \mathbf{v}_1 + 3\mathbf{v}_2$$

$$AQ = \begin{pmatrix} 6 & 5 \\ 3 & 1 \end{pmatrix}$$

$$\begin{aligned} (Q^{-1}AQ)_{11} &= 0 \cdot 6 + 1 \cdot 3 = 3 \\ (Q^{-1}AQ)_{12} &= 0 \cdot 5 + 1 \cdot 1 = 1 \\ (Q^{-1}AQ)_{21} &= 1 \cdot 6 + (-2) \cdot 3 = 0 \\ (Q^{-1}AQ)_{22} &= 1 \cdot 5 + (-2) \cdot 1 = 3 \end{aligned}$$

$$Q^{-1}AQ = \begin{pmatrix} 3 & 1 \\ 0 & 3 \end{pmatrix} = J = \begin{pmatrix} \lambda & 1 \\ 0 & \lambda \end{pmatrix}$$

Solution in the canonical plane

$$\dot z_1 = 3z_1 + z_2, \qquad \dot z_2 = 3z_2$$

Second equation: $\frac{d}{dt}(e^{-\lambda t}z_2) = e^{-\lambda t}(\dot z_2 - \lambda z_2) = 0$, so $e^{-\lambda t}z_2$ is constant: $z_2 = c_2 e^{3t}$. First equation with this z_2: $\dot z_1 - \lambda z_1 = c_2 e^{3t}$. The integrating factor is $e^{-\lambda t}$: by the product rule $\frac{d}{dt}(e^{-\lambda t}z_1) = e^{-\lambda t}(\dot z_1 - \lambda z_1) = c_2$. Barrow's rule between 0 and t:

$$\left[e^{-\lambda s}z_1(s)\right]_{s=0}^{s=t} = \int_0^t c_2\, ds = c_2 t \implies e^{-\lambda t}z_1(t) - z_1(0) = c_2 t$$

$$z_1(t) = (c_1 + c_2 t)e^{3t}, \qquad z_2(t) = c_2 e^{3t} \qquad (c_1 = z_1(0),\ c_2 = z_2(0))$$

Back to the original coordinates

$\mathbf{x} = Q\mathbf{z} = z_1\mathbf{v}_1 + z_2\mathbf{v}_2$, hence (Topic 4, Eq. 8.5)

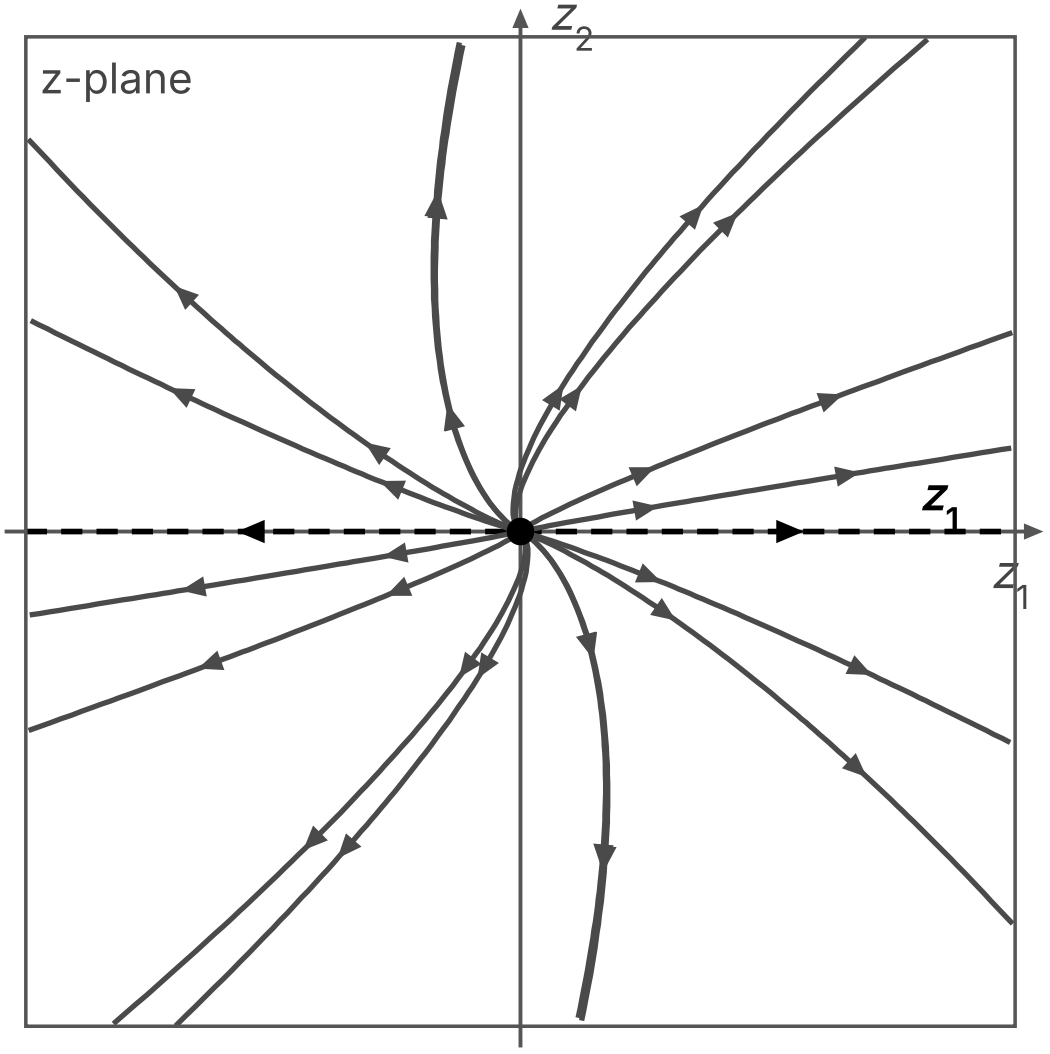
$$\mathbf{x}(t) = e^{3t}\left[c_1\begin{pmatrix}2\\1\end{pmatrix} + c_2\left(t\begin{pmatrix}2\\1\end{pmatrix} + \begin{pmatrix}1\\0\end{pmatrix}\right)\right]$$

$$x(t) = 2c_1 e^{3t} + 2c_2 t e^{3t} + c_2 e^{3t}, \qquad y(t) = c_1 e^{3t} + c_2 t e^{3t}$$

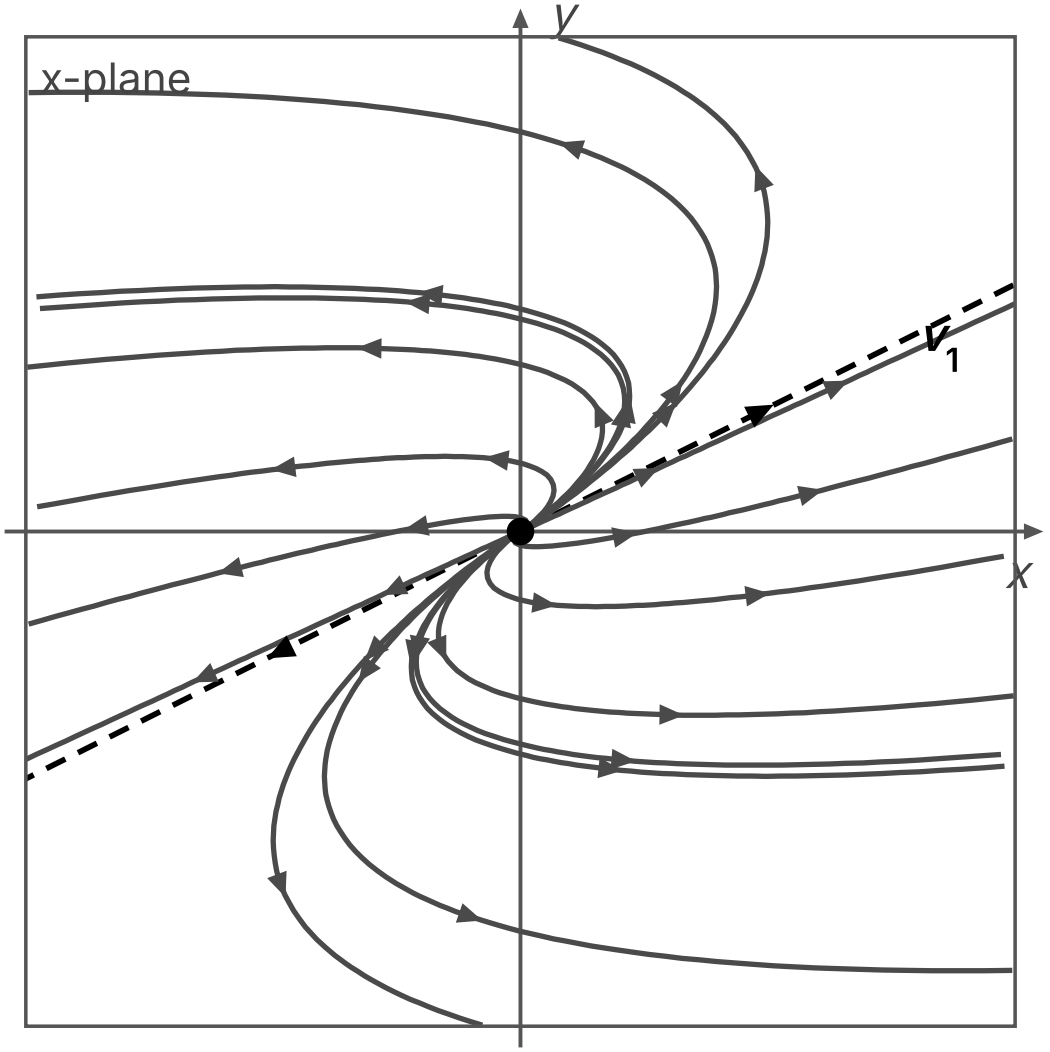
Phase portrait

$\lambda = 3$ is repeated with a single eigendirection and $A \neq \lambda I$: the origin is an **unstable improper node** (Topic 4 Sec. 8.8 and 8.9). In the $\mathbf{z}$-plane the z_1-axis is invariant ($z_2 \equiv 0$, $z_1 = c_1 e^{3t}$, arrows away from the origin); the other orbits satisfy $z_1 = z_2\left[K + \frac{1}{\lambda}\ln|z_2|\right]$, are S-shaped, tangent to the z_1-axis at the origin and parallel to it far away. Since $\dot z_1 = \lambda z_1 + z_2 = z_2$ on the z_2-axis, orbits cross it moving to the right in the upper half-plane and to the left in the lower one.

Orientation in the $\mathbf{x}$-plane, from $N\mathbf{x} = (A - \lambda I)\mathbf{x} = z_2\mathbf{v}_1$ (Topic 4 Sec. 8.9): at $(0,1)^T$, $N(0,1)^T = \begin{pmatrix}-4\\-2\end{pmatrix} = -2\,\mathbf{v}_1$, so there the orbit drifts along $-\mathbf{v}_1$. The orbits are tangent to the eigenline at the origin and bend into an S around it.



Canonical $\mathbf{z}$-plane (unstable improper node).



Original (x, y)-plane (unstable improper node); dashed: the eigenline with $\mathbf{v}_1 = (2, 1)$.

(ii) $A = \begin{pmatrix} -6 & 2 \\ -2 & -2 \end{pmatrix}$

Characteristic equation $\det(A - \lambda I) = 0$ (Topic 4, Eq. 27.8):

$$\det(A - \lambda I) = \begin{vmatrix} -6 - \lambda & 2 \\ -2 & -2 - \lambda \end{vmatrix} = (-6 - \lambda)(-2 - \lambda) - (2)(-2)$$

$$= \lambda^2 - (-6 + (-2))\lambda + (-6 \cdot (-2) - 2 \cdot (-2)) = \lambda^2 - (-8)\lambda + (16) = \lambda^2 + 8\lambda + 16$$

$\operatorname{tr} A = -8$, $\det A = 16$, discriminant $\Delta = (\operatorname{tr} A)^2 - 4 \det A = 64 - 64 = 0$.

$\Delta = 0$: the characteristic polynomial is a perfect square, $(\lambda + 4)^2 = \lambda^2 + 8\lambda + 16$, so

$$\lambda_1 = \lambda_2 = \lambda = \tfrac{1}{2}\operatorname{tr} A = -4 \quad (m_a = 2)$$

Geometric multiplicity: $N = A - \lambda I = \begin{pmatrix} -2 & 2 \\ -2 & 2 \end{pmatrix}$.

$N \neq O$ and $\det N = (-2)(2) - (2)(-2) = 0$, so N has rank 1: $m_g = 1$ and there is a single eigendirection (improper node). By Cayley-Hamilton $N^2 = O$ (Topic 4, Sec. 8.4):

$$\begin{aligned}(N^2)_{11} &= (-2)\cdot(-2) + 2\cdot(-2) = 0\\ (N^2)_{12} &= (-2)\cdot 2 + 2\cdot 2 = 0\\ (N^2)_{21} &= (-2)\cdot(-2) + 2\cdot(-2) = 0\\ (N^2)_{22} &= (-2)\cdot 2 + 2\cdot 2 = 0\end{aligned}$$

Eigenvector

For $\lambda = -4$: $(A - (-4)I)\mathbf{v_1} = \mathbf{0}$, that is

$$\begin{pmatrix} -2 & 2 \\ -2 & 2 \end{pmatrix}\begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\begin{aligned}-2v_1 + 2v_2 &= 0 \quad (\text{row } 1)\\ -2v_1 + 2v_2 &= 0 \quad (\text{row } 2)\end{aligned}$$

The coefficient matrix is singular ($\det(A - \lambda I) = 0$), so the rows are proportional (row 2 is identical to row 1) and express a single condition. Row 1 reads $-2v_1 + 2v_2 = 0$, so $v_1 = v_2$: every eigenvector is a multiple of $(1, 1)^T$. The smallest integer representative is

$$\mathbf{v_1} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

Check: $A\mathbf{v_1} = \begin{pmatrix} -4 \\ -4 \end{pmatrix} = -4\mathbf{v_1}$.

Generalised eigenvector

Solve $(A - \lambda I)\mathbf{v}_2 = \mathbf{v}_1$, i.e. $N\mathbf{v}_2 = \mathbf{v}_1$ (Eq. 8.2). The system is solvable because $\operatorname{im} N = \ker N$ (Sec. 8.4):

$$\begin{pmatrix} -2 & 2 \\ -2 & 2 \end{pmatrix}\begin{pmatrix} p \\ q \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$-2p + 2q = 1$$
$$-2p + 2q = 1$$

The two equations are proportional (the right-hand sides are in the same ratio as the rows, because $\mathbf{v}_1 \in \operatorname{im} N$), so one is enough. The solution is not unique: $\mathbf{v}_2 + s\mathbf{v}_1$ also works for every s, because $N\mathbf{v}_1 = \mathbf{0}$.

Take $p = 0$. Row 1 gives $2q = 1$, so $q = \frac{1}{2}$:

$$\mathbf{v}_2 = \begin{pmatrix} 0 \\ \frac{1}{2} \end{pmatrix}$$

Check: $N\mathbf{v}_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \mathbf{v}_1$, equivalently $A\mathbf{v}_2 = \lambda\mathbf{v}_2 + \mathbf{v}_1 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$ (Eq. 8.2).

Change of coordinates

$$Q = [\mathbf{v}_1 \mid \mathbf{v}_2] = \begin{pmatrix} 1 & 0 \\ 1 & \frac{1}{2} \end{pmatrix}$$

$\det Q = (1)(\frac{1}{2}) - (0)(1) = \frac{1}{2}$, which is not zero, so Q is invertible. For a 2×2 matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad-bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$ (Robinson, Appendix B):

$$Q^{-1} = \frac{1}{\frac{1}{2}} \begin{pmatrix} \frac{1}{2} & 0 \\ -1 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ -2 & 2 \end{pmatrix}$$

$$A\mathbf{v}_1 = \begin{pmatrix} -4 \\ -4 \end{pmatrix} = (-4)\mathbf{v}_1, \qquad A\mathbf{v}_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \mathbf{v}_1 + (-4)\mathbf{v}_2$$

$$AQ = \begin{pmatrix} -4 & 1 \\ -4 & -1 \end{pmatrix}$$

$$\begin{aligned}
(Q^{-1}AQ)_{11} &= 1 \cdot (-4) + 0 \cdot (-4) = -4 \\
(Q^{-1}AQ)_{12} &= 1 \cdot 1 + 0 \cdot (-1) = 1 \\
(Q^{-1}AQ)_{21} &= (-2) \cdot (-4) + 2 \cdot (-4) = 0 \\
(Q^{-1}AQ)_{22} &= (-2) \cdot 1 + 2 \cdot (-1) = -4
\end{aligned}$$

$$Q^{-1}AQ = \begin{pmatrix} -4 & 1 \\ 0 & -4 \end{pmatrix} = J = \begin{pmatrix} \lambda & 1 \\ 0 & \lambda \end{pmatrix}$$

Solution in the canonical plane

$$\dot{z}_1 = -4z_1 + z_2, \qquad \dot{z}_2 = -4z_2$$

Second equation: $\frac{d}{dt}(e^{-\lambda t}z_2) = e^{-\lambda t}(\dot{z}_2 - \lambda z_2) = 0$, so $e^{-\lambda t}z_2$ is constant: $z_2 = c_2 e^{-4t}$. First equation with this z_2: $\dot{z}_1 - \lambda z_1 = c_2 e^{-4t}$. The integrating factor is $e^{-\lambda t}$: by the product rule $\frac{d}{dt}(e^{-\lambda t}z_1) = e^{-\lambda t}(\dot{z}_1 - \lambda z_1) = c_2$. Barrow's rule between 0 and t:

$$\left[e^{-\lambda s}z_1(s)\right]_{s=0}^{s=t} = \int_0^t c_2\, ds = c_2 t \implies e^{-\lambda t}z_1(t) - z_1(0) = c_2 t$$

$$z_1(t) = (c_1 + c_2 t)e^{-4t}, \qquad z_2(t) = c_2 e^{-4t} \qquad (c_1 = z_1(0),\ c_2 = z_2(0))$$

Back to the original coordinates

$\mathbf{x} = Q\mathbf{z} = z_1\mathbf{v}_1 + z_2\mathbf{v}_2$, hence (Topic 4, Eq. 8.5)

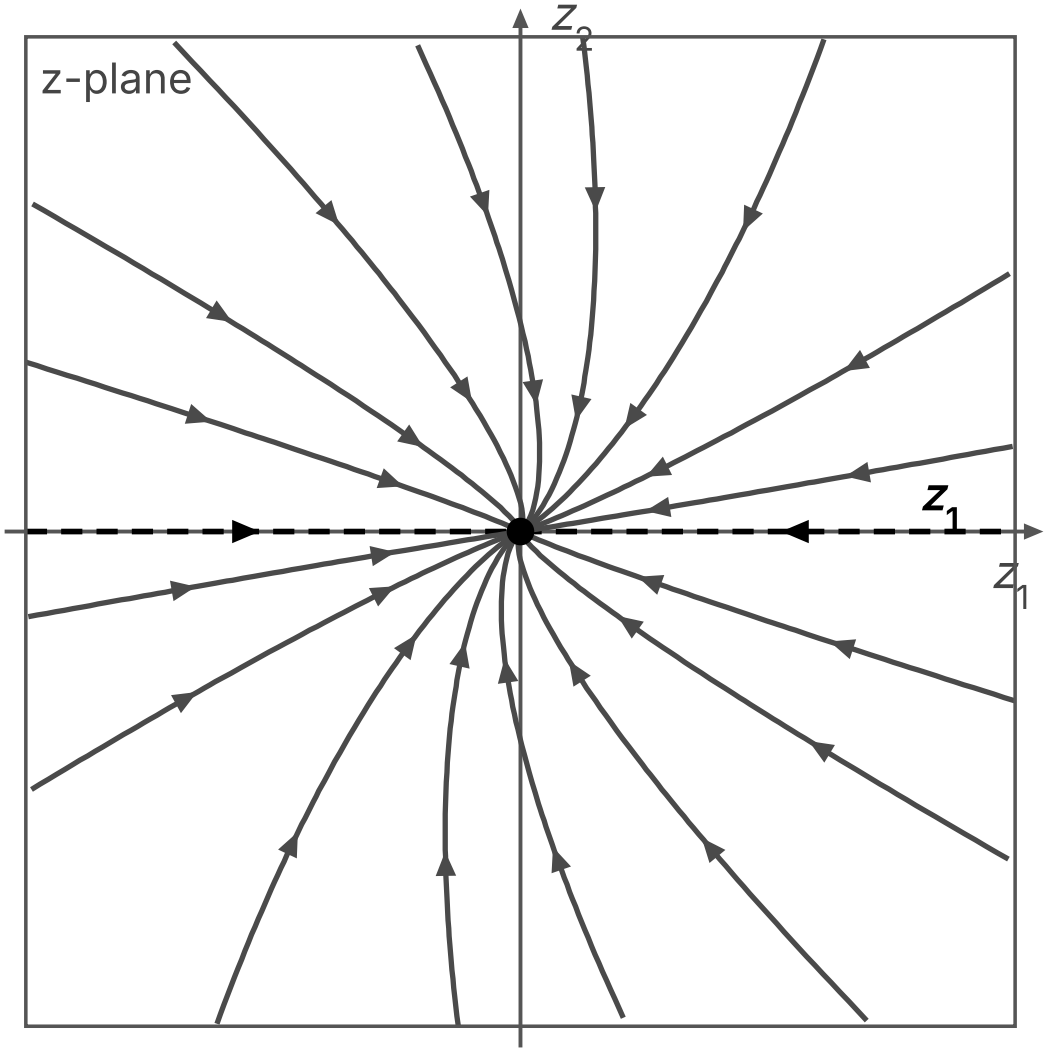
$$\mathbf{x}(t) = e^{-4t}\left[c_1\begin{pmatrix}1\\1\end{pmatrix} + c_2\left(t\begin{pmatrix}1\\1\end{pmatrix} + \begin{pmatrix}0\\\frac{1}{2}\end{pmatrix}\right)\right]$$

$$x(t) = c_1 e^{-4t} + c_2 t e^{-4t}, \qquad y(t) = c_1 e^{-4t} + c_2 t e^{-4t} + \frac{c_2 e^{-4t}}{2}$$

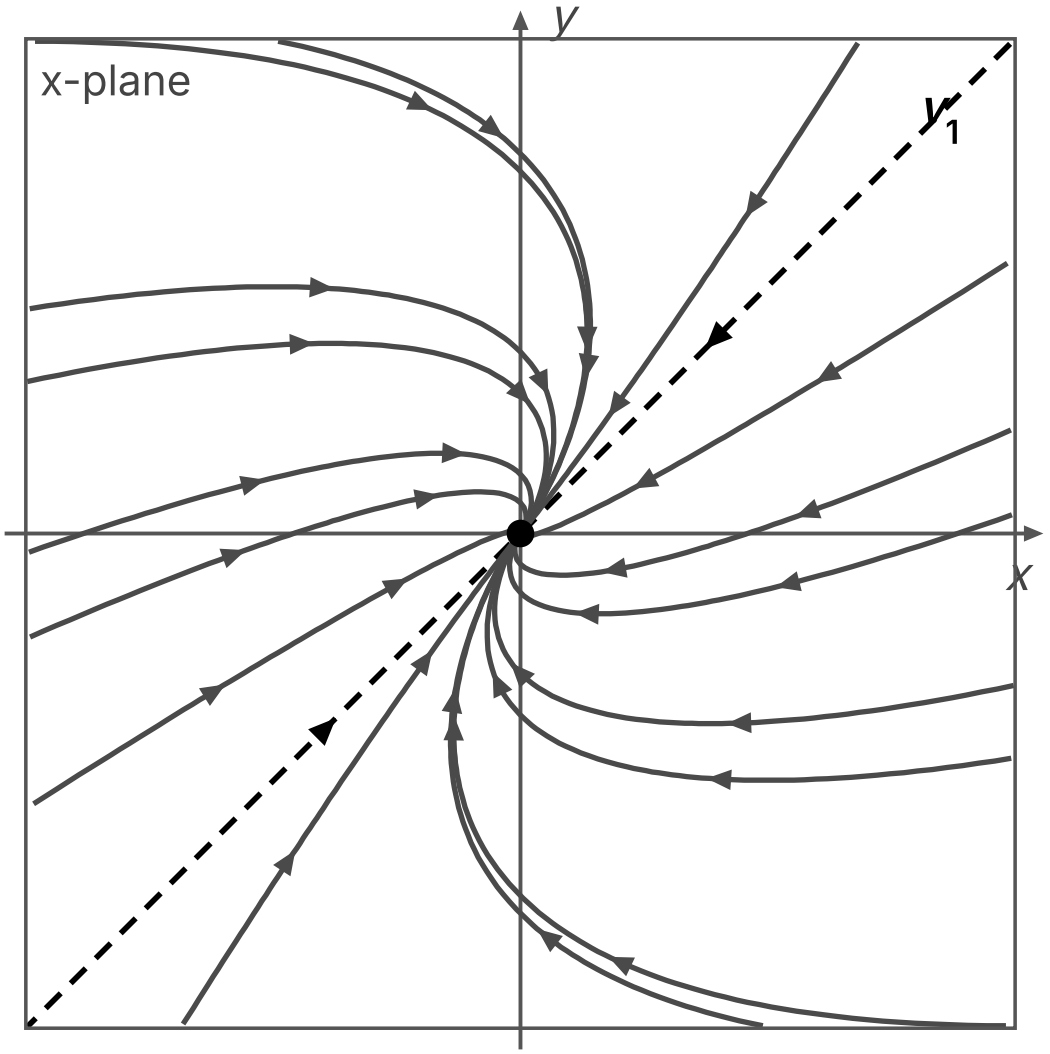
Phase portrait

$\lambda = -4$ is repeated with a single eigendirection and $A \neq \lambda I$: the origin is an **stable improper node** (Topic 4 Sec. 8.8 and 8.9). In the $\mathbf{z}$-plane the z_1-axis is invariant ($z_2 \equiv 0$, $z_1 = c_1 e^{-4t}$, arrows towards the origin); the other orbits satisfy $z_1 = z_2\left[K + \frac{1}{\lambda}\ln|z_2|\right]$, are S-shaped, tangent to the z_1-axis at the origin and parallel to it far away. Since $\dot{z}_1 = \lambda z_1 + z_2 = z_2$ on the z_2-axis, orbits cross it moving to the right in the upper half-plane and to the left in the lower one.

Orientation in the $\mathbf{x}$-plane, from $N\mathbf{x} = (A - \lambda I)\mathbf{x} = z_2\mathbf{v}_1$ (Topic 4 Sec. 8.9): at $(0,1)^T$, $N(0,1)^T = \begin{pmatrix}2\\2\end{pmatrix} = 2\,\mathbf{v}_1$, so there the orbit drifts along $+\mathbf{v}_1$. The orbits are tangent to the eigenline at the origin and bend into an S around it.



Canonical $\mathbf{z}$-plane (stable improper node).



Original (x, y)-plane (stable improper node); dashed: the eigenline with $\mathbf{v}_1 = (1, 1)$.

(iii) $A = \begin{pmatrix} -3 & -1 \\ 1 & -5 \end{pmatrix}$

Characteristic equation $\det(A - \lambda I) = 0$ (Topic 4, Eq. 27.8):

$$\det(A - \lambda I) = \begin{vmatrix} -3 - \lambda & -1 \\ 1 & -5 - \lambda \end{vmatrix} = (-3 - \lambda)(-5 - \lambda) - (-1)(1)$$

$$= \lambda^2 - (-3 + (-5))\lambda + (-3 \cdot (-5) - (-1) \cdot 1) = \lambda^2 - (-8)\lambda + (16) = \lambda^2 + 8\lambda + 16$$

$\operatorname{tr} A = -8$, $\det A = 16$, discriminant $\Delta = (\operatorname{tr} A)^2 - 4 \det A = 64 - 64 = 0$.

$\Delta = 0$: the characteristic polynomial is a perfect square, $(\lambda + 4)^2 = \lambda^2 + 8\lambda + 16$, so

$$\lambda_1 = \lambda_2 = \lambda = \tfrac{1}{2}\operatorname{tr} A = -4 \quad (m_a = 2)$$

Geometric multiplicity: $N = A - \lambda I = \begin{pmatrix} 1 & -1 \\ 1 & -1 \end{pmatrix}$.

$N \neq O$ and $\det N = (1)(-1) - (-1)(1) = 0$, so N has rank 1: $m_g = 1$ and there is a single eigendirection (improper node). By Cayley-Hamilton $N^2 = O$ (Topic 4, Sec. 8.4):

$$\begin{aligned}(N^2)_{11} &= 1 \cdot 1 + (-1) \cdot 1 = 0 \\ (N^2)_{12} &= 1 \cdot (-1) + (-1) \cdot (-1) = 0 \\ (N^2)_{21} &= 1 \cdot 1 + (-1) \cdot 1 = 0 \\ (N^2)_{22} &= 1 \cdot (-1) + (-1) \cdot (-1) = 0\end{aligned}$$

Eigenvector

For $\lambda = -4$: $(A - (-4)I)\mathbf{v_1} = \mathbf{0}$, that is

$$\begin{pmatrix} 1 & -1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\begin{aligned}1v_1 - 1v_2 &= 0 \quad (\text{row } 1) \\ 1v_1 - 1v_2 &= 0 \quad (\text{row } 2)\end{aligned}$$

The coefficient matrix is singular ($\det(A - \lambda I) = 0$), so the rows are proportional (row 2 is identical to row 1) and express a single condition. Row 1 reads $1v_1 + (-1)v_2 = 0$, so $v_1 = v_2$: every eigenvector is a multiple of $(1, 1)^T$. The smallest integer representative is

$$\mathbf{v_1} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

Check: $A\mathbf{v_1} = \begin{pmatrix} -4 \\ -4 \end{pmatrix} = -4\mathbf{v_1}$.

Generalised eigenvector

Solve $(A - \lambda I)\mathbf{v}_2 = \mathbf{v}_1$, i.e. $N\mathbf{v}_2 = \mathbf{v}_1$ (Eq. 8.2). The system is solvable because $\operatorname{im} N = \ker N$ (Sec. 8.4):

$$\begin{pmatrix} 1 & -1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} p \\ q \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$1p - 1q = 1$$
$$1p - 1q = 1$$

The two equations are proportional (the right-hand sides are in the same ratio as the rows, because $\mathbf{v}_1 \in \operatorname{im} N$), so one is enough. The solution is not unique: $\mathbf{v}_2 + s\mathbf{v}_1$ also works for every s, because $N\mathbf{v}_1 = \mathbf{0}$.

Take $p = 0$. Row 1 gives $-1q = 1$, so $q = -1$:

$$\mathbf{v}_2 = \begin{pmatrix} 0 \\ -1 \end{pmatrix}$$

Check: $N\mathbf{v}_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \mathbf{v}_1$, equivalently $A\mathbf{v}_2 = \lambda\mathbf{v}_2 + \mathbf{v}_1 = \begin{pmatrix} 1 \\ 5 \end{pmatrix}$ (Eq. 8.2).

Change of coordinates

$$Q = [\mathbf{v}_1 \mid \mathbf{v}_2] = \begin{pmatrix} 1 & 0 \\ 1 & -1 \end{pmatrix}$$

$\det Q = (1)(-1) - (0)(1) = -1$, which is not zero, so Q is invertible. For a 2×2 matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad-bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$ (Robinson, Appendix B):

$$Q^{-1} = \frac{1}{-1}\begin{pmatrix} -1 & 0 \\ -1 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 1 & -1 \end{pmatrix}$$

$$A\mathbf{v}_1 = \begin{pmatrix} -4 \\ -4 \end{pmatrix} = (-4)\mathbf{v}_1, \qquad A\mathbf{v}_2 = \begin{pmatrix} 1 \\ 5 \end{pmatrix} = \mathbf{v}_1 + (-4)\mathbf{v}_2$$

$$AQ = \begin{pmatrix} -4 & 1 \\ -4 & 5 \end{pmatrix}$$

$$\begin{aligned} (Q^{-1}AQ)_{11} &= 1 \cdot (-4) + 0 \cdot (-4) = -4 \\ (Q^{-1}AQ)_{12} &= 1 \cdot 1 + 0 \cdot 5 = 1 \\ (Q^{-1}AQ)_{21} &= 1 \cdot (-4) + (-1) \cdot (-4) = 0 \\ (Q^{-1}AQ)_{22} &= 1 \cdot 1 + (-1) \cdot 5 = -4 \end{aligned}$$

$$Q^{-1}AQ = \begin{pmatrix} -4 & 1 \\ 0 & -4 \end{pmatrix} = J = \begin{pmatrix} \lambda & 1 \\ 0 & \lambda \end{pmatrix}$$

Solution in the canonical plane

$$\dot{z}_1 = -4z_1 + z_2, \qquad \dot{z}_2 = -4z_2$$

Second equation: $\frac{d}{dt}(e^{-\lambda t}z_2) = e^{-\lambda t}(\dot{z}_2 - \lambda z_2) = 0$, so $e^{-\lambda t}z_2$ is constant: $z_2 = c_2 e^{-4t}$. First equation with this z_2: $\dot{z}_1 - \lambda z_1 = c_2 e^{-4t}$. The integrating factor is $e^{-\lambda t}$: by the product rule $\frac{d}{dt}(e^{-\lambda t}z_1) = e^{-\lambda t}(\dot{z}_1 - \lambda z_1) = c_2$. Barrow's rule between 0 and t:

$$\left[e^{-\lambda s}z_1(s)\right]_{s=0}^{s=t} = \int_0^t c_2\, ds = c_2 t \implies e^{-\lambda t}z_1(t) - z_1(0) = c_2 t$$

$$z_1(t) = (c_1 + c_2 t)e^{-4t}, \qquad z_2(t) = c_2 e^{-4t} \qquad (c_1 = z_1(0),\ c_2 = z_2(0))$$

Back to the original coordinates

$\mathbf{x} = Q\mathbf{z} = z_1\mathbf{v}_1 + z_2\mathbf{v}_2$, hence (Topic 4, Eq. 8.5)

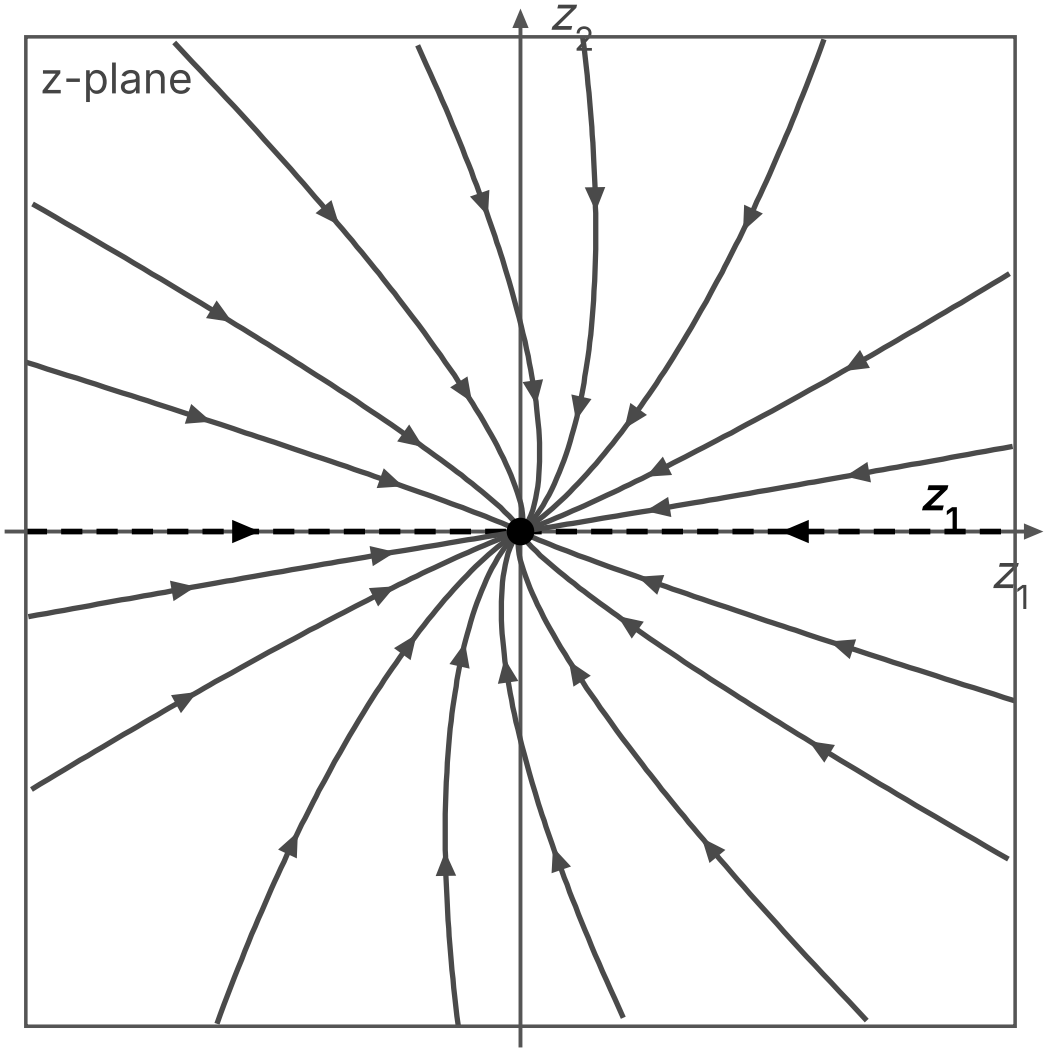
$$\mathbf{x}(t) = e^{-4t}\left[c_1\begin{pmatrix}1\\1\end{pmatrix} + c_2\left(t\begin{pmatrix}1\\1\end{pmatrix} + \begin{pmatrix}0\\-1\end{pmatrix}\right)\right]$$

$$x(t) = c_1 e^{-4t} + c_2 t e^{-4t}, \qquad y(t) = c_1 e^{-4t} + c_2 t e^{-4t} - c_2 e^{-4t}$$

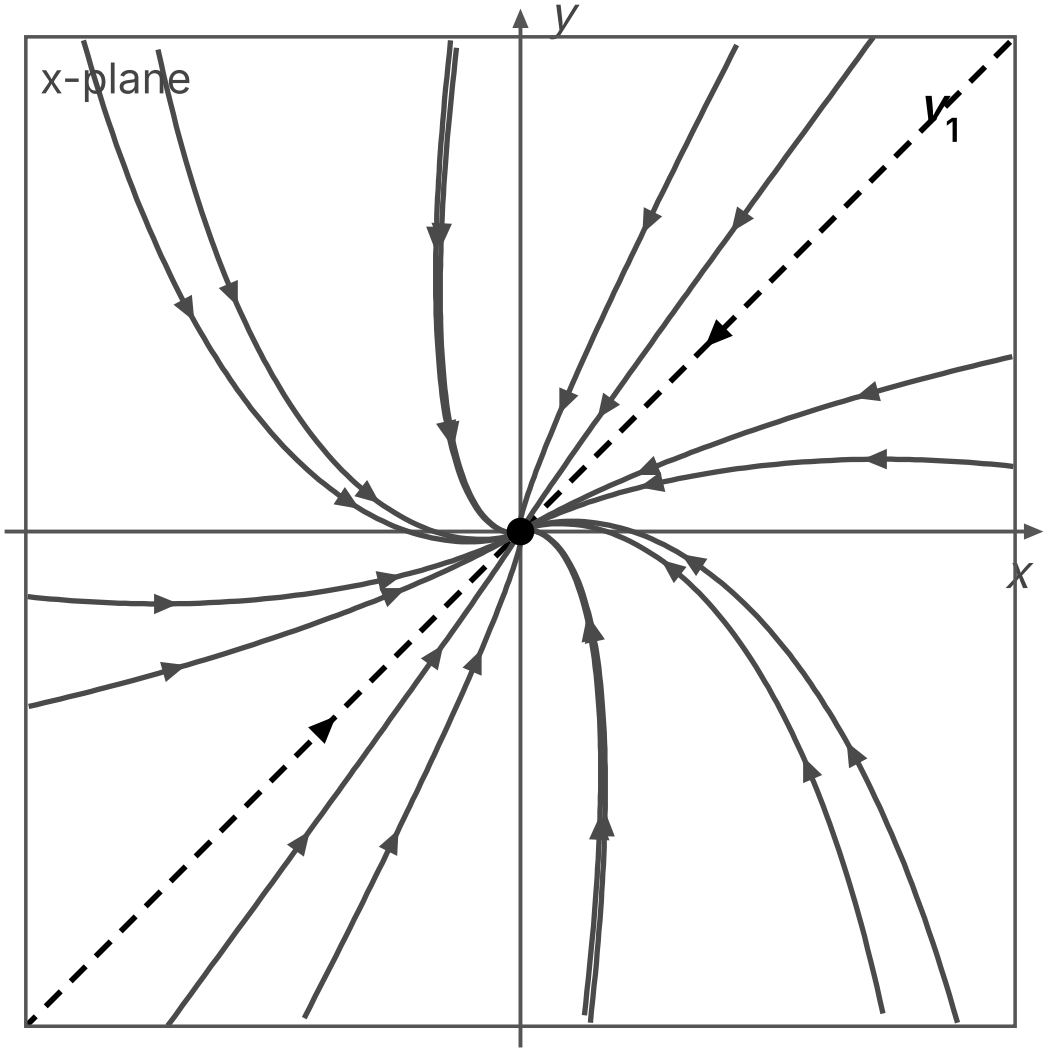
Phase portrait

$\lambda = -4$ is repeated with a single eigendirection and $A \neq \lambda I$: the origin is an **stable improper node** (Topic 4 Sec. 8.8 and 8.9). In the $\mathbf{z}$-plane the z_1-axis is invariant ($z_2 \equiv 0$, $z_1 = c_1 e^{-4t}$, arrows towards the origin); the other orbits satisfy $z_1 = z_2\left[K + \frac{1}{\lambda}\ln|z_2|\right]$, are S-shaped, tangent to the z_1-axis at the origin and parallel to it far away. Since $\dot{z}_1 = \lambda z_1 + z_2 = z_2$ on the z_2-axis, orbits cross it moving to the right in the upper half-plane and to the left in the lower one.

Orientation in the $\mathbf{x}$-plane, from $N\mathbf{x} = (A - \lambda I)\mathbf{x} = z_2\mathbf{v}_1$ (Topic 4 Sec. 8.9): at $(0,1)^T$, $N(0,1)^T = \begin{pmatrix}-1\\-1\end{pmatrix} = -1\,\mathbf{v}_1$, so there the orbit drifts along $-\mathbf{v}_1$. The orbits are tangent to the eigenline at the origin and bend into an S around it.



Canonical $\mathbf{z}$-plane (stable improper node).



Original (x, y)-plane (stable improper node); dashed: the eigenline with $\mathbf{v}_1 = (1, 1)$.

(iv) $A = \begin{pmatrix} 13 & 0 \\ 0 & 13 \end{pmatrix}$

Characteristic equation $\det(A - \lambda I) = 0$ (Topic 4, Eq. 27.8):

$$\det(A - \lambda I) = \begin{vmatrix} 13 - \lambda & 0 \\ 0 & 13 - \lambda \end{vmatrix} = (13 - \lambda)(13 - \lambda) - (0)(0)$$

$$= \lambda^2 - (13 + 13)\lambda + (13 \cdot 13 - 0 \cdot 0) = \lambda^2 - (26)\lambda + (169) = \lambda^2 - 26\lambda + 169$$

$\operatorname{tr} A = 26$, $\det A = 169$, discriminant $\Delta = (\operatorname{tr} A)^2 - 4 \det A = 676 - 676 = 0$.

$\Delta = 0$: the characteristic polynomial is a perfect square, $(\lambda - 13)^2 = \lambda^2 - 26\lambda + 169$, so

$$\lambda_1 = \lambda_2 = \lambda = \tfrac{1}{2}\operatorname{tr} A = 13 \quad (m_a = 2)$$

Geometric multiplicity: $N = A - \lambda I = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$.

$N = O$: every vector is an eigenvector ($m_g = 2$) and $A = \lambda I$ (Topic 4, Sec. 8.3). The system is already uncoupled in every basis.

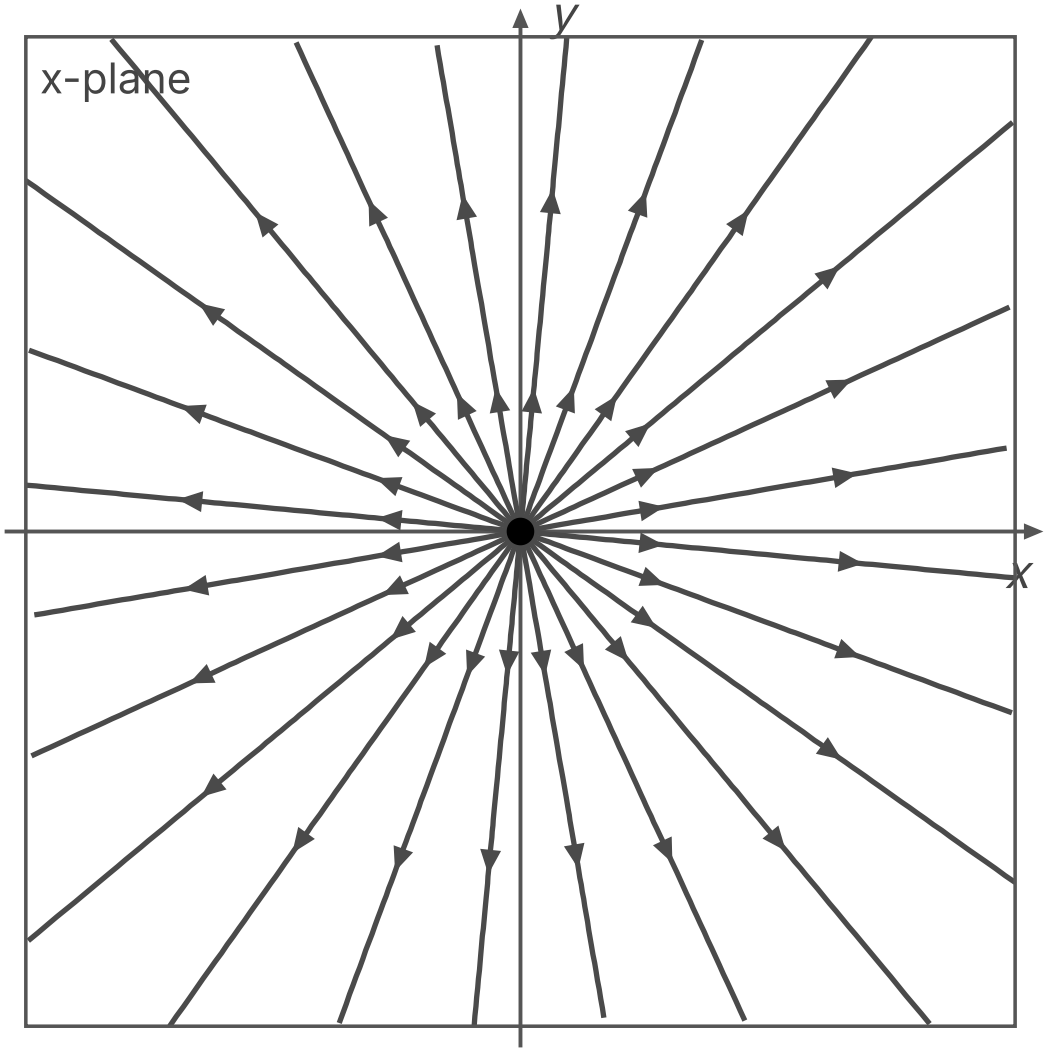
$$\dot{x} = 13x, \qquad \dot{y} = 13y$$

Each equation is separable, $\frac{d}{dt}(e^{-\lambda t}x_i) = 0$, so $x_i = c_i e^{13t}$ and

$$\mathbf{x}(t) = e^{13t}\mathbf{x}_0 = c_1 e^{13t}\begin{pmatrix} 1 \\ 0 \end{pmatrix} + c_2 e^{13t}\begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

Phase portrait

$A = 13I$: every line through the origin is invariant and every solution moves along its own ray, $\mathbf{x}(t) = e^{13t}\mathbf{x}_0$, away from the origin (the origin is a **unstable star**). The picture is the same in every basis, so there is no separate canonical plane.



Unstable star: straight rays out of the origin.

(v) $A = \begin{pmatrix} 7 & -4 \\ 1 & 3 \end{pmatrix}$

Characteristic equation $\det(A - \lambda I) = 0$ (Topic 4, Eq. 27.8):

$$\det(A - \lambda I) = \begin{vmatrix} 7 - \lambda & -4 \\ 1 & 3 - \lambda \end{vmatrix} = (7 - \lambda)(3 - \lambda) - (-4)(1)$$

$$= \lambda^2 - (7 + 3)\lambda + (7 \cdot 3 - (-4) \cdot 1) = \lambda^2 - (10)\lambda + (25) = \lambda^2 - 10\lambda + 25$$

$\operatorname{tr} A = 10$, $\det A = 25$, discriminant $\Delta = (\operatorname{tr} A)^2 - 4 \det A = 100 - 100 = 0$.

$\Delta = 0$: the characteristic polynomial is a perfect square, $(\lambda - 5)^2 = \lambda^2 - 10\lambda + 25$, so

$$\lambda_1 = \lambda_2 = \lambda = \tfrac{1}{2}\operatorname{tr} A = 5 \quad (m_a = 2)$$

Geometric multiplicity: $N = A - \lambda I = \begin{pmatrix} 2 & -4 \\ 1 & -2 \end{pmatrix}$.

$N \neq O$ and $\det N = (2)(-2) - (-4)(1) = 0$, so N has rank 1: $m_g = 1$ and there is a single eigendirection (improper node). By Cayley-Hamilton $N^2 = O$ (Topic 4, Sec. 8.4):

$$\begin{aligned}(N^2)_{11} &= 2 \cdot 2 + (-4) \cdot 1 = 0 \\ (N^2)_{12} &= 2 \cdot (-4) + (-4) \cdot (-2) = 0 \\ (N^2)_{21} &= 1 \cdot 2 + (-2) \cdot 1 = 0 \\ (N^2)_{22} &= 1 \cdot (-4) + (-2) \cdot (-2) = 0\end{aligned}$$

Eigenvector

For $\lambda = 5$: $(A - (5)I)\mathbf{v_1} = \mathbf{0}$, that is

$$\begin{pmatrix} 2 & -4 \\ 1 & -2 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\begin{aligned}2v_1 - 4v_2 &= 0 \quad (\text{row } 1) \\ 1v_1 - 2v_2 &= 0 \quad (\text{row } 2)\end{aligned}$$

The coefficient matrix is singular ($\det(A - \lambda I) = 0$), so the rows are proportional (row 2 equals $\tfrac{1}{2}$ times row 1) and express a single condition. Row 1 reads $2v_1 + (-4)v_2 = 0$, so $v_1 = 2\,v_2$: every eigenvector is a multiple of $(2, 1)^T$. The smallest integer representative is

$$\mathbf{v_1} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$$

Check: $A\mathbf{v_1} = \begin{pmatrix} 10 \\ 5 \end{pmatrix} = 5\mathbf{v_1}$.

Generalised eigenvector

Solve $(A - \lambda I)\mathbf{v}_2 = \mathbf{v}_1$, i.e. $N\mathbf{v}_2 = \mathbf{v}_1$ (Eq. 8.2). The system is solvable because $\operatorname{im} N = \ker N$ (Sec. 8.4):

$$\begin{pmatrix} 2 & -4 \\ 1 & -2 \end{pmatrix} \begin{pmatrix} p \\ q \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$$

$$2p - 4q = 2$$
$$1p - 2q = 1$$

The two equations are proportional (the right-hand sides are in the same ratio as the rows, because $\mathbf{v}_1 \in \operatorname{im} N$), so one is enough. The solution is not unique: $\mathbf{v}_2 + s\mathbf{v}_1$ also works for every s, because $N\mathbf{v}_1 = \mathbf{0}$.

Take $q = 0$. Row 1 gives $2p = 2$, so $p = 1$:

$$\mathbf{v}_2 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

Check: $N\mathbf{v}_2 = \begin{pmatrix} 2 \\ 1 \end{pmatrix} = \mathbf{v}_1$, equivalently $A\mathbf{v}_2 = \lambda\mathbf{v}_2 + \mathbf{v}_1 = \begin{pmatrix} 7 \\ 1 \end{pmatrix}$ (Eq. 8.2).

Change of coordinates

$$Q = [\mathbf{v}_1 \mid \mathbf{v}_2] = \begin{pmatrix} 2 & 1 \\ 1 & 0 \end{pmatrix}$$

$\det Q = (2)(0) - (1)(1) = -1$, which is not zero, so Q is invertible. For a 2×2 matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad-bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$ (Robinson, Appendix B):

$$Q^{-1} = \frac{1}{-1}\begin{pmatrix} 0 & -1 \\ -1 & 2 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 1 & -2 \end{pmatrix}$$

$$A\mathbf{v}_1 = \begin{pmatrix} 10 \\ 5 \end{pmatrix} = 5\mathbf{v}_1, \qquad A\mathbf{v}_2 = \begin{pmatrix} 7 \\ 1 \end{pmatrix} = \mathbf{v}_1 + 5\mathbf{v}_2$$

$$AQ = \begin{pmatrix} 10 & 7 \\ 5 & 1 \end{pmatrix}$$

$$(Q^{-1}AQ)_{11} = 0 \cdot 10 + 1 \cdot 5 = 5$$
$$(Q^{-1}AQ)_{12} = 0 \cdot 7 + 1 \cdot 1 = 1$$
$$(Q^{-1}AQ)_{21} = 1 \cdot 10 + (-2) \cdot 5 = 0$$
$$(Q^{-1}AQ)_{22} = 1 \cdot 7 + (-2) \cdot 1 = 5$$

$$Q^{-1}AQ = \begin{pmatrix} 5 & 1 \\ 0 & 5 \end{pmatrix} = J = \begin{pmatrix} \lambda & 1 \\ 0 & \lambda \end{pmatrix}$$

Solution in the canonical plane

$$\dot{z}_1 = 5z_1 + z_2, \qquad \dot{z}_2 = 5z_2$$

Second equation: $\frac{d}{dt}(e^{-\lambda t}z_2) = e^{-\lambda t}(\dot{z}_2 - \lambda z_2) = 0$, so $e^{-\lambda t}z_2$ is constant: $z_2 = c_2 e^{5t}$. First equation with this z_2: $\dot{z}_1 - \lambda z_1 = c_2 e^{5t}$. The integrating factor is $e^{-\lambda t}$: by the product rule $\frac{d}{dt}(e^{-\lambda t}z_1) = e^{-\lambda t}(\dot{z}_1 - \lambda z_1) = c_2$. Barrow's rule between 0 and t:

$$\left[e^{-\lambda s}z_1(s)\right]_{s=0}^{s=t} = \int_0^t c_2\, ds = c_2 t \implies e^{-\lambda t}z_1(t) - z_1(0) = c_2 t$$

$$z_1(t) = (c_1 + c_2 t)e^{5t}, \qquad z_2(t) = c_2 e^{5t} \qquad (c_1 = z_1(0),\ c_2 = z_2(0))$$

Back to the original coordinates

$\mathbf{x} = Q\mathbf{z} = z_1\mathbf{v}_1 + z_2\mathbf{v}_2$, hence (Topic 4, Eq. 8.5)

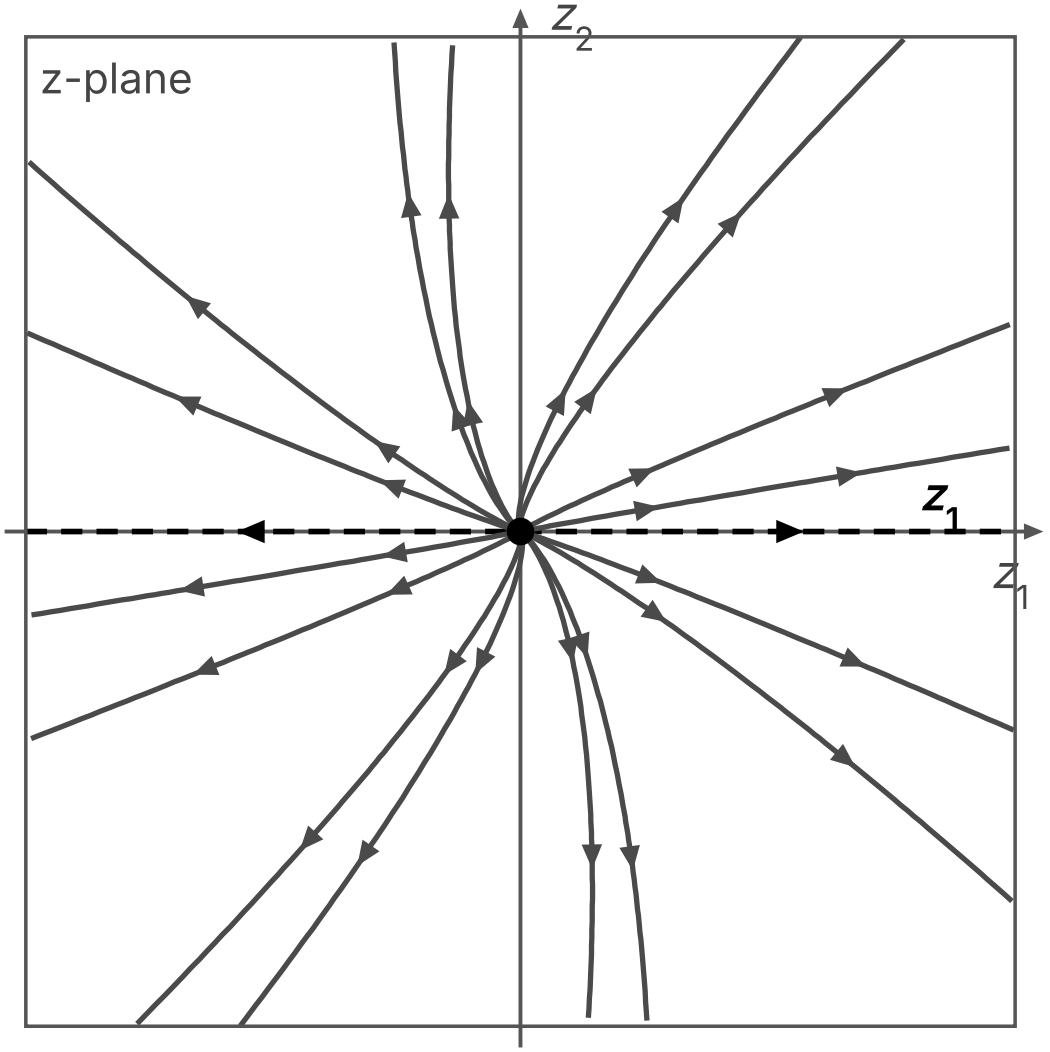
$$\mathbf{x}(t) = e^{5t}\left[c_1\begin{pmatrix} 2 \\ 1 \end{pmatrix} + c_2\left(t\begin{pmatrix} 2 \\ 1 \end{pmatrix} + \begin{pmatrix} 1 \\ 0 \end{pmatrix}\right)\right]$$

$$x(t) = 2c_1 e^{5t} + 2c_2 t e^{5t} + c_2 e^{5t}, \qquad y(t) = c_1 e^{5t} + c_2 t e^{5t}$$

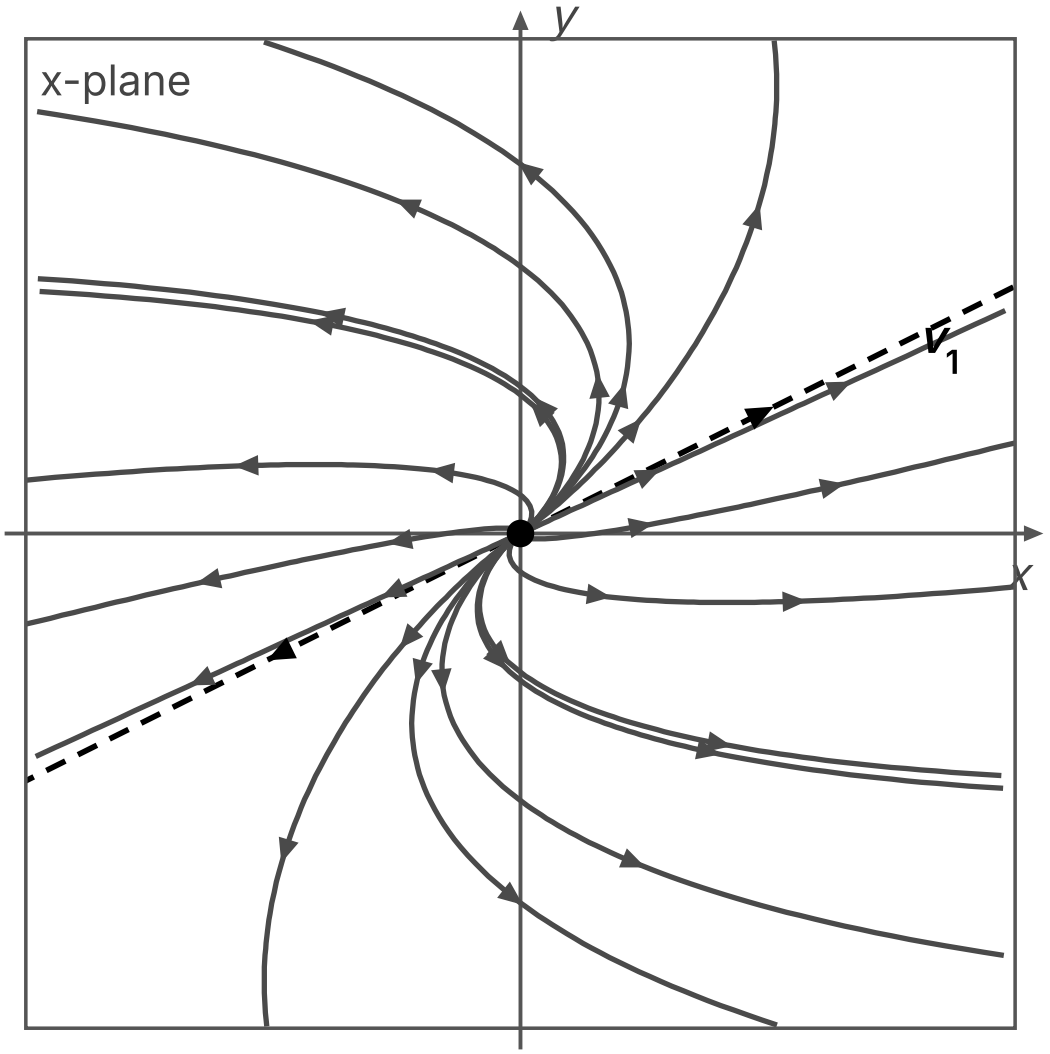
Phase portrait

$\lambda = 5$ is repeated with a single eigendirection and $A \neq \lambda I$: the origin is an **unstable improper node** (Topic 4 Sec. 8.8 and 8.9). In the $\mathbf{z}$-plane the z_1-axis is invariant ($z_2 \equiv 0$, $z_1 = c_1 e^{5t}$, arrows away from the origin); the other orbits satisfy $z_1 = z_2\left[K + \frac{1}{\lambda}\ln|z_2|\right]$, are S-shaped, tangent to the z_1-axis at the origin and parallel to it far away. Since $\dot{z}_1 = \lambda z_1 + z_2 = z_2$ on the z_2-axis, orbits cross it moving to the right in the upper half-plane and to the left in the lower one.

Orientation in the $\mathbf{x}$-plane, from $N\mathbf{x} = (A - \lambda I)\mathbf{x} = z_2\mathbf{v}_1$ (Topic 4 Sec. 8.9): at $(0,1)^T$, $N(0,1)^T = \begin{pmatrix} -4 \\ -2 \end{pmatrix} = -2\,\mathbf{v}_1$, so there the orbit drifts along $-\mathbf{v}_1$. The orbits are tangent to the eigenline at the origin and bend into an S around it.



Canonical $\mathbf{z}$-plane (unstable improper node).



Original (x, y)-plane (unstable improper node); dashed: the eigenline with $\mathbf{v}_1 = (2, 1)$.

PHASE 4: INTERPRETATION AND CHECKS

Item	λ	Type
(i)	3	unstable improper node
(ii)	-4	stable improper node
(iii)	-4	stable improper node
(iv)	13	unstable star
(v)	5	unstable improper node

- (i) is exactly the worked Example 8.1 of Topic 4 (Robinson Example 30.1): $\lambda = 3$, $\mathbf{v}_1 = (2, 1)^T$, $\mathbf{v}_2 = (1, 0)^T$ and $J = \begin{pmatrix} 3 & 1 \\ 0 & 3 \end{pmatrix}$ – the result reproduces it step by step.

- In each improper node $Q^{-1}AQ = J$ was multiplied out entry by entry; $N^2 = O$ was checked (Cayley-Hamilton).
- The general solution $\mathbf{x} = e^{\lambda t}[c_1\mathbf{v}_1 + c_2(t\mathbf{v}_1 + \mathbf{v}_2)]$ contains the secular term $te^{\lambda t}\mathbf{v}_1$, which makes all orbits tangent to $\mathbf{v}_1$ at both ends (S-shaped).
- (iv) is a multiple of the identity: every vector is an eigenvector and the orbits are rays; the canonical form is A itself, and any basis (for instance $Q = I$) works.
- A repeated eigenvalue is a borderline case between nodes ($\Delta > 0$) and spirals ($\Delta < 0$).

EXERCISE 7

The Nilpotent Case: Zero as a Repeated Eigenvalue

Original statement · probls_ch4-ii_2627.pdf, p. 2

Nilpotent Matrix • Shear • Line of Equilibria

STATEMENT

Suppose that a non-zero matrix A has zero as a repeated eigenvalue, with eigenvector $\mathbf{v}$. Then we can change to coordinates referred to $\mathbf{v}_2$ and $\mathbf{v}_1 = A\mathbf{v}_2$, where $\mathbf{v}_2$ is any vector in a different direction to $\mathbf{v}$, so that $\tilde{\mathbf{x}} = \tilde{x}\mathbf{v}_1 + \tilde{y}\mathbf{v}_2$. The equation $d\mathbf{x}/dt = A\mathbf{x}$ becomes $\dfrac{d\tilde{\mathbf{x}}}{dt} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \tilde{\mathbf{x}}$, hence,

$$\frac{d\tilde{x}}{dt} = \tilde{y} \quad \text{and} \quad \frac{d\tilde{y}}{dt} = 0.$$

- (i) Solve the equations for $\tilde{x}(t)$ and $\tilde{y}(t)$, and hence write down the general solution for $\mathbf{x}(t)$.
- (ii) Draw the phase diagram in the $(\tilde{x}, \tilde{y})$ and (x, y) planes.
- (iii) Draw the phase diagram for the equation $\dfrac{d\mathbf{x}}{dt} = \begin{pmatrix} -1 & 1 \\ -1 & 1 \end{pmatrix} \mathbf{x}$.

PHASE 1: SETUP AND HYPOTHESES

$A \neq O$ is 2×2 with the double eigenvalue 0: $\operatorname{tr} A = 0$, $\det A = 0$, so by Cayley-Hamilton $A^2 = O$ ($A^2 - (\operatorname{tr} A)A + (\det A)I = O$). Since $A \neq O$, $\ker A$ is one-dimensional, spanned by $\mathbf{v}$: $m_g = 1$ (Topic 4 Sec. 8.4 with $\lambda = 0$, $N = A$).

Why the proposed basis works: take $\mathbf{v}_2 \notin \operatorname{span}\{\mathbf{v}\}$. Then $A\mathbf{v}_2 \neq \mathbf{0}$ (otherwise $\mathbf{v}_2$ would be in $\ker A$), and $\mathbf{v}_1 = A\mathbf{v}_2$ satisfies $A\mathbf{v}_1 = A^2\mathbf{v}_2 = \mathbf{0}$, so $\mathbf{v}_1 \in \ker A = \operatorname{span}\{\mathbf{v}\}$ and $\mathbf{v}_1 \neq \mathbf{0}$ is an eigenvector. $\mathbf{v}_1, \mathbf{v}_2$ are independent (not parallel), and

$$A\mathbf{v}_1 = \mathbf{0} = 0 \cdot \mathbf{v}_1 + 0 \cdot \mathbf{v}_2, \qquad A\mathbf{v}_2 = \mathbf{v}_1 = 1 \cdot \mathbf{v}_1 + 0 \cdot \mathbf{v}_2 \implies Q^{-1}AQ = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$

PHASE 2: FORMULATION

With $Q = [\mathbf{v}_1 \mid \mathbf{v}_2]$ and $\mathbf{x} = Q\tilde{\mathbf{x}} = \tilde{x}\mathbf{v}_1 + \tilde{y}\mathbf{v}_2$ the canonical system is $\dot{\tilde{x}} = \tilde{y}$, $\dot{\tilde{y}} = 0$ (the Jordan block of Topic 4 Eq. 8.3 with $\lambda = 0$).

METHOD OF THE CLASS NOTES [NOTES, P. 10]

Case 3.2 of the notes with $\lambda = 0$: $\mathbf{v}_1 = (A - 0I)\mathbf{v}_2 = A\mathbf{v}_2$, exactly the change of coordinates of the statement. The solution of the notes becomes $z_2 = C$, $z_1 = Ct + D$ and $\mathbf{x} = (Ct + D)\mathbf{v}_1 + C\mathbf{v}_2$: horizontal lines in the $\tilde{\mathbf{x}}$-plane, with the line of $\mathbf{v}_1$ ($C = 0$) made of equilibria.

PHASE 3: DERIVATION

(i) Solution

The second equation $\dot{\tilde{y}} = 0$ integrates to $\tilde{y}(t) = c_2$ (a constant). Substituting in the first, $\dot{\tilde{x}} = c_2$, and Barrow's rule between 0 and t gives $\tilde{x}(t) - \tilde{x}(0) = \int_0^t c_2\, ds = c_2 t$:

$$\tilde{x}(t) = c_1 + c_2 t, \qquad \tilde{y}(t) = c_2 \qquad (c_1 = \tilde{x}(0),\ c_2 = \tilde{y}(0))$$

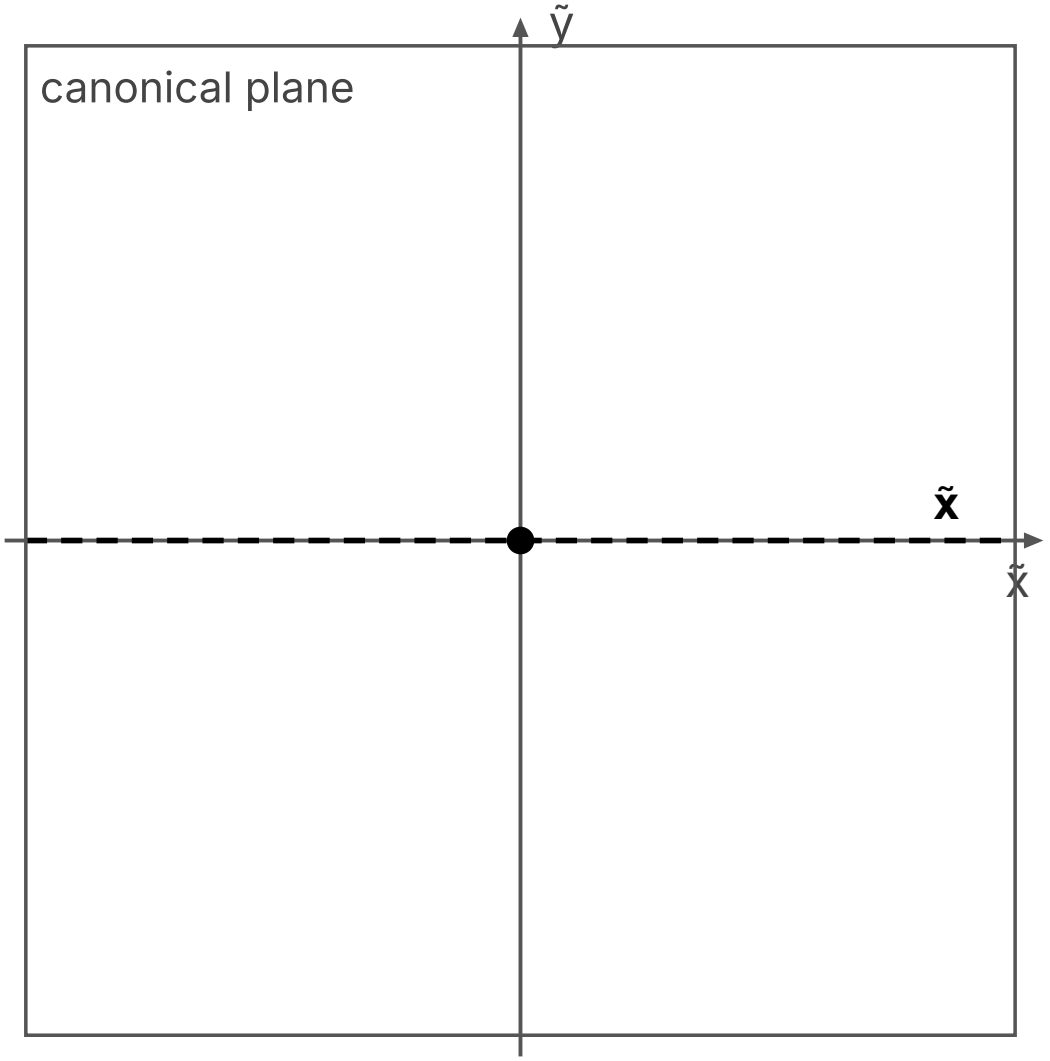
Back to the original coordinates, $\mathbf{x} = \tilde{x}\mathbf{v}_1 + \tilde{y}\mathbf{v}_2$:

$$\boxed{\mathbf{x}(t) = (c_1 + c_2 t)\,\mathbf{v}_1 + c_2\,\mathbf{v}_2}$$

Check: $\dot{\mathbf{x}} = c_2\mathbf{v}_1$ and $A\mathbf{x} = (c_1 + c_2 t)A\mathbf{v}_1 + c_2 A\mathbf{v}_2 = \mathbf{0} + c_2\mathbf{v}_1$. They agree. Unlike the case $\lambda \neq 0$ there is no exponential: the secular term grows only **linearly**.

(ii) Phase diagrams

In the $(\tilde{x}, \tilde{y})$ plane the orbits are the **horizontal lines** $\tilde{y} = c_2$ (since $\tilde{y}$ is constant). On the line $\tilde{y} = c_2 > 0$ the abscissa $\tilde{x} = c_1 + c_2 t$ increases at constant speed c_2: motion to the right; for $c_2 < 0$ to the left. The $\tilde{x}$-axis ($c_2 = 0$) consists of equilibria. There are no curved orbits: the speed is proportional to the height $\tilde{y}$ (a *shear* flow).



$(\tilde{x}, \tilde{y})$-plane: horizontal lines $\tilde{y} = c_2$ traversed to the right above the axis and to the left below it; the $\tilde{x}$-axis is all equilibria.

In the (x, y) plane, $\mathbf{x} = \tilde{x}\mathbf{v}_1 + \tilde{y}\mathbf{v}_2$ maps horizontal lines to **lines parallel to $\mathbf{v}_1$**, offset from the line $\operatorname{span}\{\mathbf{v}_1\}$ of equilibria by $c_2\mathbf{v}_2$; they are traversed along $+\mathbf{v}_1$ on the side of $\operatorname{span}\{\mathbf{v}_1\}$ where $\mathbf{v}_2$ points and along $-\mathbf{v}_1$ on the other side. The (x, y)-diagram of the following item shows a concrete example.

(iii) The matrix $\begin{pmatrix} -1 & 1 \\ -1 & 1 \end{pmatrix}$

$\operatorname{tr} A = -1 + 1 = 0$ and $\det A = (-1)(1) - (1)(-1) = 0$, so the characteristic equation is $\lambda^2 = 0$: the double eigenvalue 0, and $A \neq O$. Check $A^2 = O$:

$$(A^2)_{11} = (-1)\cdot(-1) + 1\cdot(-1) = 0$$
$$(A^2)_{12} = (-1)\cdot 1 + 1\cdot 1 = 0$$
$$(A^2)_{21} = (-1)\cdot(-1) + 1\cdot(-1) = 0$$
$$(A^2)_{22} = (-1)\cdot 1 + 1\cdot 1 = 0$$

Eigenvector: $A\mathbf{v} = \mathbf{0}$ gives $-v_1 + v_2 = 0$ (both rows are the same equation up to sign), so $\mathbf{v} = (1,1)^T$. Following the exercise, take any $\mathbf{v}_2$ not parallel to $\mathbf{v}$, for instance $\mathbf{v}_2 = (1,0)^T$, and set $\mathbf{v}_1 = A\mathbf{v}_2$:

$$\mathbf{v}_1 = A\mathbf{v}_2 = \begin{pmatrix} -1 & 1 \\ -1 & 1 \end{pmatrix}\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} -1 \\ -1 \end{pmatrix}$$

$\mathbf{v}_1 = (-1,-1)^T$ is indeed an eigenvector ($A\mathbf{v}_1 = (1-1, 1-1)^T = \mathbf{0}$), a multiple of $\mathbf{v}$.

$$Q = [\mathbf{v}_1 \mid \mathbf{v}_2] = \begin{pmatrix} -1 & 1 \\ -1 & 0 \end{pmatrix}$$

$\det Q = (-1)(0) - (1)(-1) = 1$, which is not zero, so Q is invertible. For a 2×2 matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad-bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$ (Robinson, Appendix B):

$$Q^{-1} = \frac{1}{1}\begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix}$$

$$AQ = [A\mathbf{v}_1 \mid A\mathbf{v}_2] = [\mathbf{0} \mid \mathbf{v}_1] = \begin{pmatrix} 0 & -1 \\ 0 & -1 \end{pmatrix}$$

$$(Q^{-1}AQ)_{11} = 0\cdot 0 + (-1)\cdot 0 = 0$$
$$(Q^{-1}AQ)_{12} = 0\cdot(-1) + (-1)\cdot(-1) = 1$$
$$(Q^{-1}AQ)_{21} = 1\cdot 0 + (-1)\cdot 0 = 0$$
$$(Q^{-1}AQ)_{22} = 1\cdot(-1) + (-1)\cdot(-1) = 0$$

$$Q^{-1}AQ = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$

By part (i) the general solution is $\mathbf{x}(t) = (c_1 + c_2 t)\mathbf{v}_1 + c_2\mathbf{v}_2$:

$$x(t) = -c_1 - c_2 t + c_2, \qquad y(t) = -c_1 - c_2 t$$

Check: $\dot{x} = -c_2$ and $-x + y = -c_2$; $\dot{y} = -c_2$ and $-x + y$ again: both equations $\dot{x} = -x + y$, $\dot{y} = -x + y$ hold.

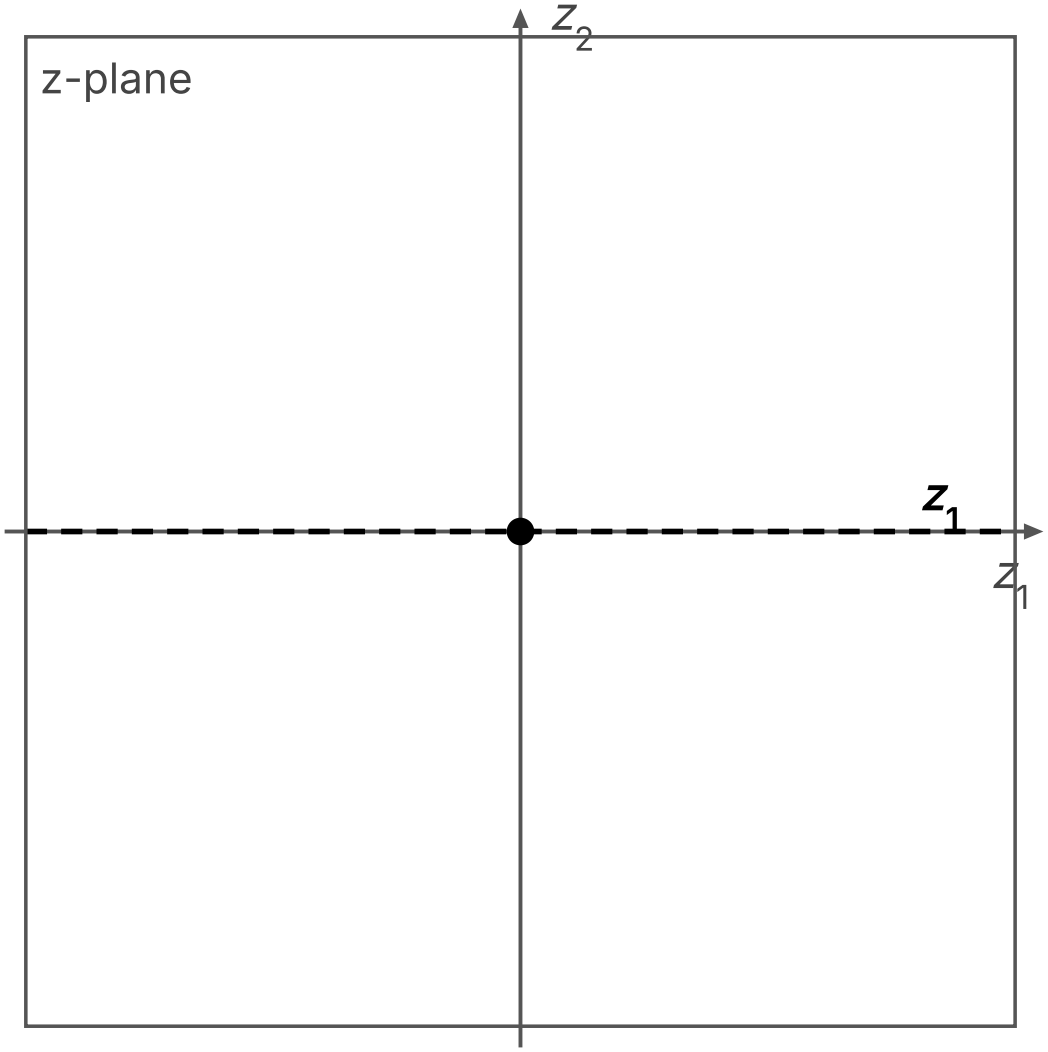
The coordinates are $\tilde{\mathbf{x}} = Q^{-1}\mathbf{x}$: $\tilde{x} = -y$, $\tilde{y} = x - y$ (first row $(0, -1)$, second row $(1, -1)$ of Q^{-1}). Every orbit is a straight line parallel to $\mathbf{v} = (1, 1)$; the line $y = x$ ($\tilde{y} = 0$) consists of equilibria.

Direction: $\dot{\mathbf{x}} = (y - x)(1, 1)^T$. Above the line $y = x$ ($y > x$) the motion is along $+(1, 1)$ (up and to the right); below it, along $-(1, 1)$ (down and to the left). The speed $|y - x|\sqrt{2}$ grows with the distance to the line of equilibria.

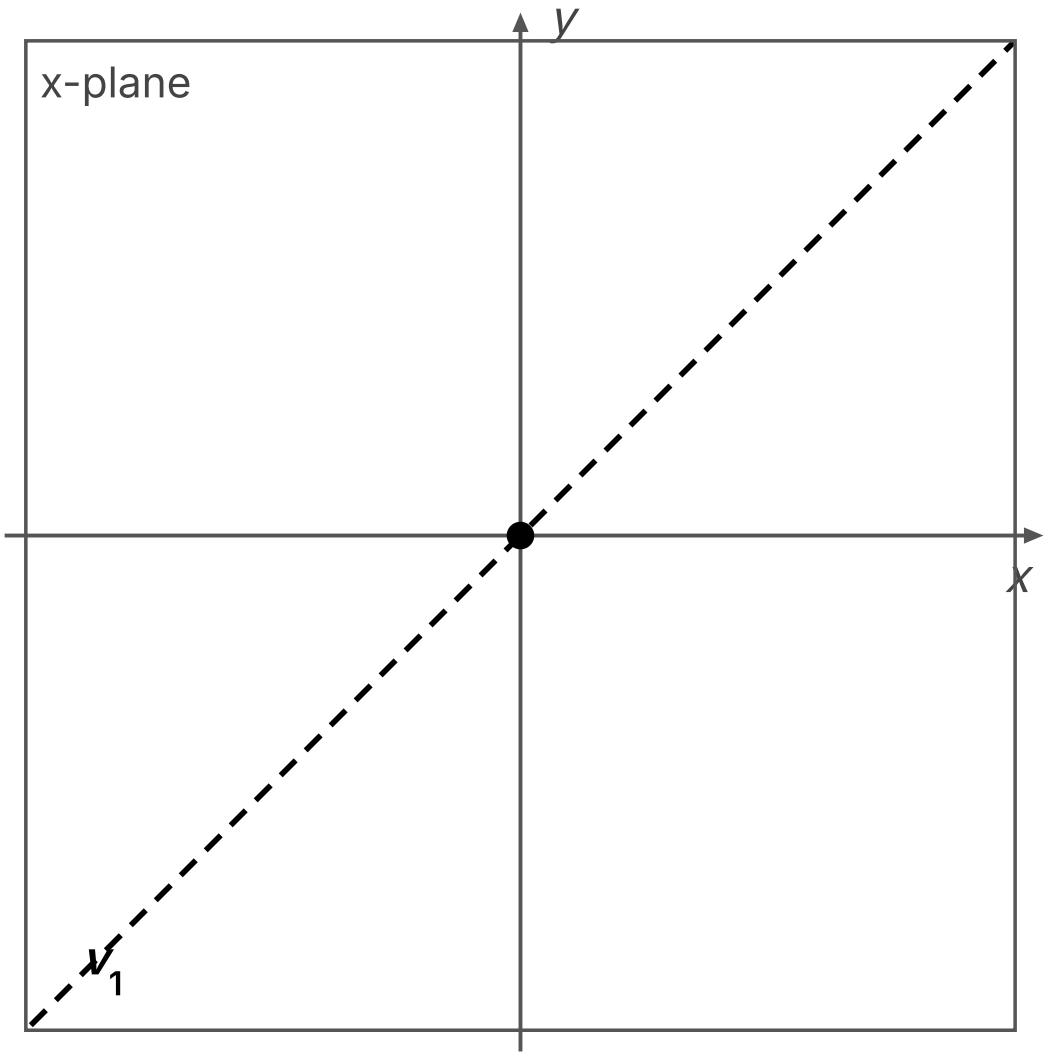
Phase portrait

$\lambda = 0$: every point of the eigenline through $\mathbf{v}_1 = (-1, -1)$ is an equilibrium. In the $\mathbf{z}$-plane $z_2 = c_2$ is constant and $z_1 = c_1 + c_2 t$ moves with constant speed c_2: horizontal lines, to the right if $z_2 > 0$ and to the left if $z_2 < 0$. Under $\mathbf{x} = Q\mathbf{z}$ they become lines parallel to $\mathbf{v}_1$: on the side of the eigenline where $\mathbf{v}_2$ points (for $z_2 > 0$) they are traversed along $+\mathbf{v}_1$, on the opposite side along $-\mathbf{v}_1$.

Orientation in the $\mathbf{x}$-plane, from $N\mathbf{x} = (A - \lambda I)\mathbf{x} = z_2\mathbf{v}_1$ (Topic 4 Sec. 8.9): at $(0, 1)^T$, $N(0, 1)^T = \begin{pmatrix} 1 \\ 1 \end{pmatrix} = -1\,\mathbf{v}_1$, so there the orbit drifts along $-\mathbf{v}_1$. The orbits are the straight lines parallel to $\mathbf{v}_1$.



Canonical $\mathbf{z}$-plane (line of equilibria (nilpotent shear)).



Original (x, y)-plane (line of equilibria (nilpotent shear)); dashed: the eigenline with $\mathbf{v}_1 = (-1, -1)$.

PHASE 4: INTERPRETATION AND CHECKS

- This is the limit $\lambda \to 0$ of the improper node $J = \begin{pmatrix} \lambda & 1 \\ 0 & \lambda \end{pmatrix}$: the factor $e^{\lambda t}$ disappears and only the shear remains (the origin of the trace-determinant plane): the flow is $\mathbf{x}(t) = \mathbf{x}_0 + t\,A\mathbf{x}_0$ (the series of e^{At} stops after the linear term because $A^2 = O$): in (iii), $\mathbf{x}(t) = \mathbf{x}_0 + t(y_0 - x_0)(1, 1)^T$, which matches the direction and speed found.
- Check at a point: $\mathbf{x}_0 = (1, 0)$: formula gives $\mathbf{x}(t) = (1, 0) + t(0 - 1)(1, 1) = (1 - t, -t)$, and the general solution with $c_2 = \tilde{y}(0) = x_0 - y_0 = 1$, $c_1 = \tilde{x}(0) = -y_0 = 0$ gives $x = -(0 + t)(1) + 1 = 1 - t$ and $y = -t$. They coincide.
- $\operatorname{tr} A = \det A = 0$ is the origin of the trace-determinant plane (Topic 4 Sec. 9.1), where all the portrait types meet.
- The solutions are polynomials in t, so every orbit not on the line of equilibria leaves to infinity linearly in time in both directions.

EXERCISE 8

Nine Qualitative Phase Portraits from Partial Data

Original statement · probls_ch4-ii_2627.pdf, p. 2

Qualitative Portraits • Classification • Nine Cases

STATEMENT

Draw the phase portrait for the equation $d\mathbf{x}/dt = A\mathbf{x}$ when the eigenvalues and eigenvectors of A are as given:

- (i) $\lambda_1 = 3$ with $\mathbf{v}_1 = (1, 1)$ and $\lambda_2 = -2$ with $\mathbf{v}_2 = (1, -2)$;
- (ii) complex conjugate eigenvalues $\lambda_{\pm} = -1 \pm 3i$, with $\dot{x} < 0$ when $x = 0$ and $y > 0$;
- (iii) a single eigenvalue $\lambda = 13$ with eigenvector $(3, 2)$, and $\dot{x} > 0$ when $x = 0$ and $y > 0$;
- (iv) $\lambda_1 = -2$ with $\mathbf{v}_1 = (2, 1)$ and $\lambda_2 = -3$ with $\mathbf{v}_2 = (1, -1)$;
- (v) a single eigenvalue $\lambda = -3$ with eigenvector $(1, -1)$, and $\dot{x} > 0$ when $x = 0$ and $y > 0$;
- (vi) $\lambda_{\pm} = \pm 2i$, where $\dot{y} < 0$ when $y = 0$ and $x > 0$;
- (vii) $\lambda_1 = 1$ with $\mathbf{v}_1 = (3, 2)$ and $\lambda_2 = 5$ with $\mathbf{v}_2 = (1, -4)$;
- (viii) $\lambda_{\pm} = 5 \pm i$, and $\dot{y} > 0$ when $y = 0$ and $x > 0$;
- (ix) a single eigenvalue $\lambda = -7$, with the matrix A being a multiple of the identity.

PHASE 1: SETUP: WHAT THE DATA DETERMINE

The data fix the **type** of the origin and the position of the eigenlines, which is all that the phase portrait of a linear system needs (Topic 4 Sec. 9, Robinson Ch. 31). A matrix consistent with the data is built in each item, to draw the portrait and to check the sign information numerically; any other matrix with the same data has the same portrait up to a linear distortion of the plane.

Data	Origin is a	Extra information used
two real eigenvalues, opposite signs	saddle	eigenlines; stable = negative eigenvalue
two real eigenvalues, same sign	node	tangent to the eigenvector of the smaller $\lvert\lambda\rvert$; stable iff $\lambda < 0$
complex $\rho \pm i\omega$, $\rho \neq 0$	spiral	stable iff $\rho < 0$; sense from the sign of $\dot{x}$ at $(0, 1)$ or $\dot{y}$ at $(1, 0)$
imaginary $\pm i\omega$	centre	closed orbits; sense from the sign condition
single eigenvalue, one eigenvector	improper node	S-shaped orbits along the eigenline; side/orientation from the sign condition

Data	Origin is a	Extra information used
single eigenvalue, $A = \lambda I$	star	straight rays

PHASE 2: FORMULATION: HOW THE SIGN CONDITIONS ARE READ

At the point $(0, 1)$ the vector field is $A(0,1)^T = (a_{12}, a_{22})^T$, so the sign of $\dot{x}$ there is the sign of a_{12}: positive means motion to the right at the top of the plane, which is clockwise; negative means counter-clockwise. At $(1, 0)$ the field is $(a_{11}, a_{21})^T$, so the sign of $\dot{y}$ there is the sign of a_{21}: negative means downwards on the right-hand side, i.e. clockwise; positive means counter-clockwise (Topic 4 Sec. 7.8).

METHOD OF THE CLASS NOTES [NOTES, PP. 9-10]

The classification of the notes decides each item. Two real eigenvalues (Case 1) give a node or a saddle, read from the sign table of λ_2/λ_1. A complex pair (Case 2) gives a spiral or a centre from the sign of ρ, with the sense of rotation read from the sign of $\dot{x}$ or $\dot{y}$ on an axis, as in the example of the notes. A repeated eigenvalue (Case 3) gives a star when $A = \lambda I$ and an improper node, with one eigenvector, otherwise.

PHASE 3: DERIVATION OF THE NINE PORTRAITS

(i) $\lambda_1 = 3$ with $\mathbf{v}_1 = (1, 1)$ and $\lambda_2 = -2$ with $\mathbf{v}_2 = (1, -2)$

Two real eigenvalues with independent eigenvectors ($\det[\mathbf{v}_1\ \mathbf{v}_2] = -3 \neq 0$). A consistent matrix is $A = Q\Lambda Q^{-1} = \begin{pmatrix} \frac{4}{3} & \frac{5}{3} \\ \frac{10}{3} & -\frac{1}{3} \end{pmatrix}$ (checked: $A\mathbf{v}_1 = \begin{pmatrix} 3 \\ 3 \end{pmatrix} = 3\mathbf{v}_1$, $A\mathbf{v}_2 = \begin{pmatrix} -2 \\ 4 \end{pmatrix} = -2\mathbf{v}_2$).

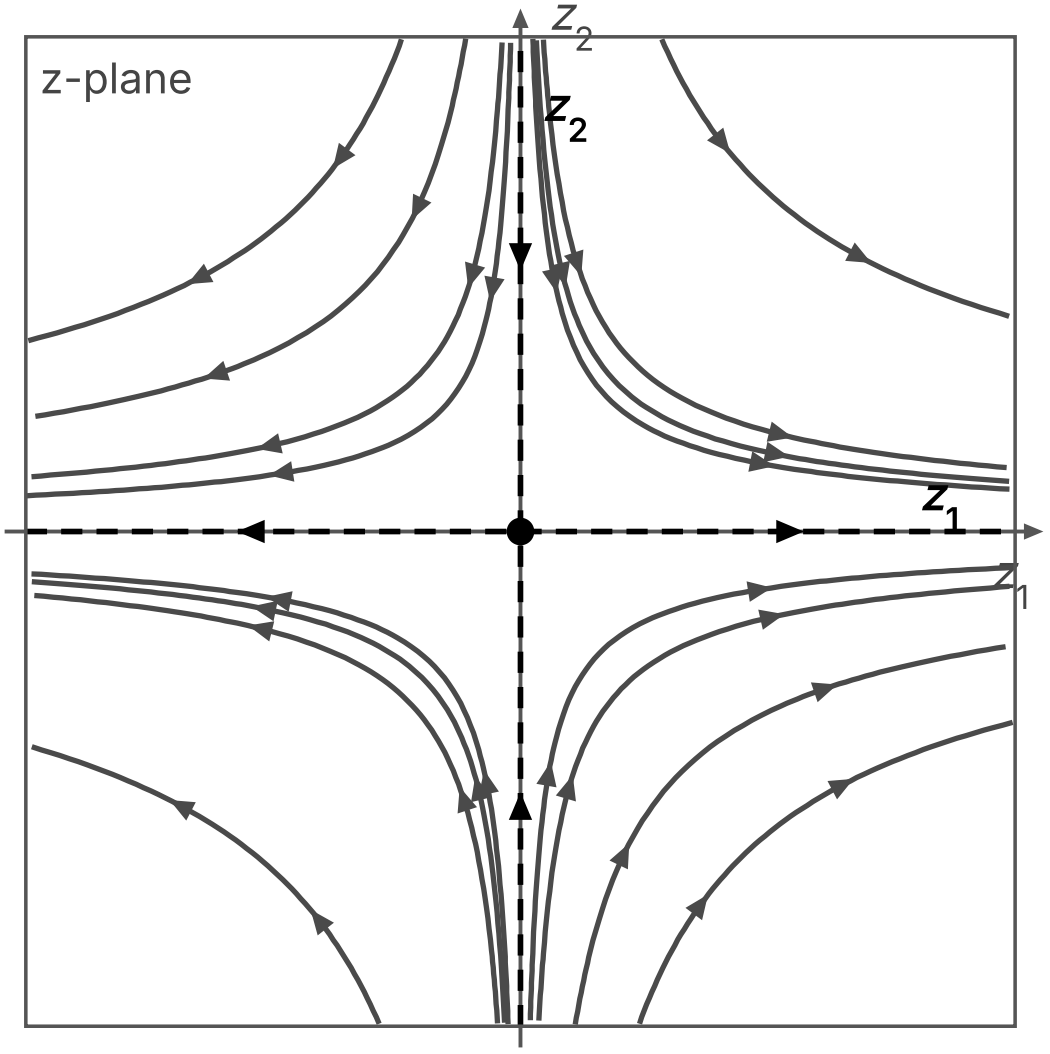
Phase portrait

$\lambda_1 = 3$ and $\lambda_2 = -2$ are of opposite signs: the origin is a **saddle** (Robinson Ch. 28, Topic 4 Sec. 6 and 9).

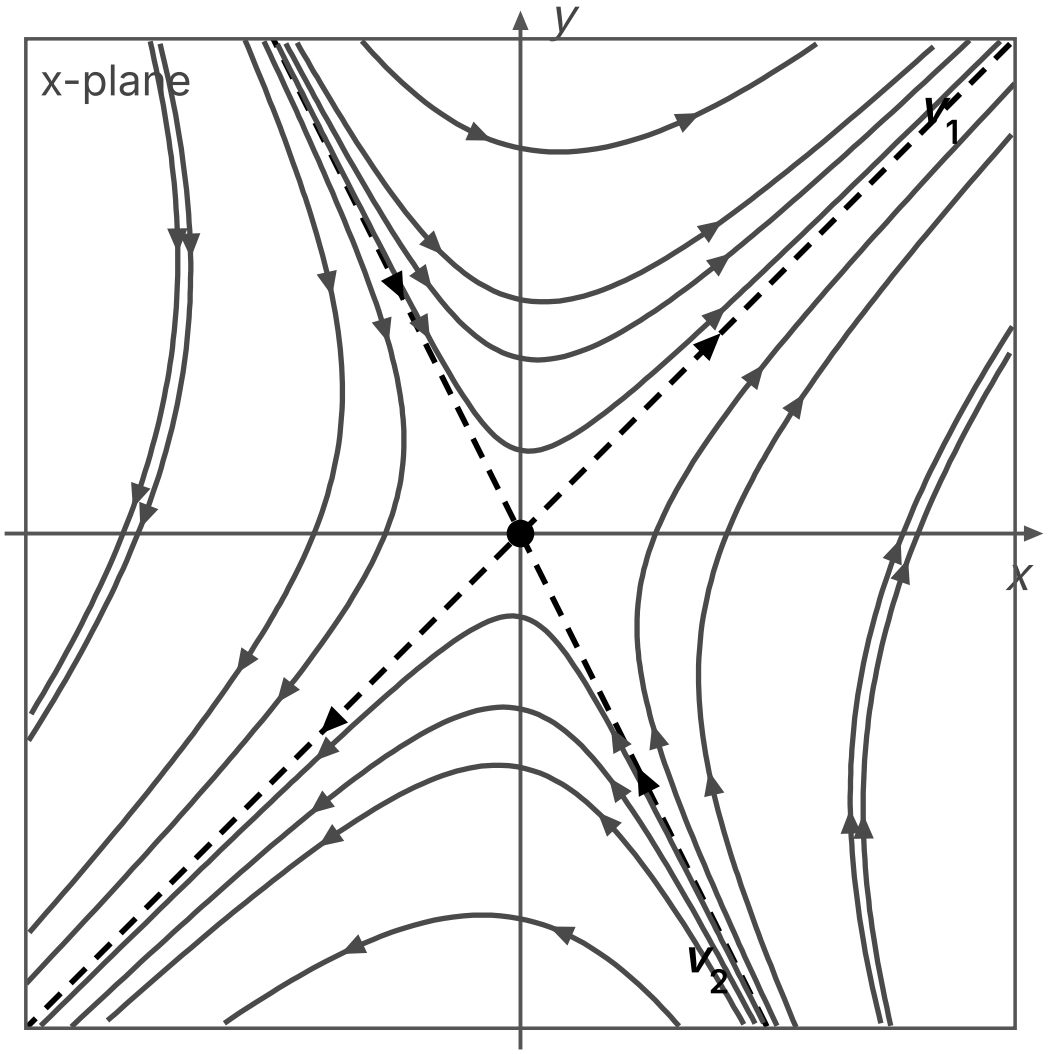
In the canonical $\mathbf{z}$-plane, eliminating t from $z_1 = c_1 e^{3t}$ and $z_2 = c_2 e^{-2t}$ (for $c_1 \neq 0$): $|z_1|/|c_1| = e^{\lambda_1 t}$, so $e^t = (|z_1|/|c_1|)^{1/\lambda_1}$ and $|z_2| = |c_2| e^{\lambda_2 t} = |c_2| \left(|z_1|/|c_1|\right)^{\lambda_2/\lambda_1}$, i.e. $z_2 = K\,|z_1|^{\lambda_2/\lambda_1} = K\,|z_1|^{-\frac{2}{3}}$ with $K = c_2 |c_1|^{-\lambda_2/\lambda_1}$ (equivalently, integrate the separable equation $\frac{dz_2}{dz_1} = \frac{\lambda_2 z_2}{\lambda_1 z_1}$ by the chain rule, as in Exercise 10). The axes $z_1 = 0$ and $z_2 = 0$ are invariant: a solution that starts on an axis stays on it.

The exponent $\lambda_2/\lambda_1 = -\frac{2}{3}$ is negative, so the orbits are hyperbola-like. Only solutions starting on the eigenline of the negative eigenvalue, the line through $(1, -2)$ (the **stable manifold**), reach the origin as $t \to +\infty$; the line through $(1, 1)$ (positive eigenvalue) is the **unstable manifold**. All other orbits come in close to the stable line, then leave along the unstable one.

The map $\mathbf{x} = Q\mathbf{z}$ sends the z_1-axis to the line through $\mathbf{v}_1 = (1, 1)$ and the z_2-axis to the line through $\mathbf{v}_2 = (1, -2)$. Straight lines stay straight and the origin stays fixed, so the $\mathbf{x}$-portrait is the canonical one sheared onto the eigenlines. On $\mathbf{v}_1$: $\dot{\mathbf{x}} = A\mathbf{v}_1 = 3\mathbf{v}_1$ points outward; on $\mathbf{v}_2$: $\dot{\mathbf{x}} = -2\mathbf{v}_2$ points inward.



Canonical $\mathbf{z}$-plane (saddle): invariant axes and orbits $z_2 = K|z_1|^{-\frac{2}{3}}$.



Original (x, y)-plane (saddle): dashed lines are the eigenlines $\mathbf{v}_1 = (1, 1)$ and $\mathbf{v}_2 = (1, -2)$.

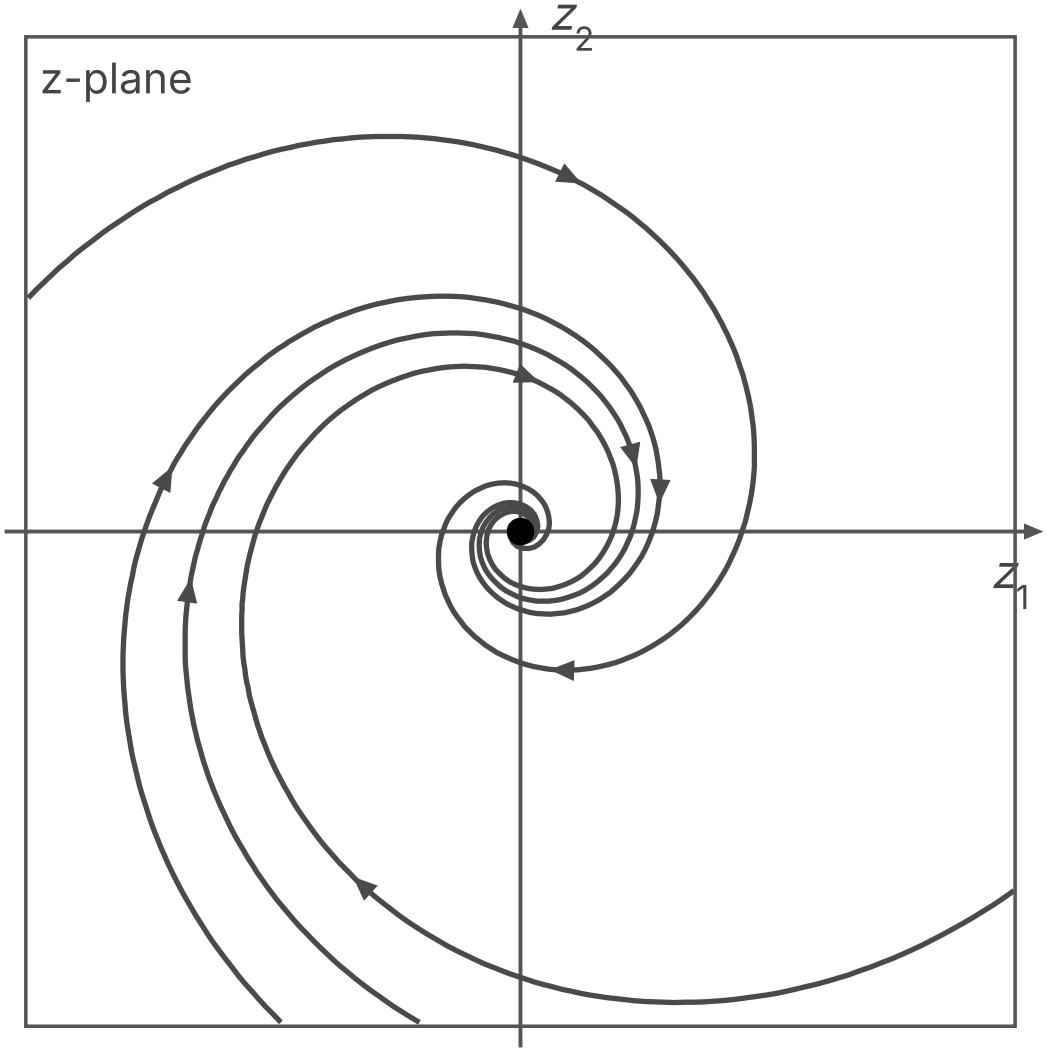
(ii) complex conjugate eigenvalues $\lambda_\pm = -1 \pm 3i$, with $\dot{x} < 0$ when $x = 0$ and $y > 0$

$\dot{x} < 0$ at $(0, 1)$ gives $a_{12} < 0$: leftwards at the top of the plane, i.e. **counter-clockwise**. A consistent matrix with eigenvalues $-1 \pm 3i$ is $A = \begin{pmatrix} -1 & -3 \\ 3 & -1 \end{pmatrix}$ (tr $A = -2$, det $A = 10$, $\lambda = -1 \pm 3i$; $a_{12} = -3 < 0$).

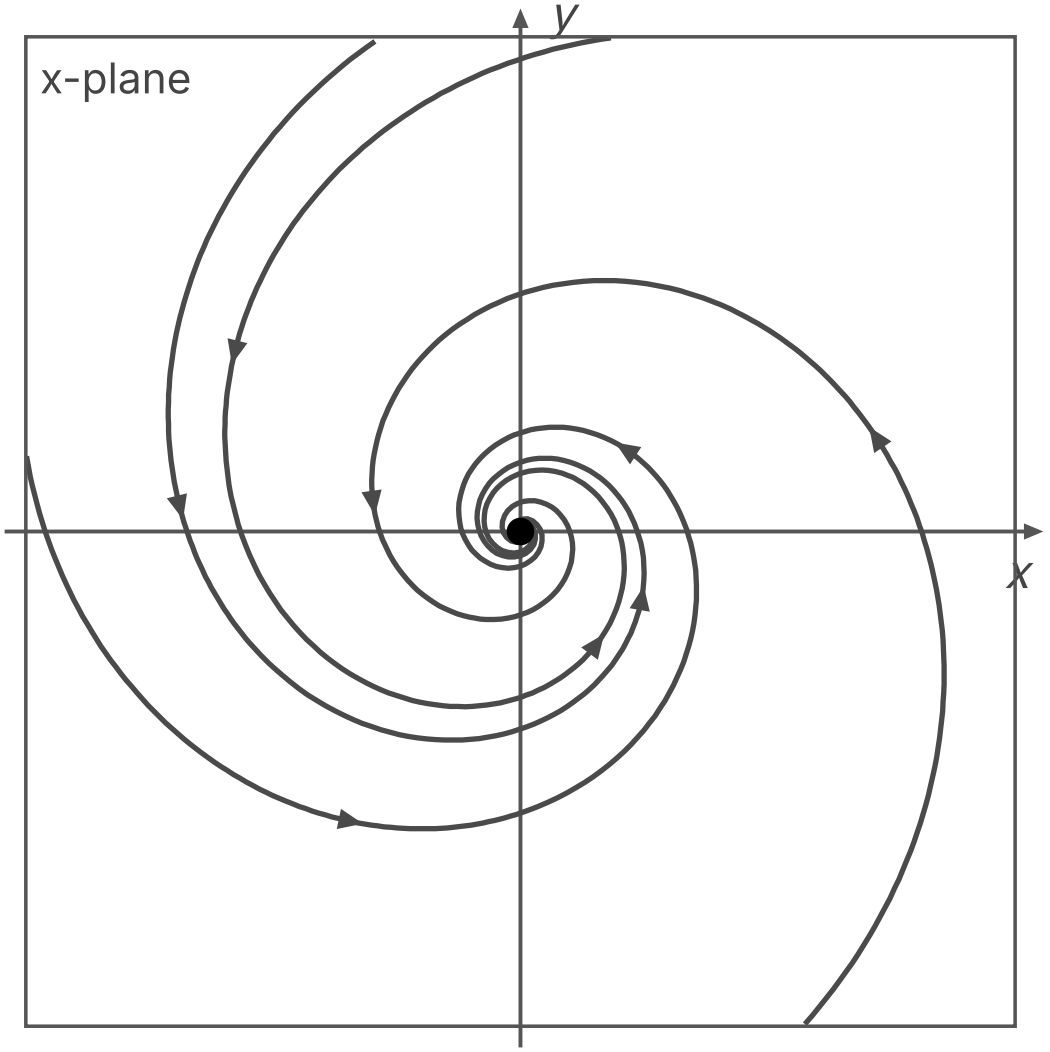
Phase portrait

$\rho = -1$ and $\omega = 3$: the origin is a **stable spiral**. In the $\mathbf{z}$-plane $r = r_0 e^{-1t}$ and $\theta = \theta_0 - 3t$: the motion is always **clockwise** ($\dot{\theta} = -\omega < 0$), and one full turn takes $T = 2\pi/\omega$ while the radius is multiplied by $e^{2\pi\rho/\omega} = e^{-\frac{2\pi}{3}}$ (shrinking).

Sense of rotation in the $\mathbf{x}$-plane, tested directly on the vector field (Topic 4 Sec. 7.8): at $(0, 1)^T$, $\dot{\mathbf{x}} = A(0, 1)^T = \begin{pmatrix} -3 \\ -1 \end{pmatrix}$, so $\dot{x} = -3$ is < 0: to the left at the top of the plane: the rotation is **counter-clockwise**. Cross-check: det $Q = -9$ is < 0, so Q reverses orientation and the clockwise motion of the $\mathbf{z}$-plane becomes counter-clockwise in the $\mathbf{x}$-plane, in agreement.



Canonical $\mathbf{z}$-plane (stable spiral): clockwise, $\dot{\theta} = -\omega$.



Original (x, y)-plane (stable spiral), counter-clockwise.

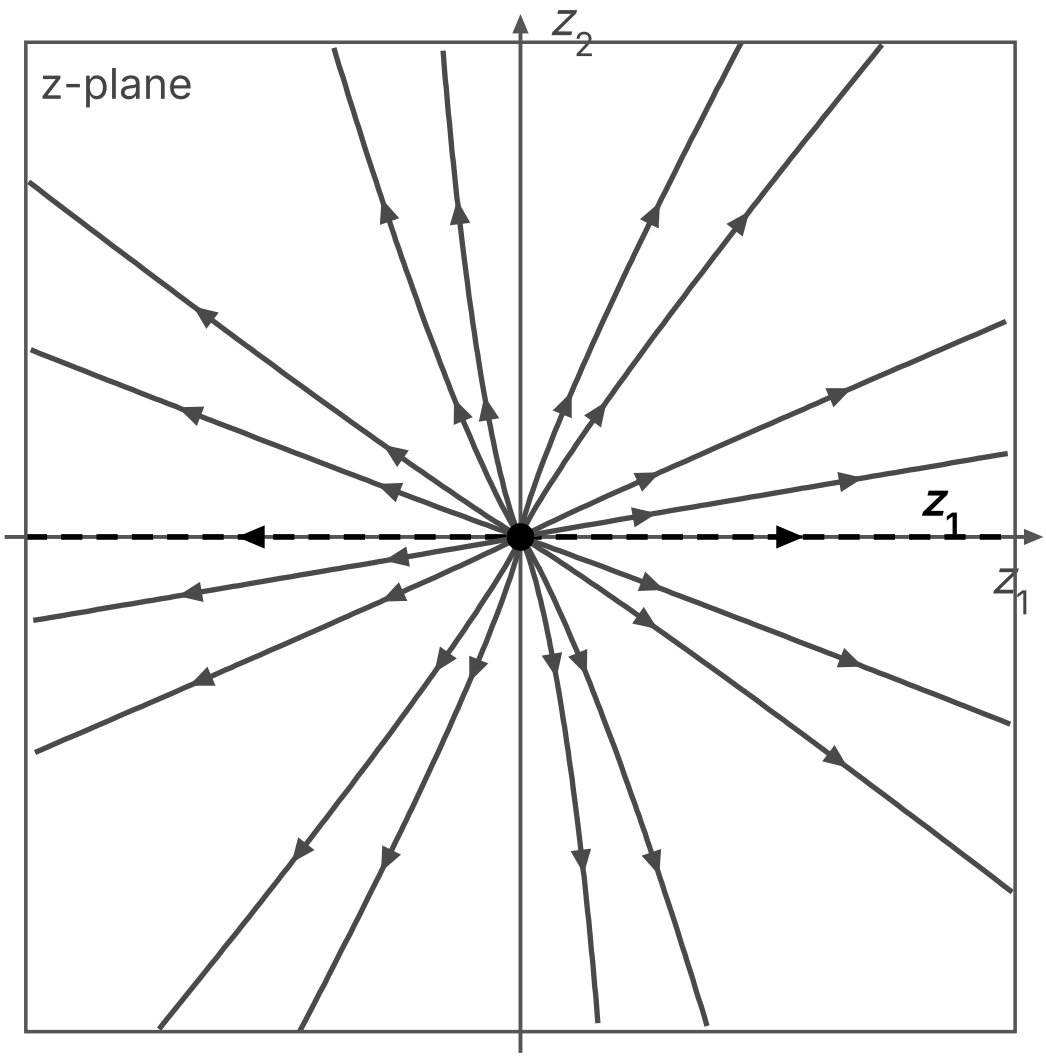
(iii) a single eigenvalue $\lambda = 13$ with eigenvector $(3, 2)$, and $\dot{x} > 0$ when $x = 0$ and $y > 0$

Single eigenvalue 13 with one eigenvector $(3, 2)^T$: $A \neq 13I$ (a star would have two independent eigenvectors), so $N = A - 13I$ is non-zero, $N^2 = O$ and $\ker N = \text{span}\{(3, 2)^T\}$. Hence $N = c\begin{pmatrix} 3 \\ 2 \end{pmatrix}(2, -3) = c\begin{pmatrix} 6 & -9 \\ 4 & -6 \end{pmatrix}$ (rows orthogonal to $(3, 2)$ so that $N(3, 2)^T = \mathbf{0}$), and $a_{12} = N_{12} = -9c$. The condition $\dot{x} > 0$ at $(0, 1)$ means $a_{12} > 0$, so $c < 0$; take $c = -1$: $N = \begin{pmatrix} -6 & 9 \\ -4 & 6 \end{pmatrix}$, $A = 13I + N = \begin{pmatrix} 7 & 9 \\ -4 & 19 \end{pmatrix}$.

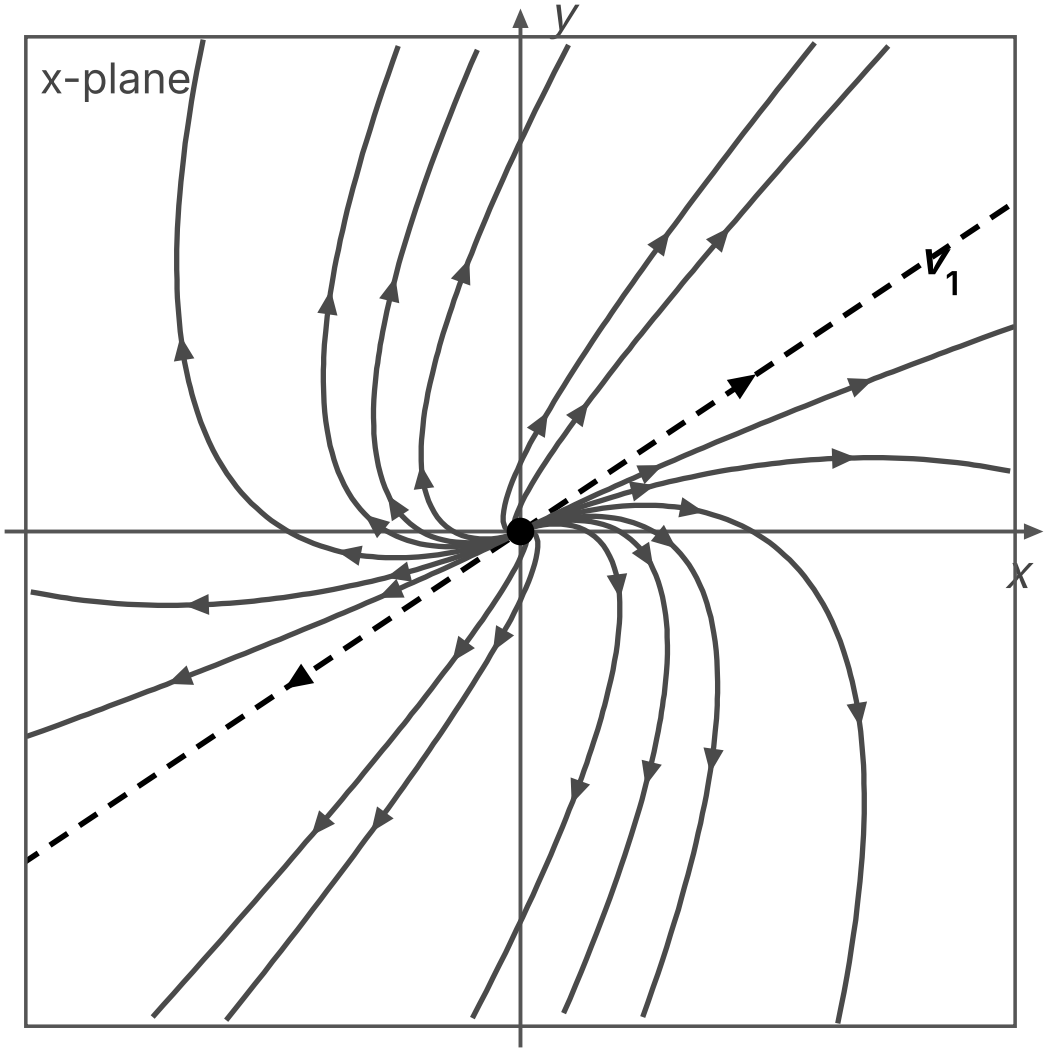
Phase portrait

$\lambda = 13$ is repeated with a single eigendirection and $A \neq \lambda I$: the origin is an **unstable improper node** (Topic 4 Sec. 8.8 and 8.9). In the **z**-plane the z_1-axis is invariant ($z_2 \equiv 0$, $z_1 = c_1 e^{13t}$, arrows away from the origin); the other orbits satisfy $z_1 = z_2\left[K + \frac{1}{\lambda}\ln|z_2|\right]$, are S-shaped, tangent to the z_1-axis at the origin and parallel to it far away. Since $\dot{z}_1 = \lambda z_1 + z_2 = z_2$ on the z_2-axis, orbits cross it moving to the right in the upper half-plane and to the left in the lower one.

Orientation in the $\mathbf{x}$-plane, from $N\mathbf{x} = (A - \lambda I)\mathbf{x} = z_2\mathbf{v}_1$ (Topic 4 Sec. 8.9): at $(0,1)^T$, $N(0,1)^T = \begin{pmatrix} 9 \\ 6 \end{pmatrix} = 3\,\mathbf{v}_1$, so there the orbit drifts along $+\mathbf{v}_1$. The orbits are tangent to the eigenline at the origin and bend into an S around it.



Canonical $\mathbf{z}$-plane (unstable improper node).



Original (x, y)-plane (unstable improper node); dashed: the eigenline with $\mathbf{v}_1 = (3, 2)$.

(iv) $\lambda_1 = -2$ **with** $\mathbf{v}_1 = (2, 1)$ **and** $\lambda_2 = -3$ **with** $\mathbf{v}_2 = (1, -1)$

Two real eigenvalues with independent eigenvectors ($\det[\mathbf{v}_1\ \mathbf{v}_2] = -3 \neq 0$). A consistent matrix is $A = Q\Lambda Q^{-1} = \begin{pmatrix} -\frac{7}{3} & \frac{2}{3} \\ \frac{1}{3} & -\frac{8}{3} \end{pmatrix}$ (checked: $A\mathbf{v}_1 = \begin{pmatrix} -4 \\ -2 \end{pmatrix} = -2\mathbf{v}_1$, $A\mathbf{v}_2 = \begin{pmatrix} -3 \\ 3 \end{pmatrix} = -3\mathbf{v}_2$).

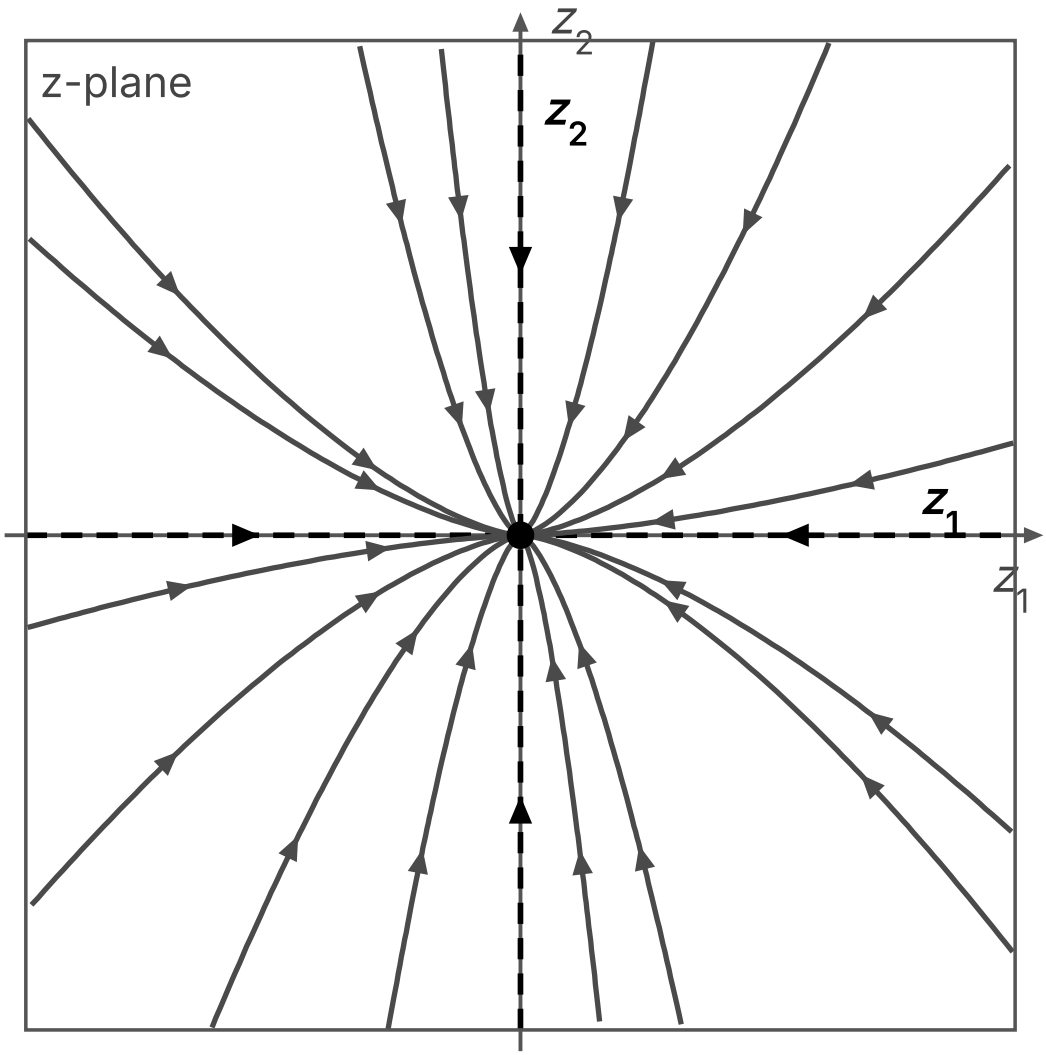
Phase portrait

$\lambda_1 = -2$ and $\lambda_2 = -3$ are both negative: the origin is a **stable node** (Robinson Ch. 28, Topic 4 Sec. 6 and 9).

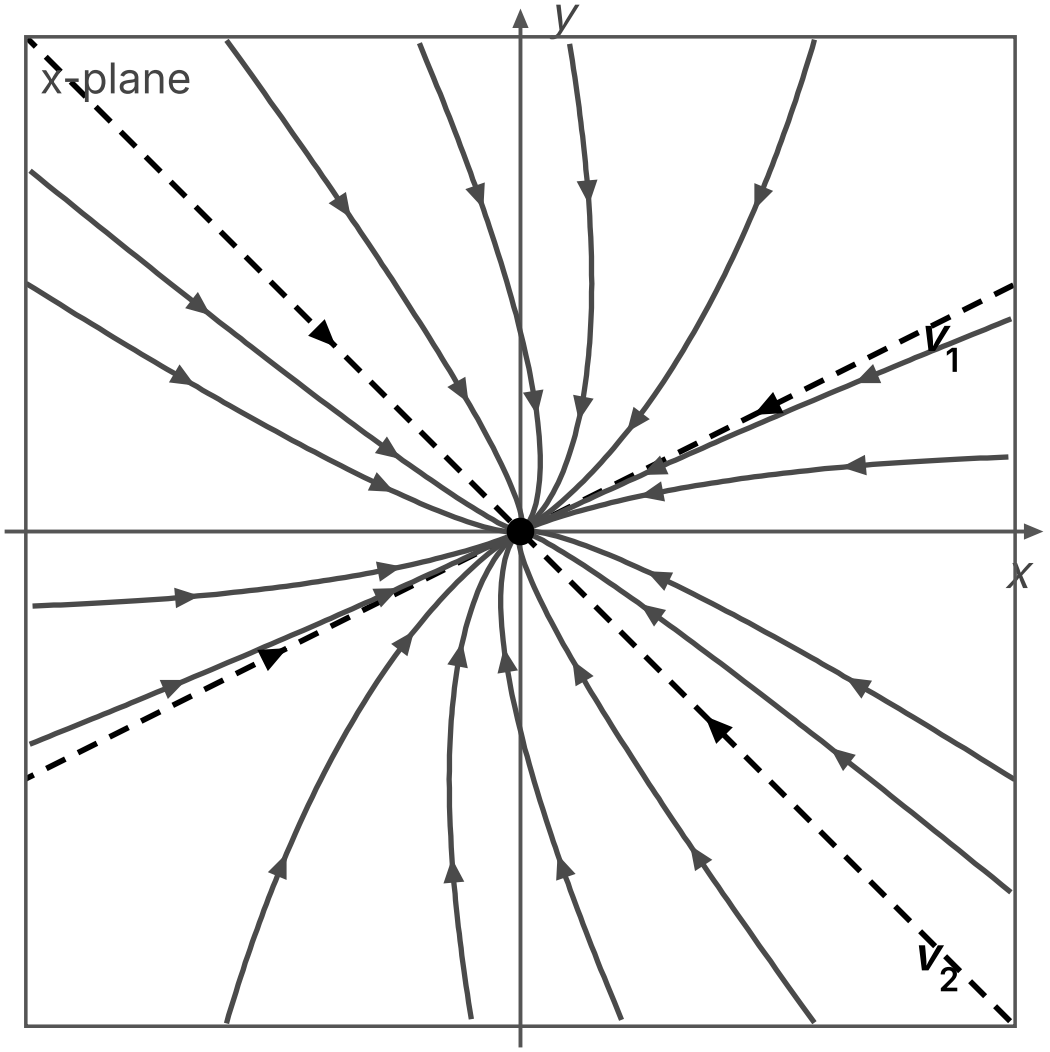
In the canonical $\mathbf{z}$-plane, eliminating t from $z_1 = c_1 e^{-2t}$ and $z_2 = c_2 e^{-3t}$ (for $c_1 \neq 0$): $|z_1|/|c_1| = e^{\lambda_1 t}$, so $e^t = (|z_1|/|c_1|)^{1/\lambda_1}$ and $|z_2| = |c_2| e^{\lambda_2 t} = |c_2| \left(|z_1|/|c_1|\right)^{\lambda_2/\lambda_1}$, i.e. $z_2 = K\,|z_1|^{\lambda_2/\lambda_1} = K\,|z_1|^{\frac{3}{2}}$ with $K = c_2 |c_1|^{-\lambda_2/\lambda_1}$ (equivalently, integrate the separable equation $\frac{dz_2}{dz_1} = \frac{\lambda_2 z_2}{\lambda_1 z_1}$ by the chain rule, as in Exercise 10). The axes $z_1 = 0$ and $z_2 = 0$ are invariant: a solution that starts on an axis stays on it.

The exponent $\lambda_2/\lambda_1 = \frac{3}{2}$ is positive, so the orbits are parabola-like curves through the origin. At the origin they are tangent to the eigenvector of the eigenvalue of **smaller modulus**, $|\lambda| = 2$, i.e. to the line through $(2, 1)$ (they arrive there as $t \to +\infty$), and far from the origin they become parallel to the other eigenvector (the fast direction). The arrows point towards the origin on both eigenlines.

The map $\mathbf{x} = Q\mathbf{z}$ sends the z_1-axis to the line through $\mathbf{v}_1 = (2, 1)$ and the z_2-axis to the line through $\mathbf{v}_2 = (1, -1)$. Straight lines stay straight and the origin stays fixed, so the $\mathbf{x}$-portrait is the canonical one sheared onto the eigenlines. On $\mathbf{v}_1$: $\dot{\mathbf{x}} = A\mathbf{v}_1 = -2\mathbf{v}_1$ points inward; on $\mathbf{v}_2$: $\dot{\mathbf{x}} = -3\mathbf{v}_2$ points inward.



Canonical $\mathbf{z}$-plane (stable node): invariant axes and orbits $z_2 = K|z_1|^{\frac{3}{2}}$.



Original (x, y)-plane (stable node): dashed lines are the eigenlines $\mathbf{v}_1 = (2, 1)$ and $\mathbf{v}_2 = (1, -1)$.

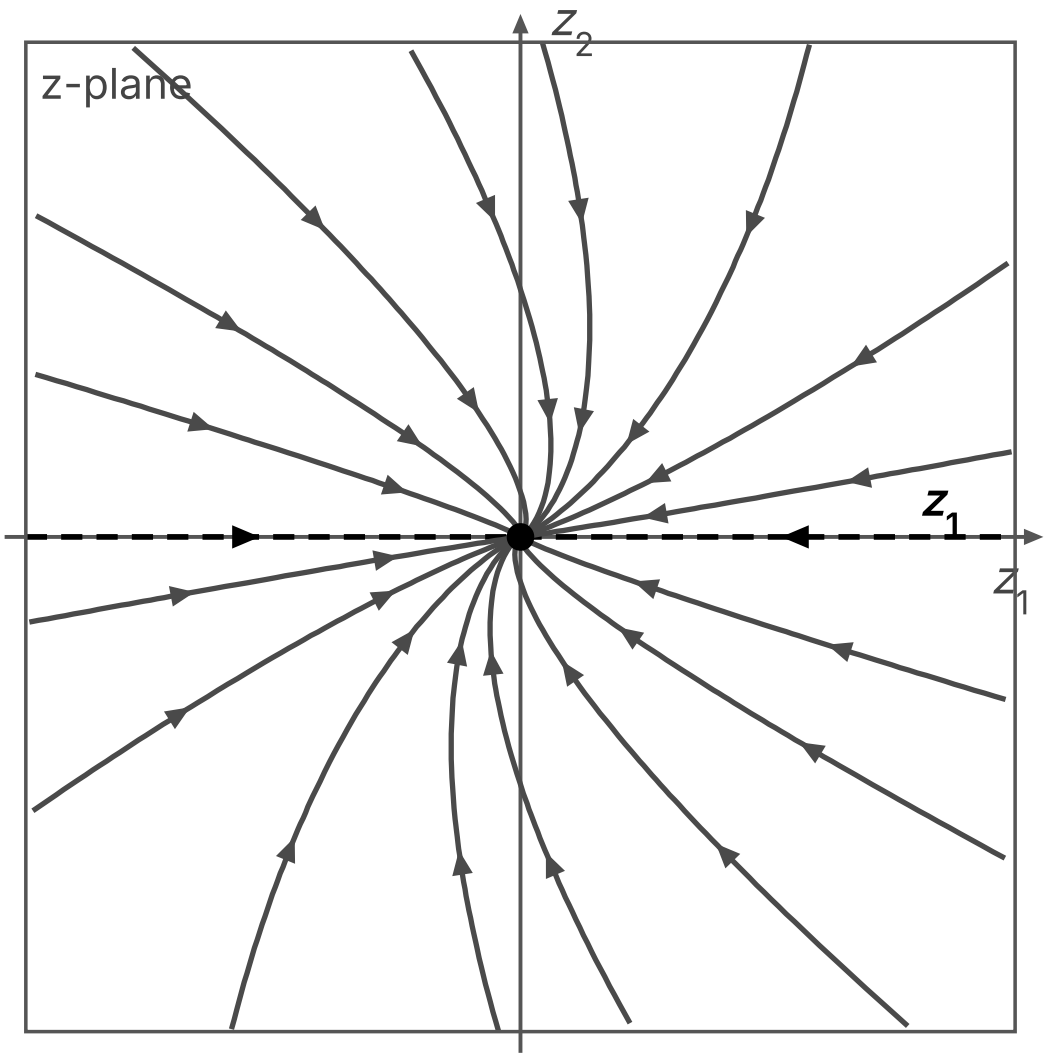
(v) a single eigenvalue $\lambda = -3$ with eigenvector $(1, -1)$, and $\dot{x} > 0$ when $x = 0$ and $y > 0$

Single eigenvalue -3 with one eigenvector $(1, -1)^T$: $A \neq -3I$ and $N = A + 3I \neq O$ with $\ker N = \text{span}\{(1, -1)^T\}$. Hence $N = c \begin{pmatrix} 1 & 1 \\ -1 & -1 \end{pmatrix}$ and $a_{12} = N_{12} = c$. The condition $\dot{x} > 0$ at $(0, 1)$ means $a_{12} > 0$, so $c > 0$; take $c = 1$: $N = \begin{pmatrix} 1 & 1 \\ -1 & -1 \end{pmatrix}$, $A = -3I + N = \begin{pmatrix} -2 & 1 \\ -1 & -4 \end{pmatrix}$.

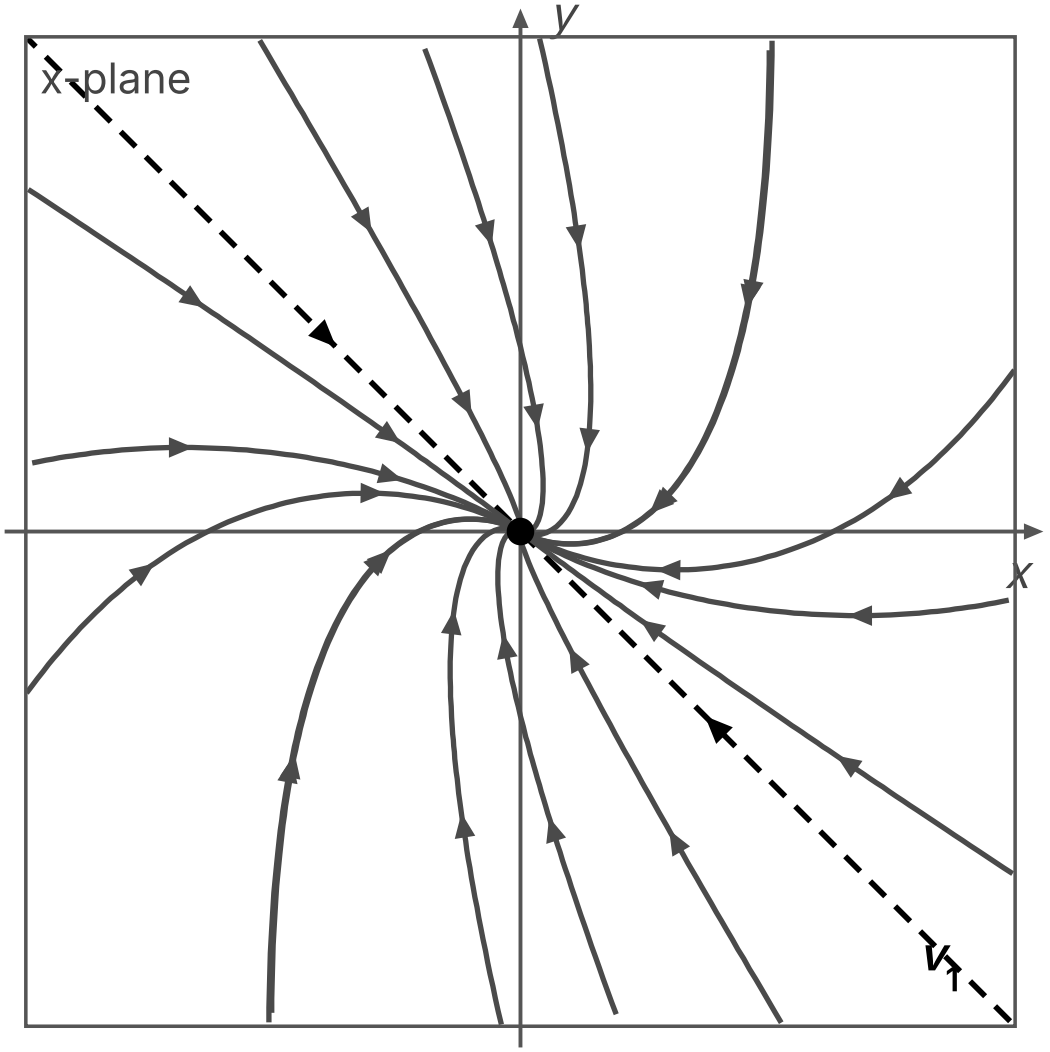
Phase portrait

$\lambda = -3$ is repeated with a single eigendirection and $A \neq \lambda I$: the origin is an **stable improper node** (Topic 4 Sec. 8.8 and 8.9). In the **z**-plane the z_1-axis is invariant ($z_2 \equiv 0$, $z_1 = c_1 e^{-3t}$, arrows towards the origin); the other orbits satisfy $z_1 = z_2 \left[K + \frac{1}{\lambda} \ln |z_2|\right]$, are S-shaped, tangent to the z_1-axis at the origin and parallel to it far away. Since $\dot{z}_1 = \lambda z_1 + z_2 = z_2$ on the z_2-axis, orbits cross it moving to the right in the upper half-plane and to the left in the lower one.

Orientation in the $\mathbf{x}$-plane, from $N\mathbf{x} = (A - \lambda I)\mathbf{x} = z_2\mathbf{v}_1$ (Topic 4 Sec. 8.9): at $(0,1)^T$, $N(0,1)^T = \begin{pmatrix} 1 \\ -1 \end{pmatrix} = 1\,\mathbf{v}_1$, so there the orbit drifts along $+\mathbf{v}_1$. The orbits are tangent to the eigenline at the origin and bend into an S around it.



Canonical $\mathbf{z}$-plane (stable improper node).



Original (x, y)-plane (stable improper node); dashed: the eigenline with $\mathbf{v}_1 = (1, -1)$.

(vi) $\lambda_{\pm} = \pm 2i$**, where** $\dot{y} < 0$ **when** $y = 0$ **and** $x > 0$

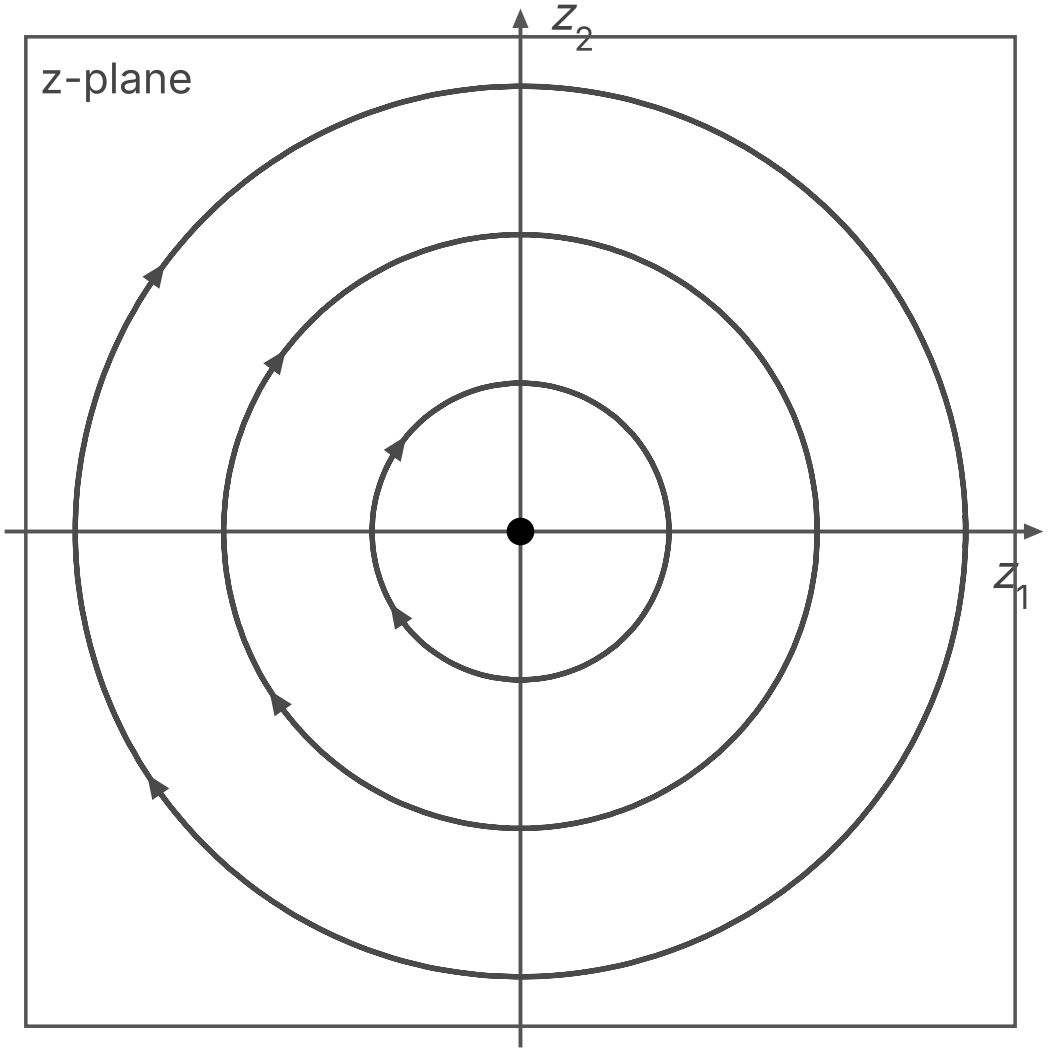
$\dot{y} = a_{21}x + a_{22}y$ at $(1, 0)$ equals a_{21}, and $\dot{y} < 0$ there, so $a_{21} < 0$: at the right-hand side of the plane the motion is downwards, i.e. **clockwise**. A consistent matrix with eigenvalues $\pm 2i$ and $a_{21} < 0$ is $A = \begin{pmatrix} 0 & 2 \\ -2 & 0 \end{pmatrix}$ ($\operatorname{tr} A = 0$, $\det A = 4$, eigenvalues $\pm 2i$; $a_{21} = -2 < 0$).

Phase portrait

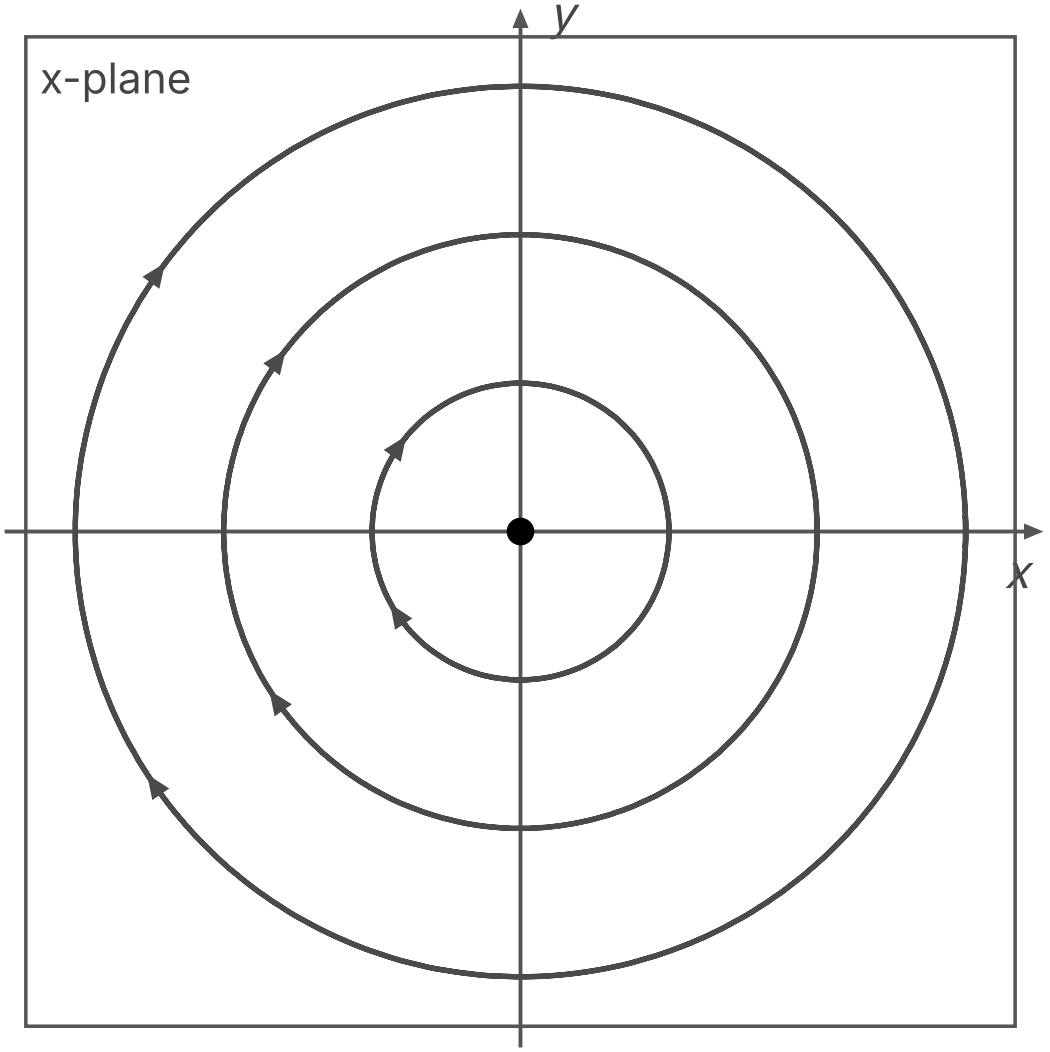
$\rho = 0$ and $\omega = 2$: the origin is a **centre**. In the $\mathbf{z}$-plane $r = r_0 e^{0t}$ and $\theta = \theta_0 - 2t$: the motion is always **clockwise** ($\dot{\theta} = -\omega < 0$), and the orbits are circles of period $T = 2\pi/\omega$ (closed orbits).

Sense of rotation in the $\mathbf{x}$-plane, tested directly on the vector field (Topic 4 Sec. 7.8): at $(0, 1)^T$, $\dot{\mathbf{x}} = A(0, 1)^T = \begin{pmatrix} 2 \\ 0 \end{pmatrix}$, so $\dot{x} = 2$ is > 0: to the right at the top of the plane: the rotation is **clockwise**.

Cross-check: $\det Q = 4$ is > 0, so Q preserves orientation and the clockwise motion of the $\mathbf{z}$-plane becomes clockwise in the $\mathbf{x}$-plane, in agreement.



Canonical $\mathbf{z}$-plane (centre): clockwise, $\dot{\theta} = -\omega$.



Original (x, y)-plane (centre), clockwise.

(vii) $\lambda_1 = 1$ **with** $\mathbf{v}_1 = (3, 2)$ **and** $\lambda_2 = 5$ **with** $\mathbf{v}_2 = (1, -4)$

Two real eigenvalues with independent eigenvectors ($\det[\mathbf{v}_1\ \mathbf{v}_2] = -14 \neq 0$). A consistent matrix is $A = Q\Lambda Q^{-1} = \begin{pmatrix} \frac{11}{7} & -\frac{6}{7} \\ -\frac{16}{7} & \frac{31}{7} \end{pmatrix}$ (checked: $A\mathbf{v}_1 = \begin{pmatrix} 3 \\ 2 \end{pmatrix} = 1\mathbf{v}_1$, $A\mathbf{v}_2 = \begin{pmatrix} 5 \\ -20 \end{pmatrix} = 5\mathbf{v}_2$).

Phase portrait

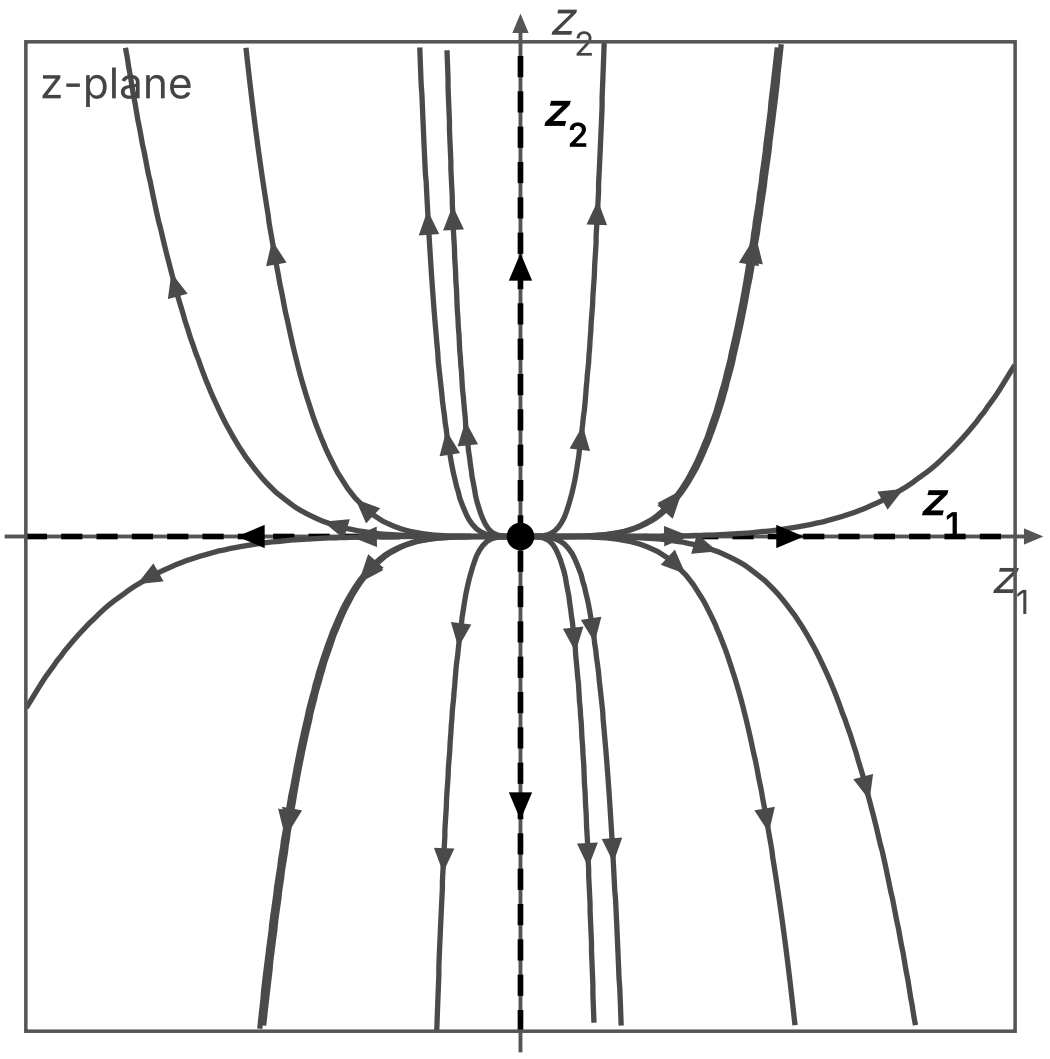
$\lambda_1 = 1$ and $\lambda_2 = 5$ are both positive: the origin is a **unstable node** (Robinson Ch. 28, Topic 4 Sec. 6 and 9).

In the canonical $\mathbf{z}$-plane, eliminating t from $z_1 = c_1 e^{1t}$ and $z_2 = c_2 e^{5t}$ (for $c_1 \neq 0$): $|z_1|/|c_1| = e^{\lambda_1 t}$, so $e^t = (|z_1|/|c_1|)^{1/\lambda_1}$ and $|z_2| = |c_2| e^{\lambda_2 t} = |c_2| \left(|z_1|/|c_1|\right)^{\lambda_2/\lambda_1}$, i.e. $z_2 = K\,|z_1|^{\lambda_2/\lambda_1} = K\,|z_1|^5$ with $K = c_2 |c_1|^{-\lambda_2/\lambda_1}$ (equivalently, integrate the separable equation $\frac{dz_2}{dz_1} = \frac{\lambda_2 z_2}{\lambda_1 z_1}$ by the chain rule, as in Exercise 10). The axes $z_1 = 0$ and $z_2 = 0$ are invariant: a solution that starts on an axis stays on it.

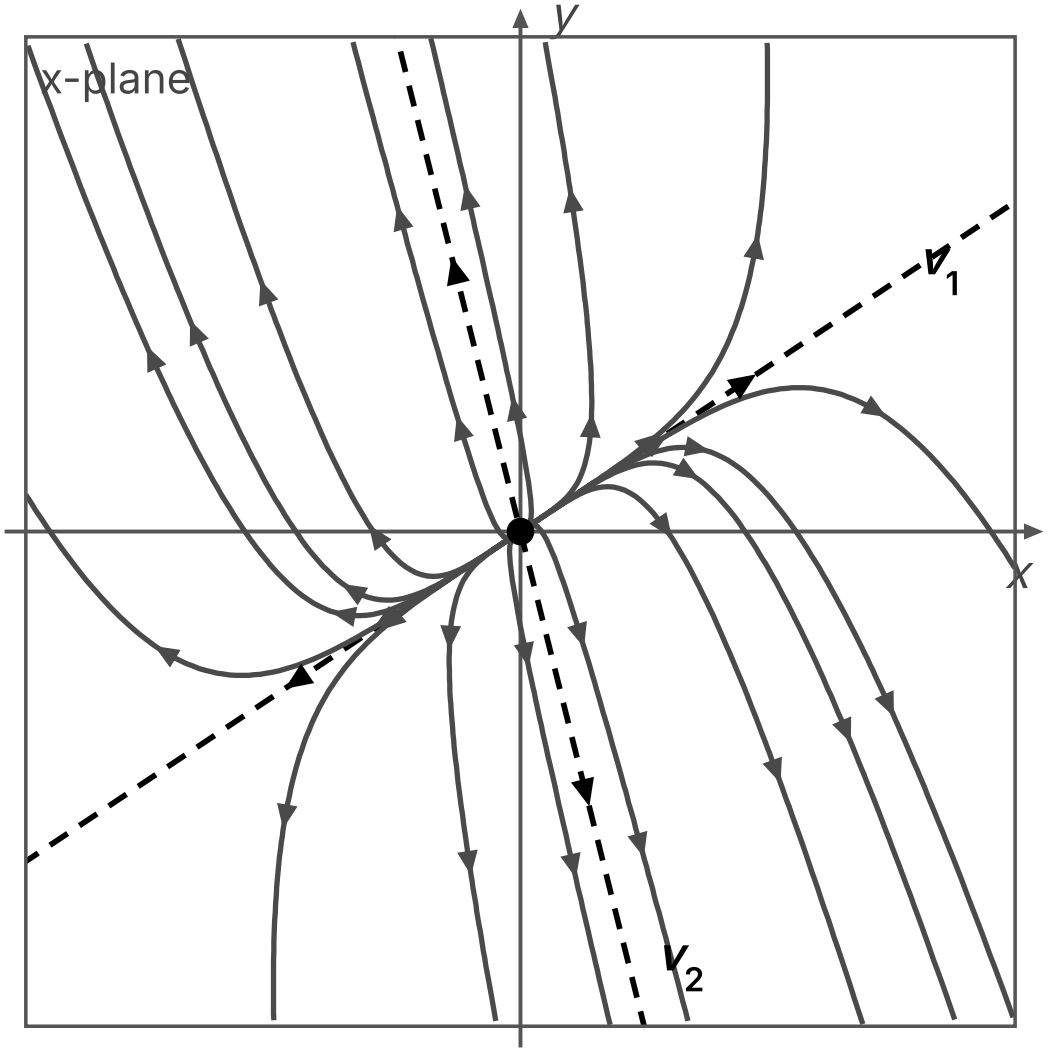
The exponent $\lambda_2/\lambda_1 = 5$ is positive, so the orbits are parabola-like curves through the origin. At the origin they are tangent to the eigenvector of the eigenvalue of **smaller modulus**, $|\lambda| = 1$, i.e. to the line through $(3, 2)$ (they leave from there (the limit $t \to -\infty$)), and far from the origin they become

parallel to the other eigenvector (the fast direction). The arrows point away from the origin on both eigenlines.

The map $\mathbf{x} = Q\mathbf{z}$ sends the z_1-axis to the line through $\mathbf{v}_1 = (3, 2)$ and the z_2-axis to the line through $\mathbf{v}_2 = (1, -4)$. Straight lines stay straight and the origin stays fixed, so the $\mathbf{x}$-portrait is the canonical one sheared onto the eigenlines. On $\mathbf{v}_1$: $\dot{\mathbf{x}} = A\mathbf{v}_1 = 1\mathbf{v}_1$ points outward; on $\mathbf{v}_2$: $\dot{\mathbf{x}} = 5\mathbf{v}_2$ points outward.



Canonical $\mathbf{z}$-plane (unstable node): invariant axes and orbits $z_2 = K|z_1|^5$.



Original (x, y)-plane (unstable node): dashed lines are the eigenlines $\mathbf{v}_1 = (3, 2)$ and $\mathbf{v}_2 = (1, -4)$.

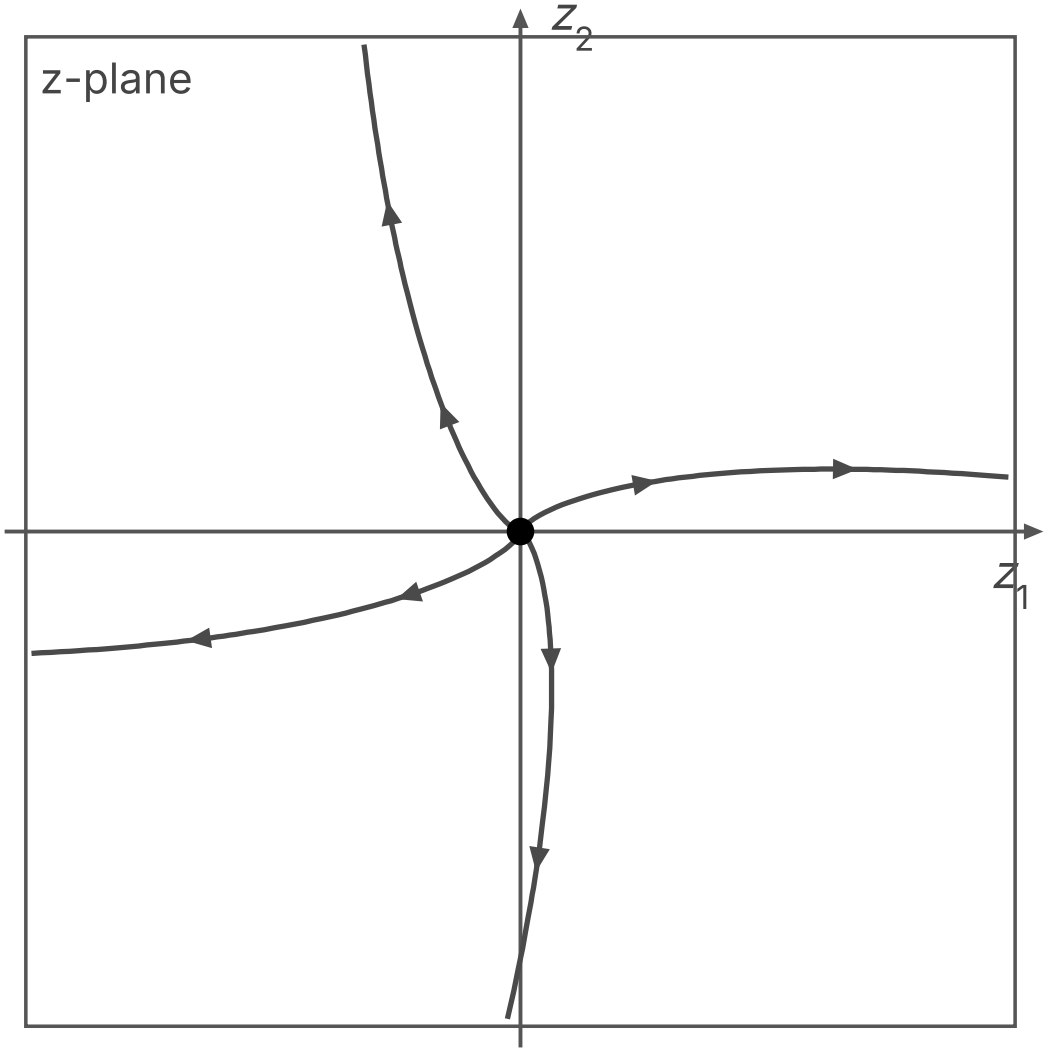
(viii) $\lambda_\pm = 5 \pm i$**, and** $\dot{y} > 0$ **when** $y = 0$ **and** $x > 0$

$\dot{y} > 0$ at $(1, 0)$ gives $a_{21} > 0$: upwards on the right-hand side, i.e. **counter-clockwise**. A consistent matrix with eigenvalues $5 \pm i$ is $A = \begin{pmatrix} 5 & -1 \\ 1 & 5 \end{pmatrix}$ (tr $A = 10$, det $A = 26$, $\lambda = 5 \pm i$; $a_{21} = 1 > 0$).

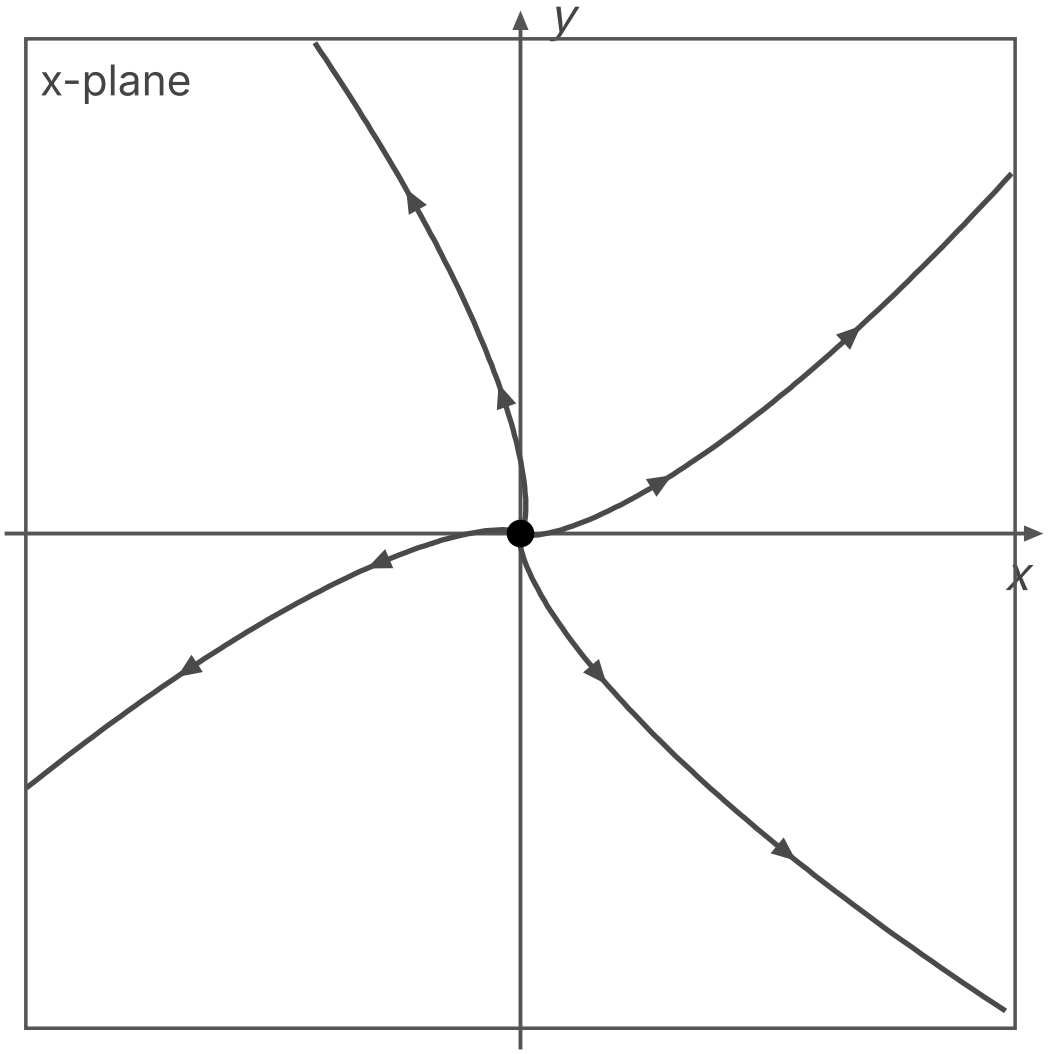
Phase portrait

$\rho = 5$ and $\omega = 1$: the origin is a **unstable spiral**. In the $\mathbf{z}$-plane $r = r_0 e^{5t}$ and $\theta = \theta_0 - 1t$: the motion is always **clockwise** ($\dot{\theta} = -\omega < 0$), and one full turn takes $T = 2\pi/\omega$ while the radius is multiplied by $e^{2\pi\rho/\omega} = e^{10\pi}$ (growing).

Sense of rotation in the $\mathbf{x}$-plane, tested directly on the vector field (Topic 4 Sec. 7.8): at $(0, 1)^T$, $\dot{\mathbf{x}} = A(0, 1)^T = \begin{pmatrix} -1 \\ 5 \end{pmatrix}$, so $\dot{x} = -1$ is < 0: to the left at the top of the plane: the rotation is **counter-clockwise**. Cross-check: det $Q = -1$ is < 0, so Q reverses orientation and the clockwise motion of the $\mathbf{z}$-plane becomes counter-clockwise in the $\mathbf{x}$-plane, in agreement.



Canonical $\mathbf{z}$-plane (unstable spiral): clockwise, $\dot{\theta} = -\omega$.



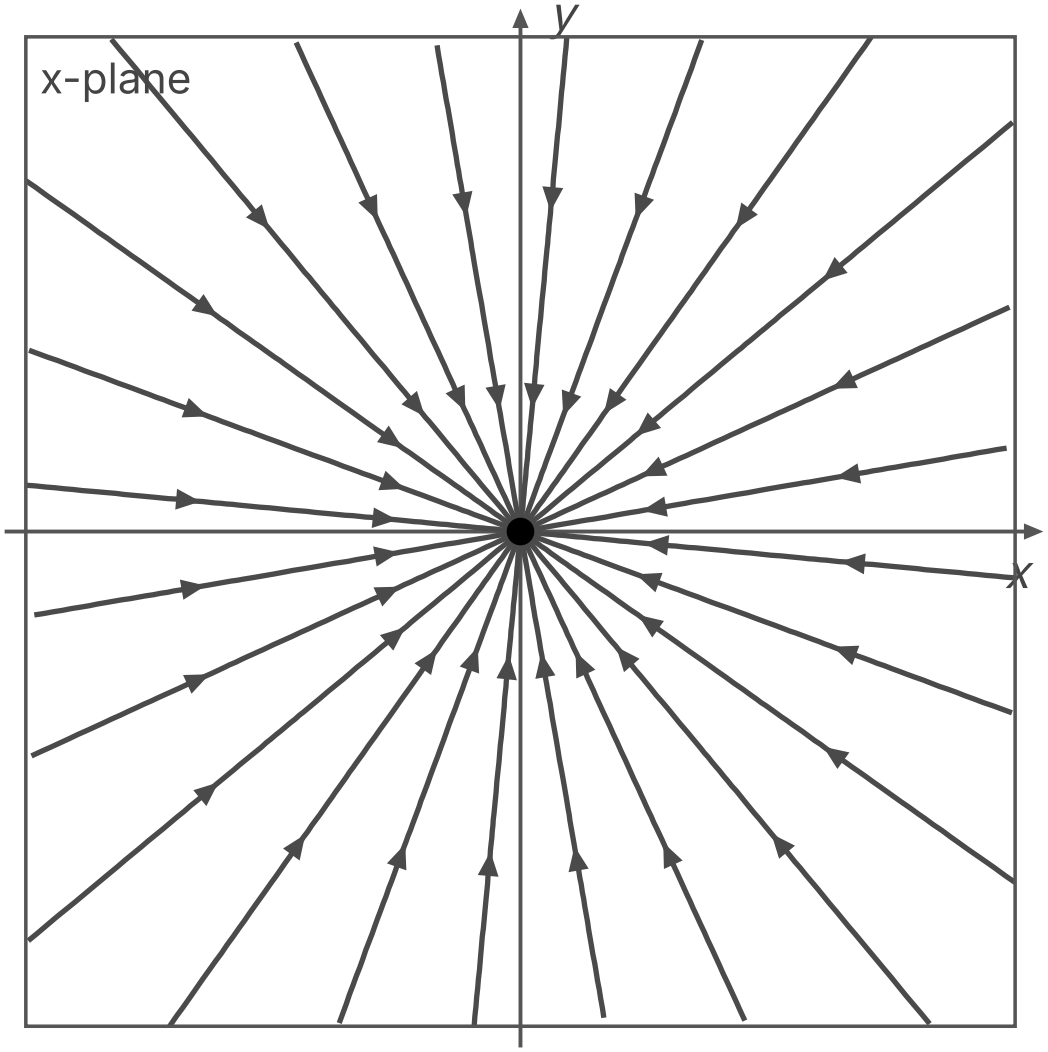
Original (x, y)-plane (unstable spiral), counter-clockwise.

(ix) a single eigenvalue $\lambda = -7$, with the matrix A being a multiple of the identity

A is a multiple of the identity with eigenvalue -7: $A = \begin{pmatrix} -7 & 0 \\ 0 & -7 \end{pmatrix}$.

Phase portrait

$A = -7I$: every line through the origin is invariant and every solution moves along its own ray, $\mathbf{x}(t) = e^{-7t}\mathbf{x}_0$, towards the origin (the origin is a **stable star**). The picture is the same in every basis, so there is no separate canonical plane.



Stable star: straight rays into the origin.

PHASE 4: INTERPRETATION AND CHECKS

Item	Origin	Stability
(i)	saddle	unstable
(ii)	spiral, counter-clockwise	stable
(iii)	improper node	unstable
(iv)	node, tangent to $\mathbf{v}_1 = (2, 1)$	stable
(v)	improper node	stable
(vi)	centre, clockwise	stable, not asymptotically
(vii)	node, tangent to $\mathbf{v}_1 = (3, 2)$	unstable
(viii)	spiral, counter-clockwise	unstable
(ix)	star	asymptotically stable

- The nine items cover the nine pictures of Robinson's classification (Ch. 31, Fig. 31.1, as reproduced in Topic 4 Fig. 4.11; the unstable star is the time reversal of the stable one): saddle (i); stable node (iv); unstable node (vii); stable spiral (ii); unstable spiral (viii); centre (vi); stable star (ix); stable improper node (v); unstable improper node (iii).
- The tangency of nodes was verified on the numerical orbits: the orbits of (iv) are tangent at the origin to $\mathbf{v}_1 = (2, 1)$ ($|\lambda| = 2 < 3$) and those of (vii) to $\mathbf{v}_1 = (3, 2)$ ($1 < 5$).
- In (iii) and (v) the sign condition only chooses the orientation of the S-shaped orbits (on which side of the eigenline the orbit moves along $+\mathbf{v}_1$); the eigenline and the type do not depend on it.
- Trace-determinant check: (i) $\det A = -6 < 0$ saddle; (iv) $\mathrm{tr} = -5$, $\det = 6$, $\Delta = 1 > 0$ stable node; (vii) $\mathrm{tr} = 6$, $\det = 5$ unstable node; (iii) $\mathrm{tr}^2 = 4\det = 676$; (ix) $\mathrm{tr}^2 = 4\det = 196$.
- Spiral (viii) has $\rho/\omega = 5$: one turn multiplies the radius by $e^{2\pi\rho/\omega} = e^{10\pi} \approx 4 \times 10^{13}$, so it is a genuine spiral that, on the scale of the figure, looks like almost radial curves (the winding is invisible after the first fraction of a turn); in (ii), $\rho/\omega = -\frac{1}{3}$ the winding is clearly visible.